

CompTIA Linux+ (XK0-005)

File and Directory Management in Linux

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 - **Lab Topology**
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Introduction

XK0-005

Create Files

Move Files

Basic Linux Commands

Hard Link

Soft Link

Welcome to the **File and Directory Management in Linux** Practice Lab. In this module, you will be provided with the instructions and devices needed to develop your hands-on skills.

Learning Outcomes

In this module, you will complete the following exercises:

- Exercise 1 - Using the Linux Terminal
- Exercise 2 - Navigate the Linux File System
- Exercise 3 - Create Files in the Linux File System
- Exercise 4 - Move Files and Create Links

After completing this module, you should be able to:

- Launch the GNOME Terminal
- Customize the Terminal

- Use Basic Commands to Navigate the Linux File System
- Create Directories in the Linux File System
- Create and View Files using Linux Commands
- Move and View Files using Linux Commands
- Create Hard Links and Soft Links

Exam Objectives

The following exam objectives are covered in this module:

1.2 Given a scenario, manage files and directories

- Copying files between systems
- File and directory operations
- Soft and hard links

Lab Duration

It will take approximately **1 hour** to complete this lab.

Help and Support

For more information on using Practice Labs, please see our **Help and Support** page. You can also raise a technical support ticket from this page.

Click **Next** to view the Lab topology used in this module.

Lab Topology

During your session, you will have access to the following lab configuration.

Depending on the exercises, you may or may not use all of the devices, but they are shown here in the layout to get an overall understanding of the topology of the lab.

- **PLABSRV01** - Ubuntu 20.04 - Stand-Alone Ubuntu Server (Console)
- **PLABSRV02** - Alma Linux 8.5 - Stand-Alone Linux Server (GUI)
- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)
- **PLABWIN10** - Windows 10 - Stand-Alone Workstation

Click **Next** to proceed to the first exercise.

Exercise 1 - Using the Linux Terminal

In this exercise, you will access the built-in terminal in the Ubuntu device and learn how to customize it.

Learning Outcomes

After completing this exercise, you should be able to:

- Launch the GNOME Terminal
- Customize the Terminal

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Launch the GNOME Terminal

In this task, you will start the GNOME Terminal and familiarize yourself with the command line interface.

Step 1

Connect to **PLABCLIENT01**.

Select the user **administrator**.

Type the following **Password** and press **Enter**:

Passw0rd

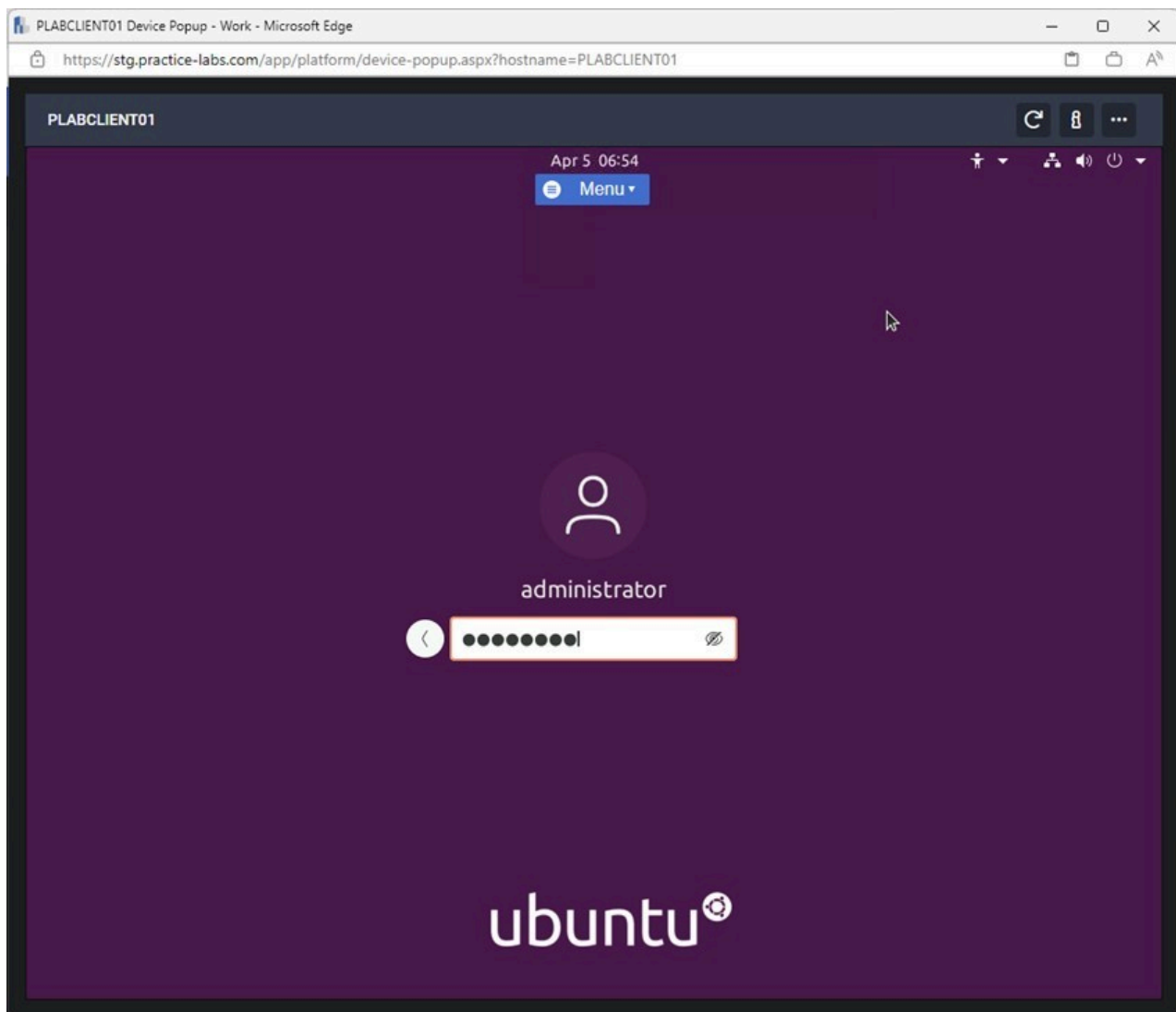


Figure 1.1 Screenshot of PLABCLIENT01: Ubuntu login screen is displayed, showing the required Password entered.

Step 2

Once logged in, click the **Show Applications** icon on the left pane.

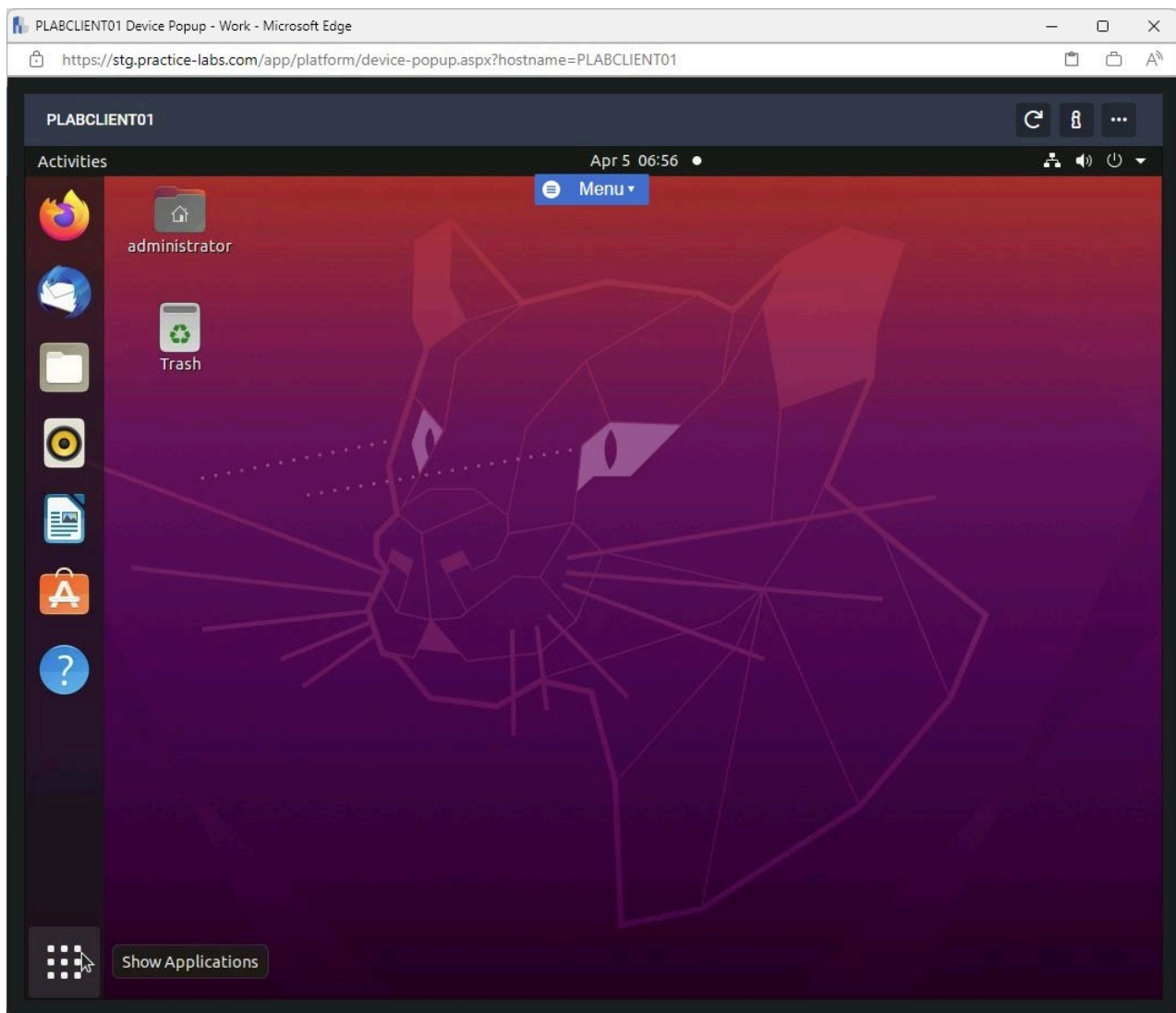


Figure 1.2 Screenshot of PLABCLIENT01: Show Applications icon on the Ubuntu desktop is selected.

Step 3

In the **Applications** window, scroll down and right-click on **Terminal** and select **Add to Favorites**.

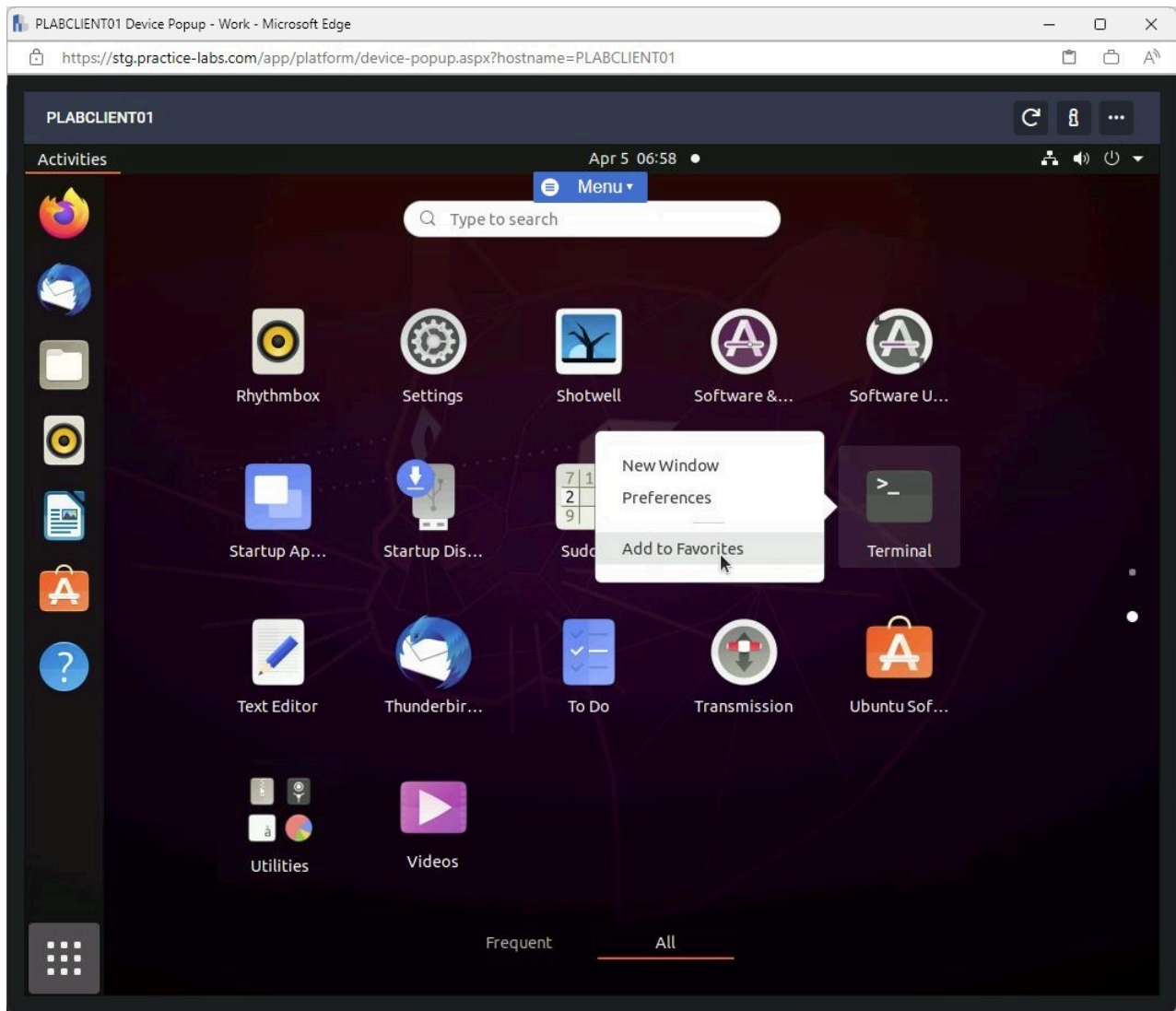


Figure 1.3 Screenshot of PLABCLIENT01: Applications window is displayed, showing Terminal right-clicked and Add to Favorites selected.

Step 4

Click the **Terminal** icon on the left pane.

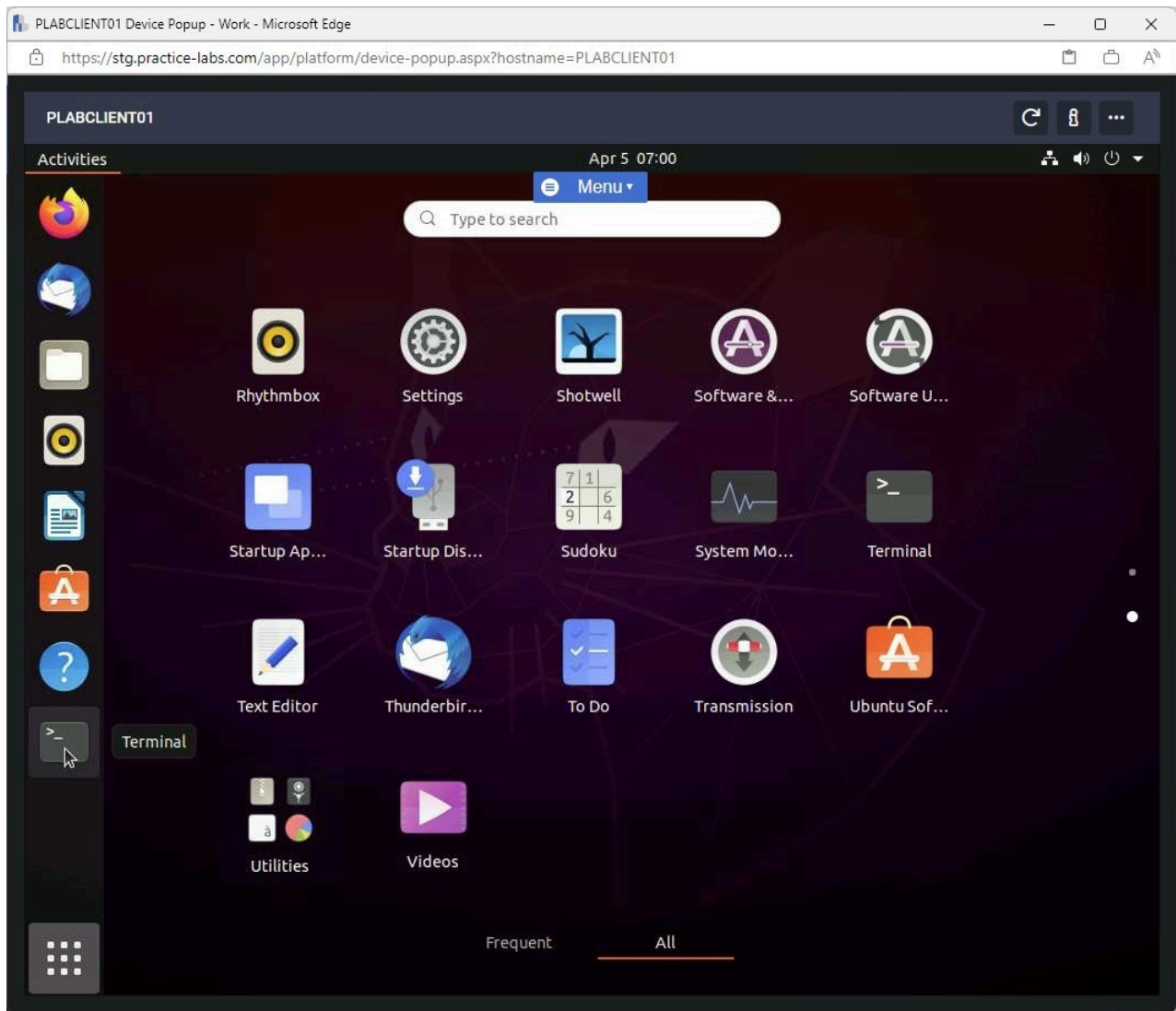


Figure 1.4 Screenshot of PLABCLIENT01: Terminal icon on the left pane is selected.

Step 5

The **Terminal** window is displayed.

Keep the **Terminal** window open.

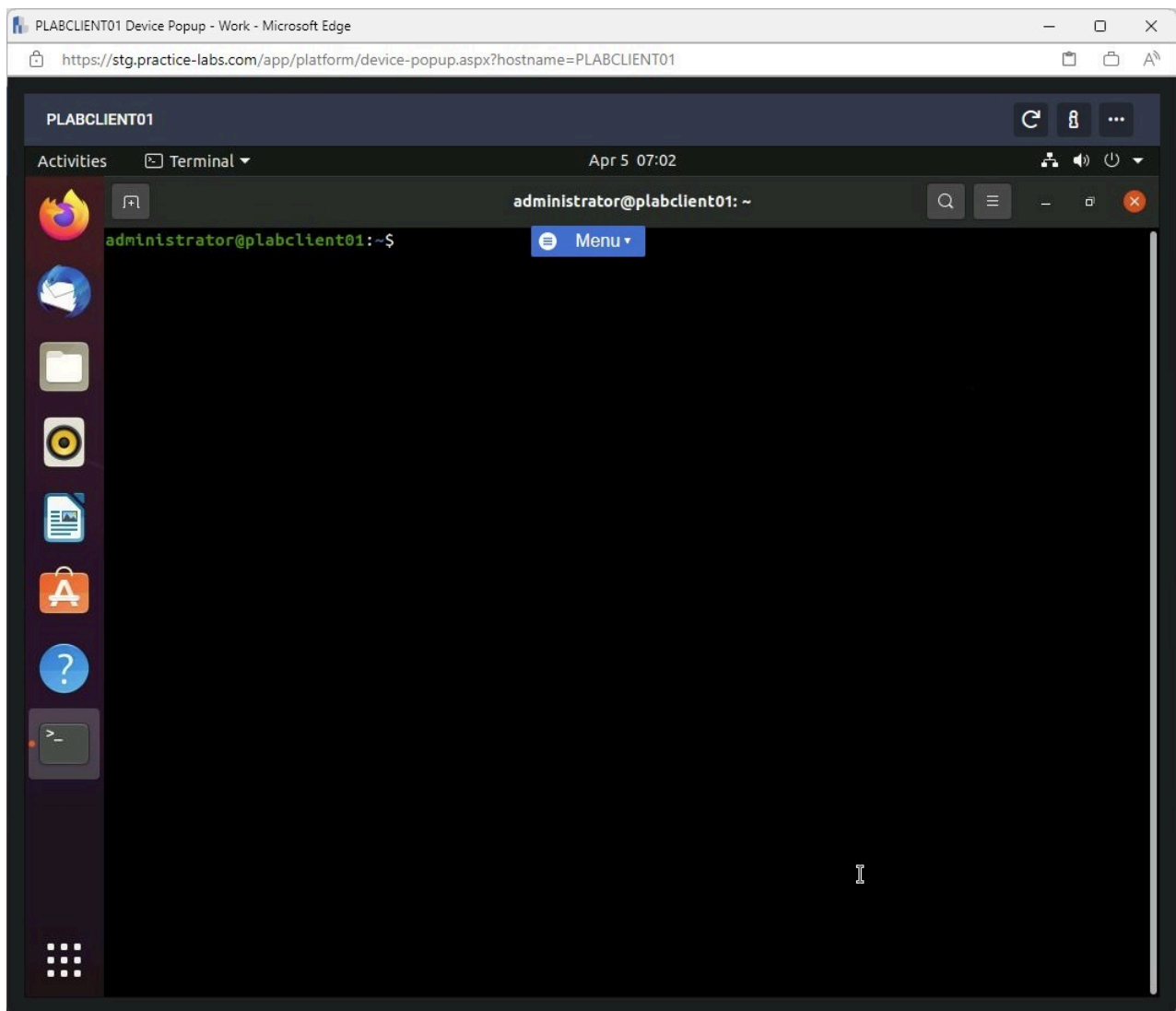


Figure 1.5 Screenshot of PLABCLIENT01: The Terminal window is displayed.

Task 2 - Customize the Terminal

In this task, you will customize the GNOME Terminal.

Step 1

Ensure you are connected to **PLABCLIENT01**, where the **Terminal** window is open.

Notice there are 3 icons on the top part of the **Terminal** window:

- Search (Magnifying glass icon on the upper-right corner)
- Menu (Three parallel horizontal lines icon on the upper-right corner)
- Tabs (Symbol with a + sign on the upper-left corner)

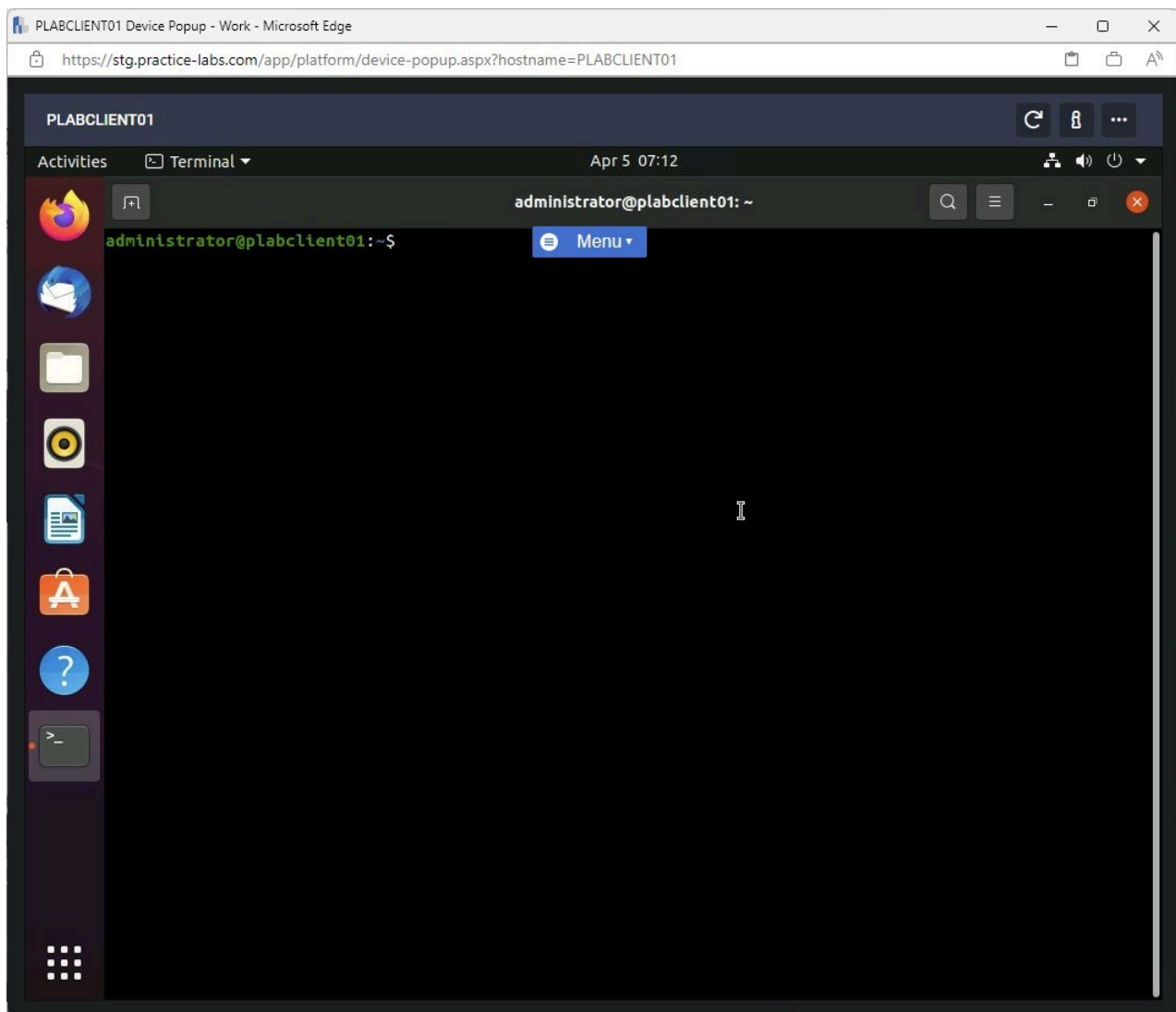


Figure 1.6 Screenshot of PLABCLIENT01: The Terminal window is displayed.

Step 2

Click on the **Menu** icon and select **Preferences** from the drop-down menu.

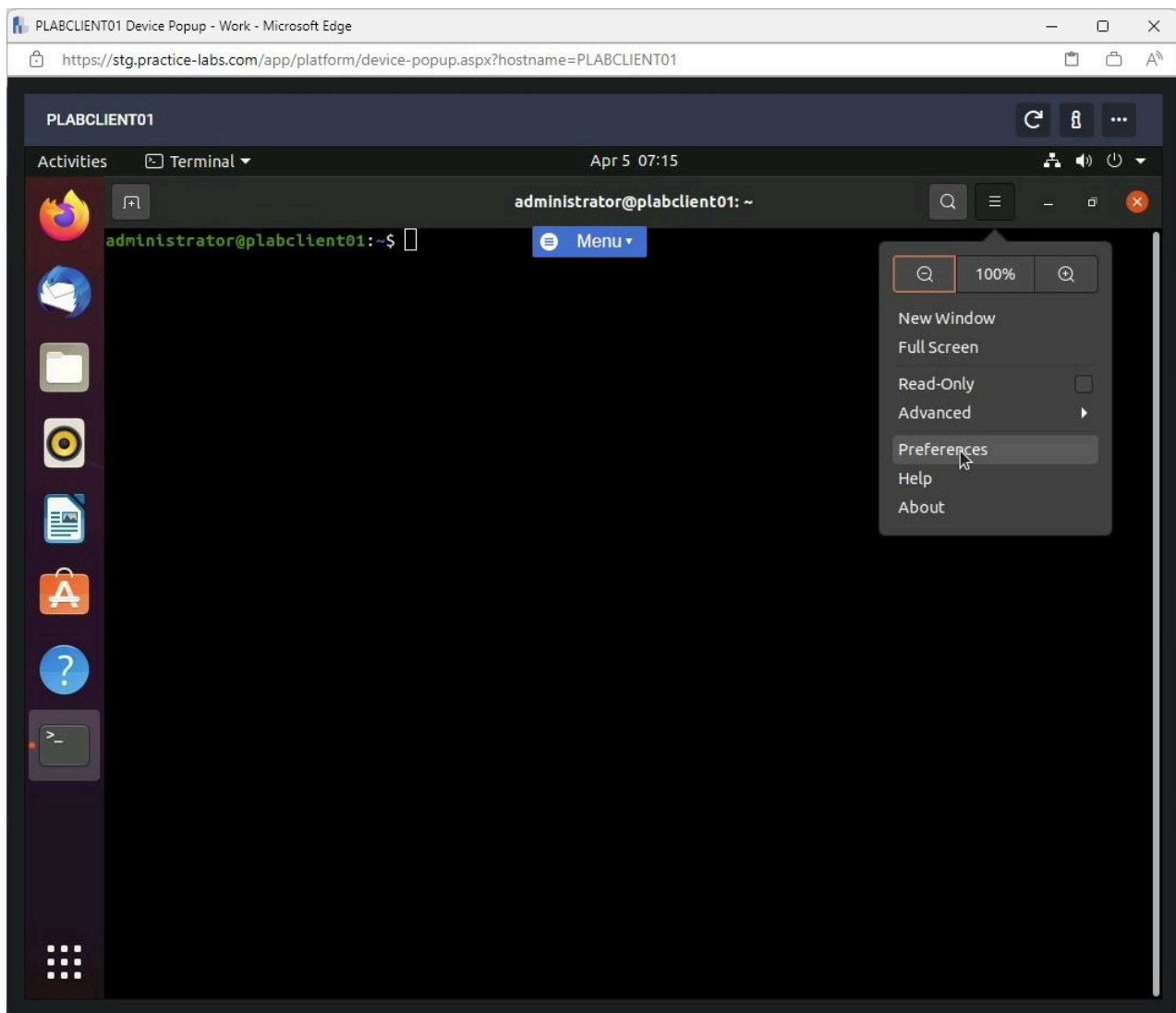


Figure 1.7 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the Menu icon accessed and Preferences selected from the drop-down menu.

Step 3

In the **Preference - General** window, click on the **Theme variant** drop-down list and select **Light**.

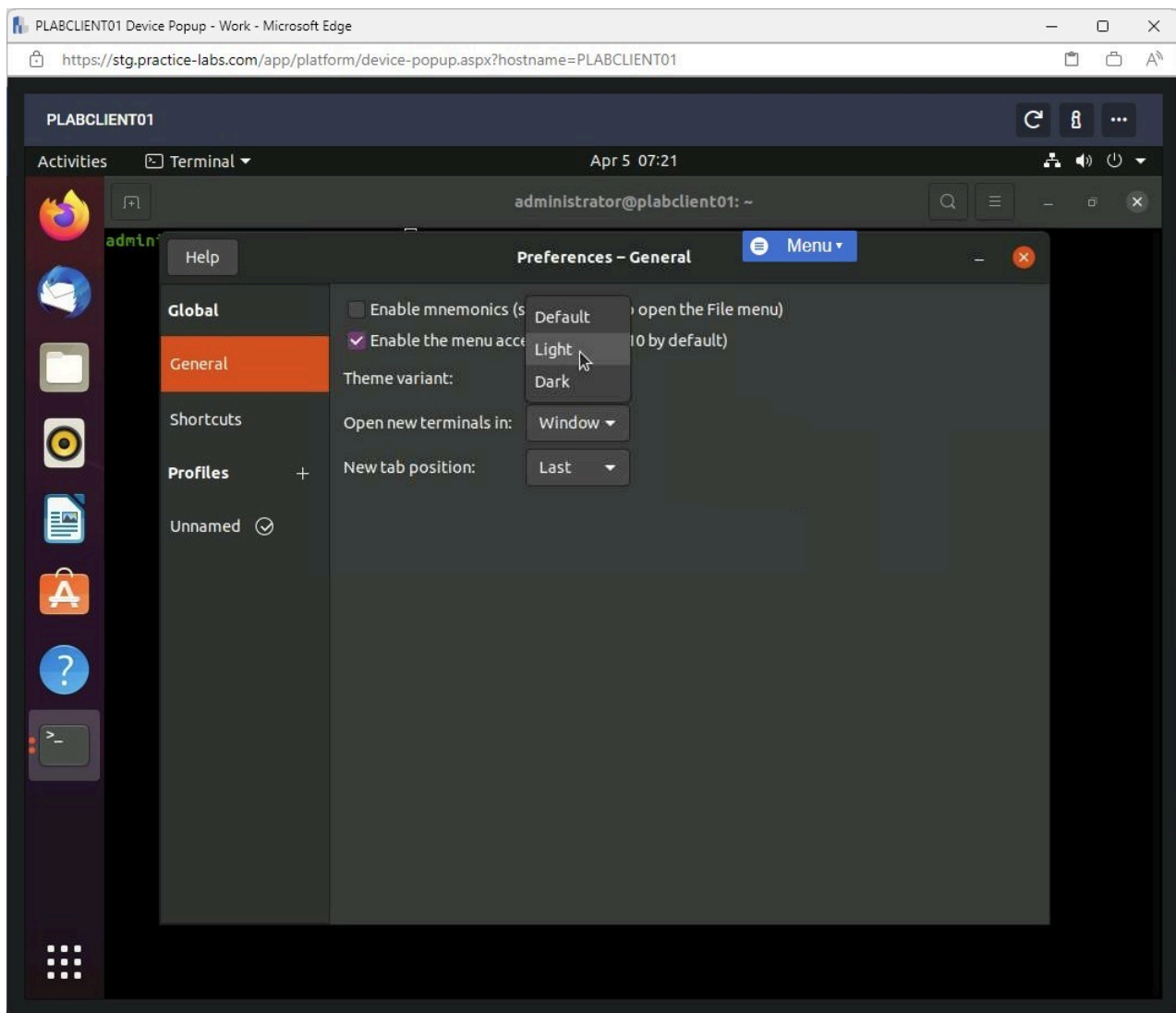


Figure 1.8 Screenshot of PLABCLIENT01: Preference - General window is displayed, showing the Theme variant drop-down accessed and Light selected.

Step 4

Select **Unnamed** on the left pane.

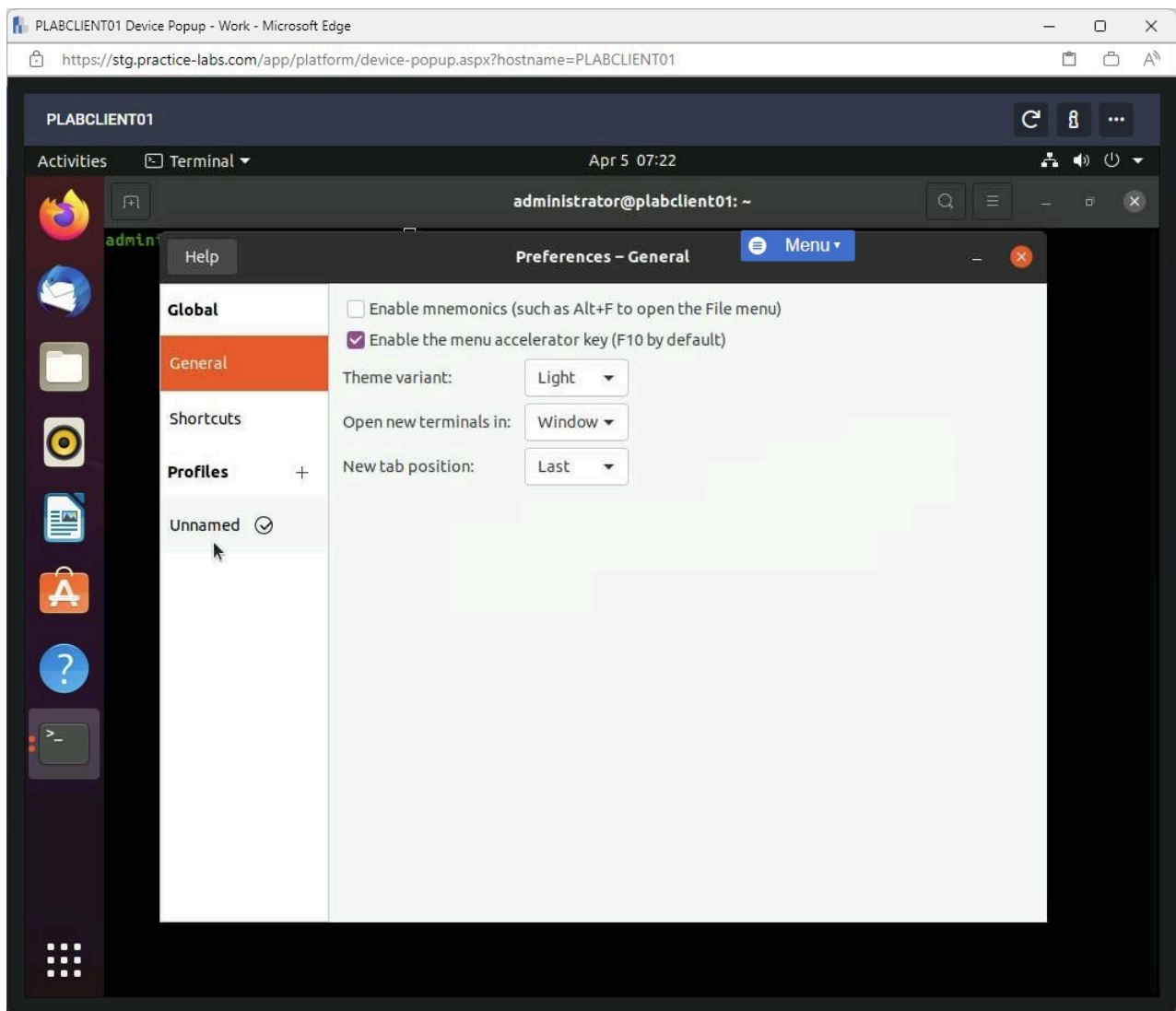


Figure 1.9 Screenshot of PLABCLIENT01: Preference - General window is displayed, showing Unnamed selected on the left pane.

Step 5

Click the drop-down next to **Unnamed** and select **Clone**.

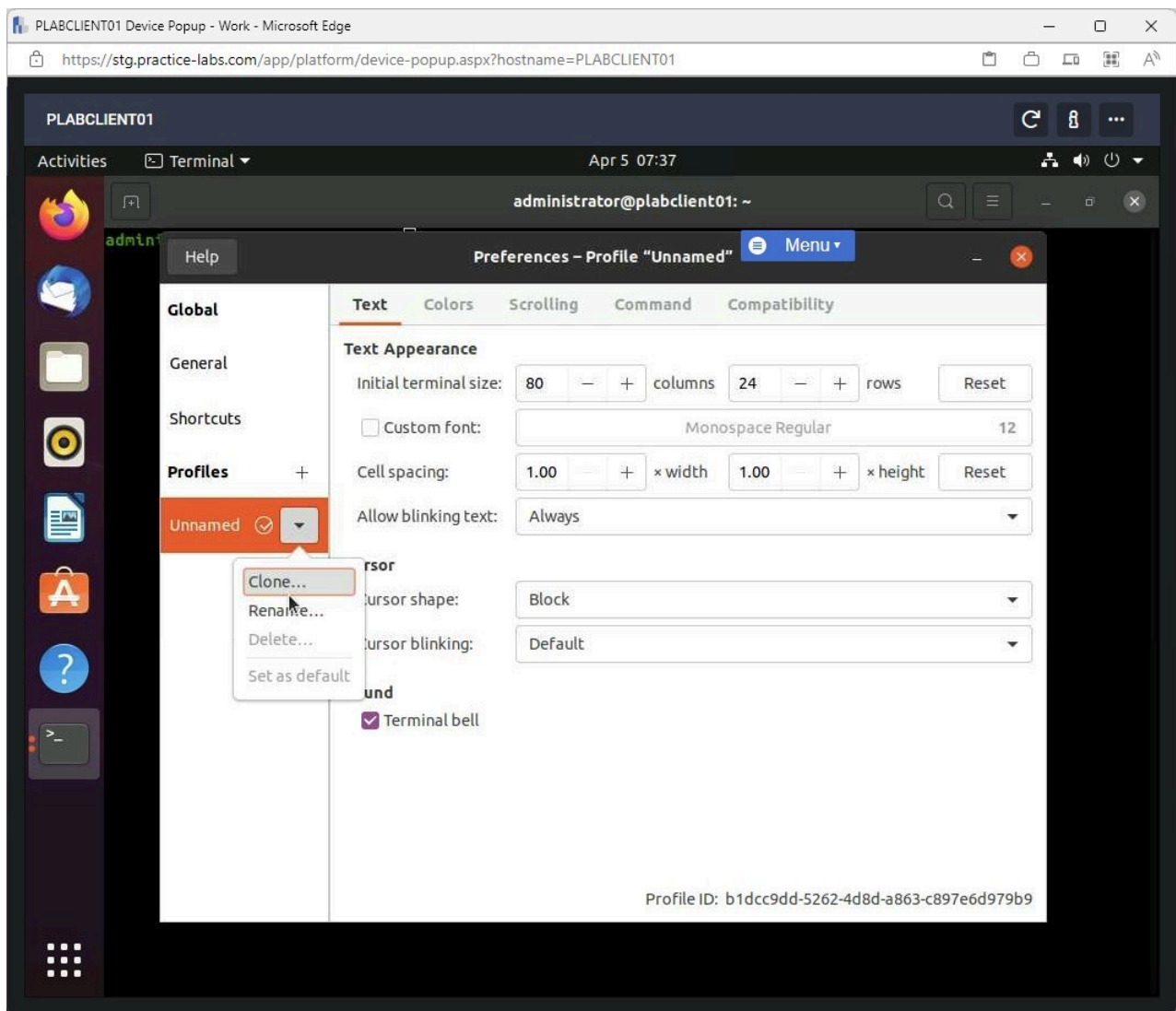


Figure 1.10 Screenshot of PLABCLIENT01: Preference - Profile “Unnamed” window is displayed, showing Unnamed drop-down accessed and Clone selected.

Step 6

In the **Clone Profile** dialog box, type the following in the textbox:

Work

Click **Clone**.

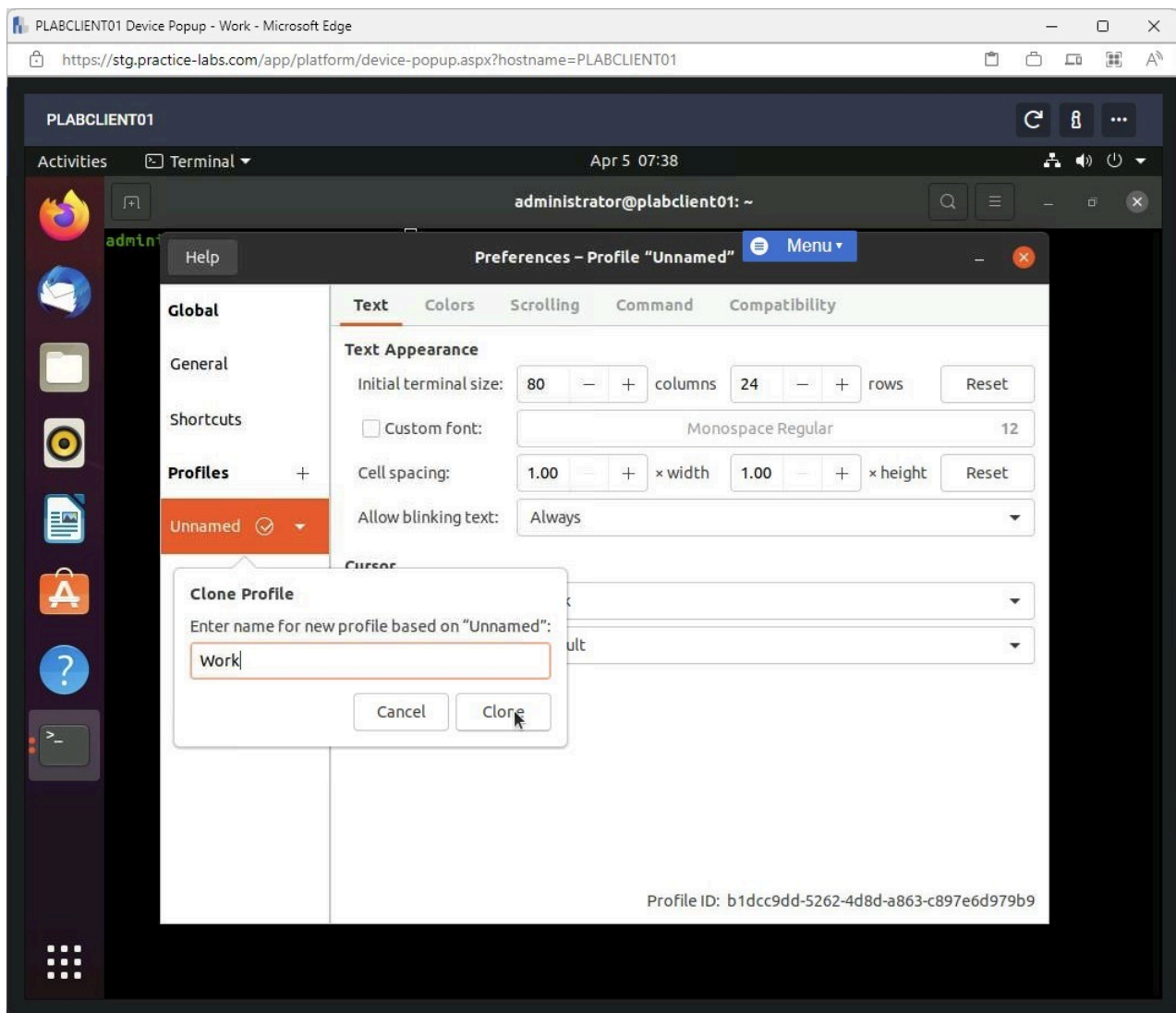


Figure 1.11 Screenshot of PLABCLIENT01: Clone Profile dialog box is displayed, showing the required Name entered and Clone selected.

Note: You have now created a profile for your customizations.

Step 7

In the **Preferences - Profile “Work”** pane, tick the **Custom font** checkbox.

Click on the **Monospace Regular** font.

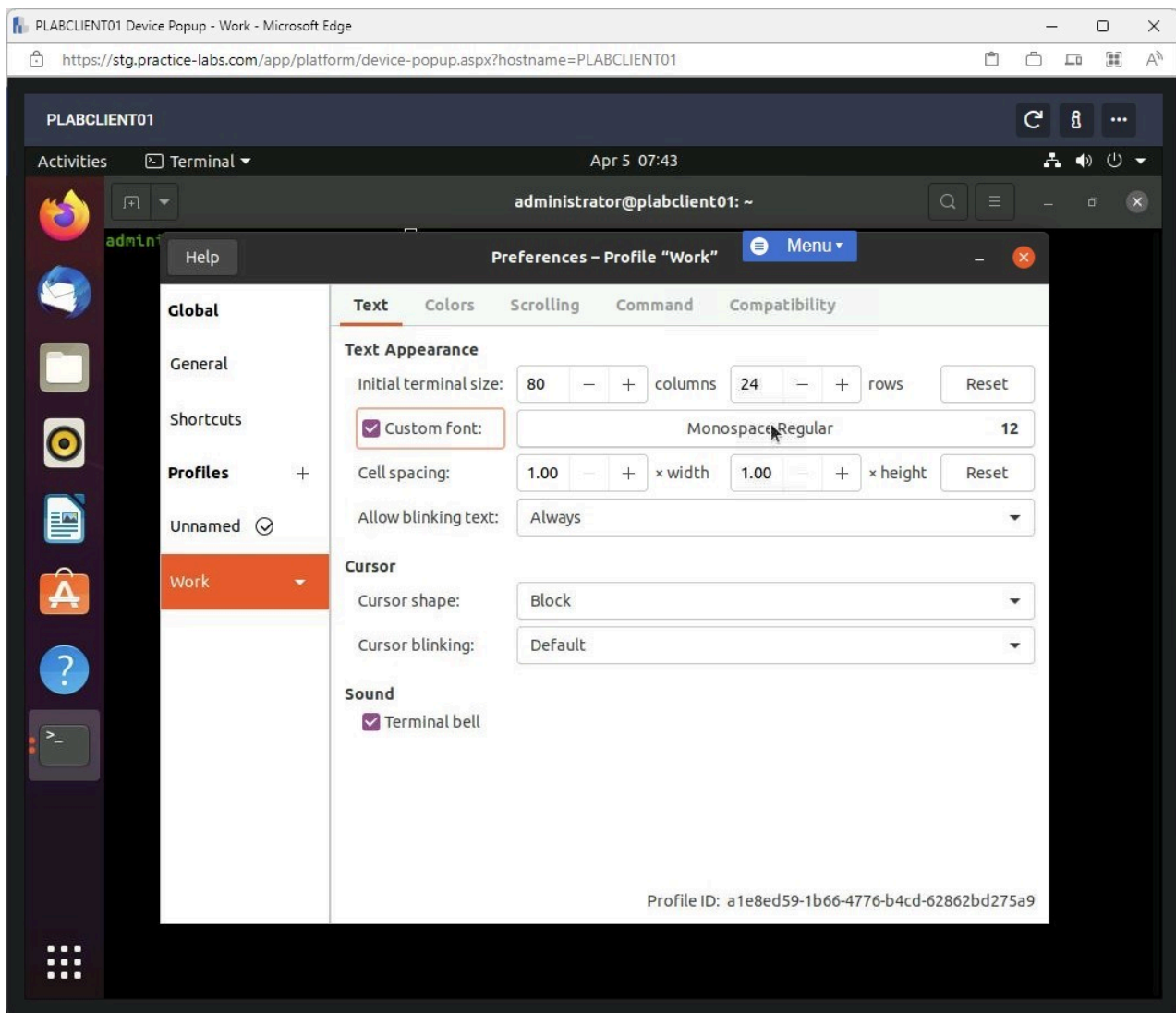


Figure 1.12 Screenshot of PLABCLIENT01: Preferences - Profile "Work" window is displayed, showing the Custom font checkbox ticked and Monospace Regular selected.

Step 8

In the **Choose A Terminal Font** dialog box, click on the + sign next to the **Size** field to change the font size to:

16

Click **Select**.

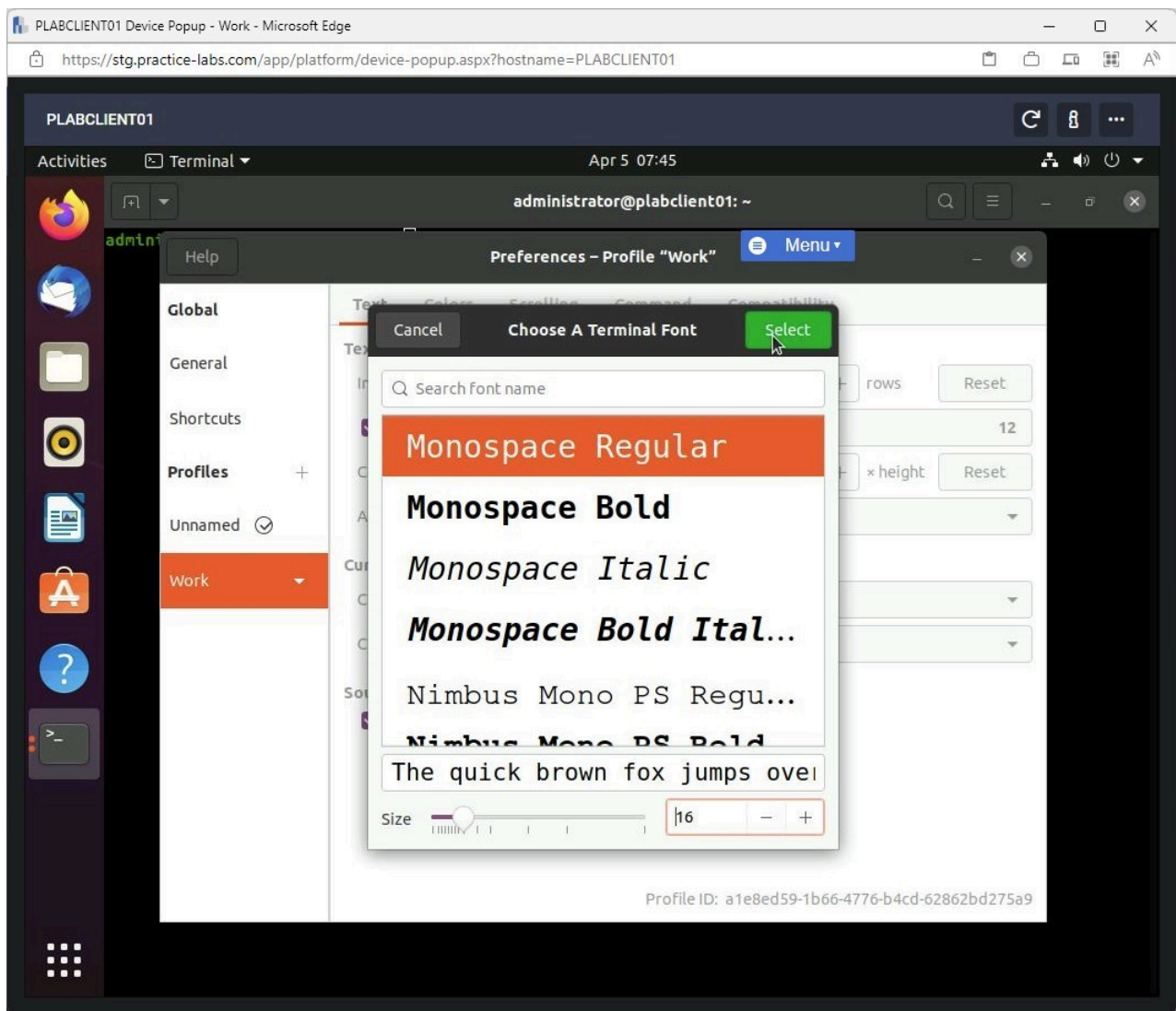


Figure 1.13 Screenshot of PLABCLIENT01: Choose A Terminal Font dialog box is displayed, showing the required settings performed and Select selected.

Step 9

Back on the **Preferences - Profile "Work"** window, untick the **Terminal bell** checkbox.

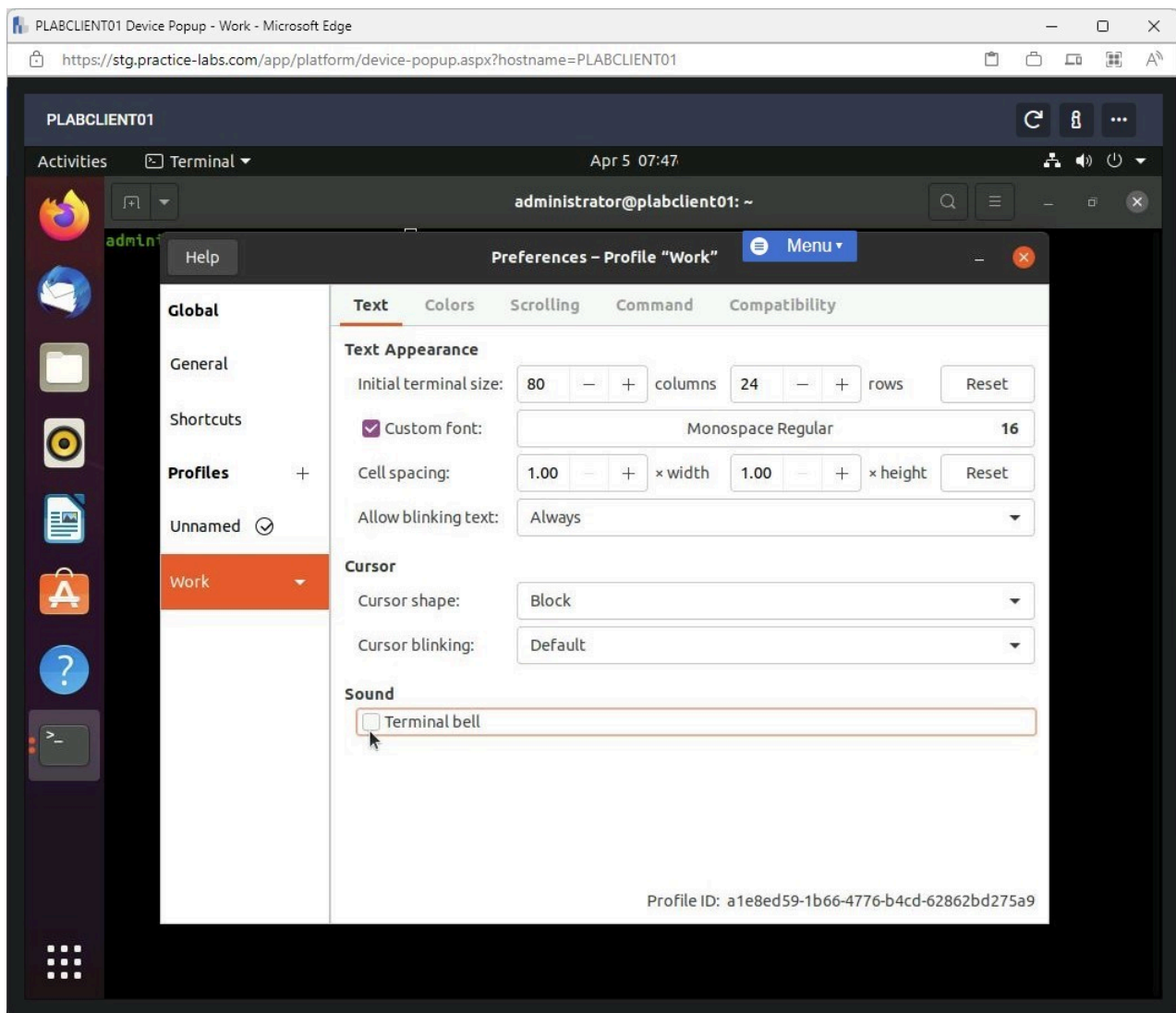


Figure 1.14 Screenshot of PLABCLIENT01: Preferences - Profile “Work” window is displayed, showing the Terminal bell checkbox unticked.

Step 10

Notice that the **Unnamed** profile has a tick next to it. This means that **Unnamed** will be considered as the default profile.

Click the drop-down next to the **Work** profile and select **Set as default**.

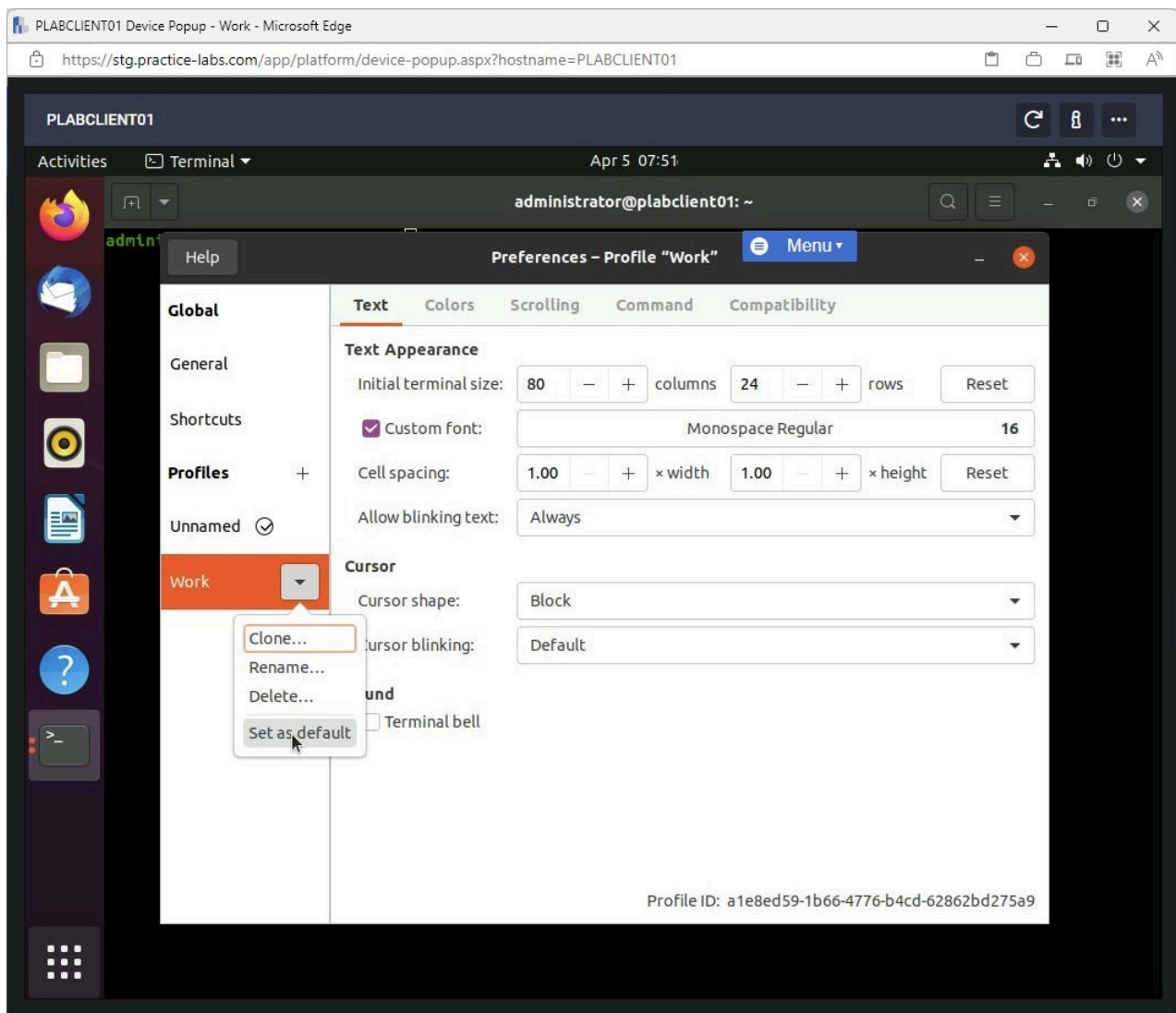


Figure 1.15 Screenshot of PLABCLIENT01: Preferences - Profile “Work” window is displayed, showing the drop-down next to the Work profile accessed and Set as default selected.

Step 11

Close the **Preferences - Profile “Work”** window.

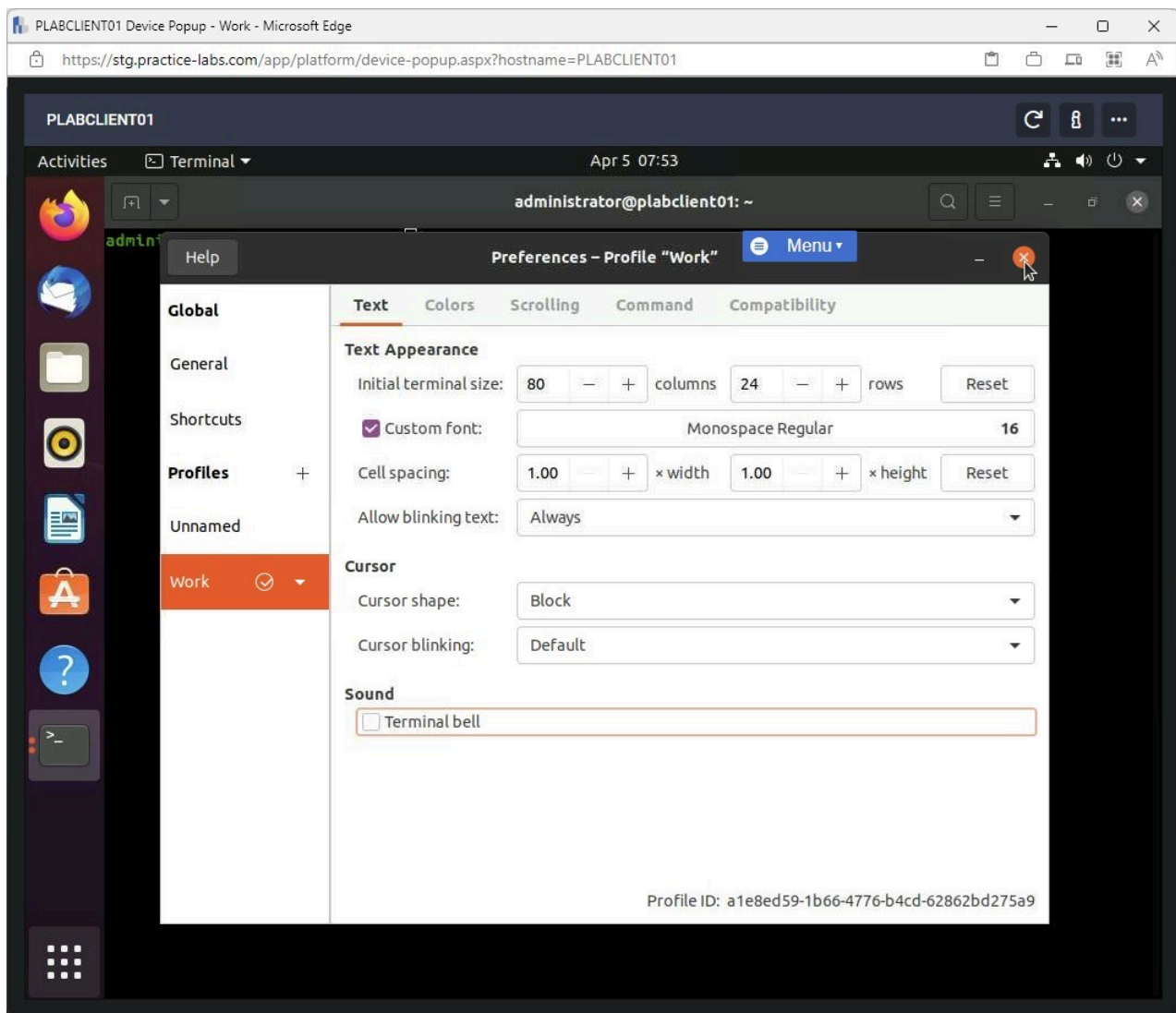


Figure 1.16 Screenshot of PLABCLIENT01: Close icon on the Preferences - Profile “Work” window is selected.

Step 12

Close the **Terminal** window.

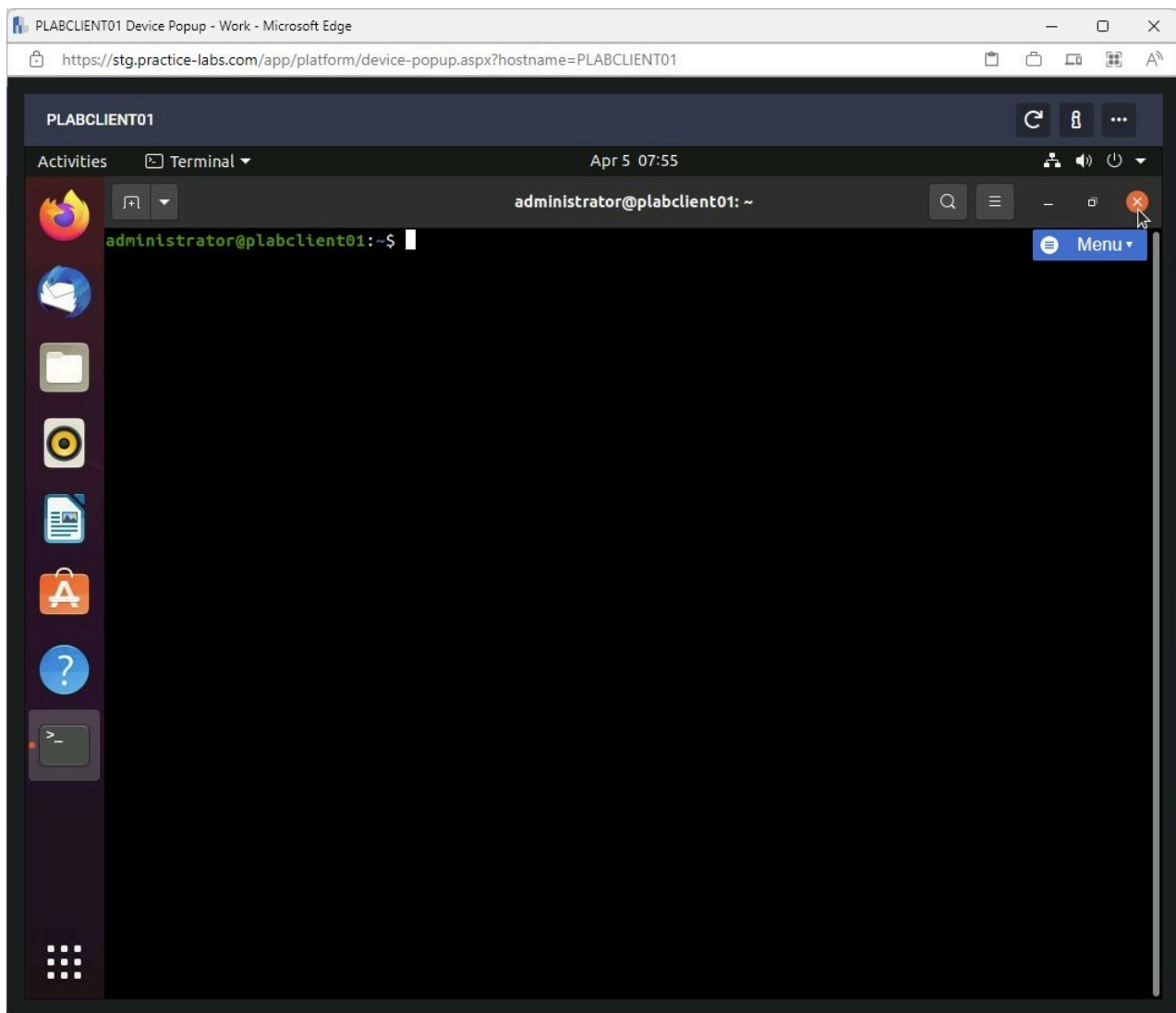


Figure 1.17 Screenshot of PLABCLIENT01: Close icon on the Terminal window is selected.

Step 13

Click the **Terminal** icon on the left (favorites) pane to relaunch it.

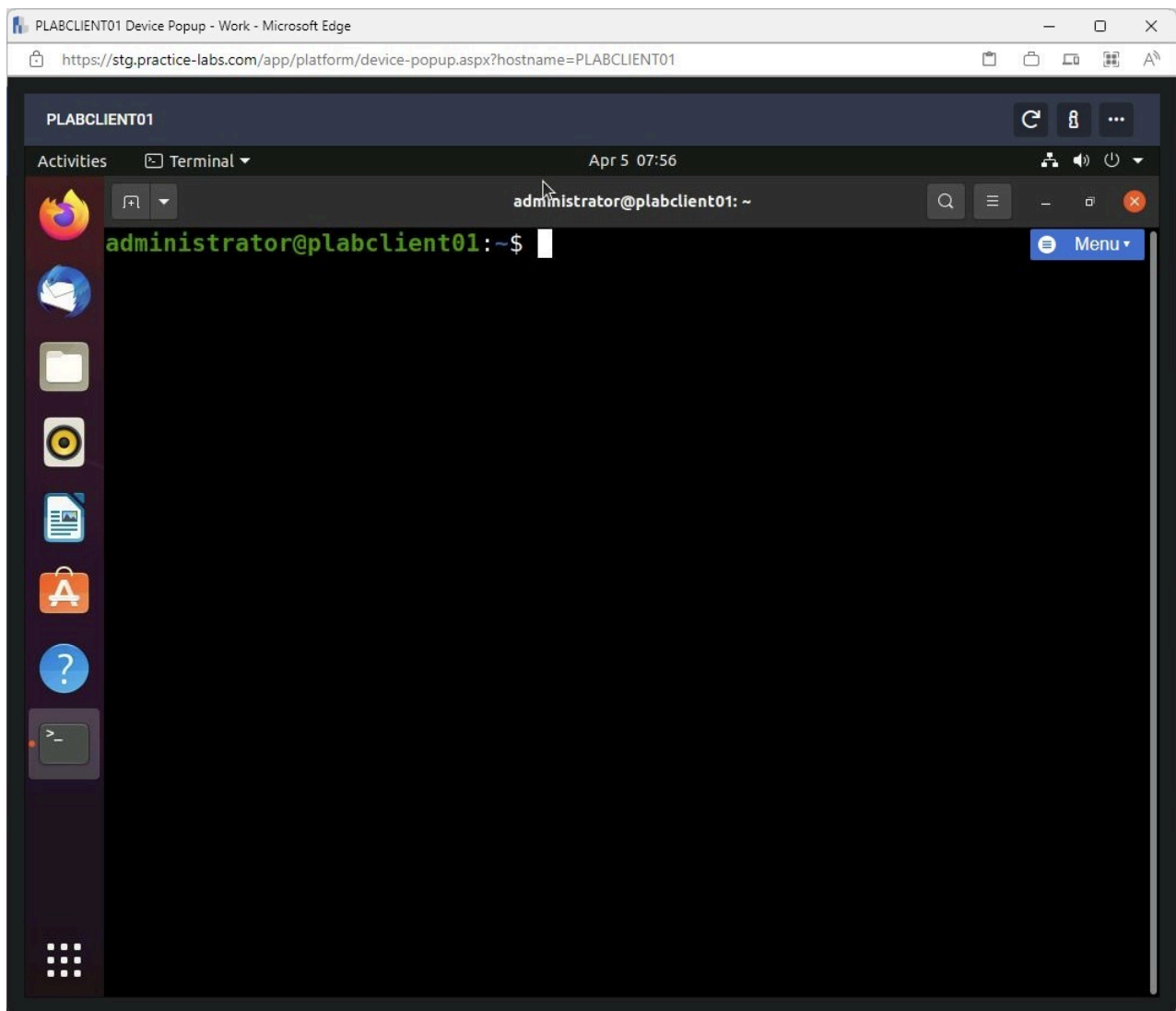


Figure 1.18 Screenshot of PLABCLIENT01: The Terminal window is displayed.

Note: Notice the text now appear with a font size of 16. If time permits, you can customize the terminal to your preferences.

Keep the **Terminal** window open.

Exercise 2 - Navigate the Linux File System

In this exercise, you will use Linux commands in the terminal to navigate the file system in Linux. You will also learn about folder hierarchy and create and remove directories in the Linux file system.

Learning Outcomes

After completing this exercise, you should be able to:

- Use Basic Commands to Navigate the Linux File System
- Create Directories in the Linux File System

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Use Basic Commands to Navigate the Linux File System

In this task, you will use basic commands to navigate the Linux file system.

Step 1

Connect to **PLABCLIENT01**, where the **Terminal** window is open.

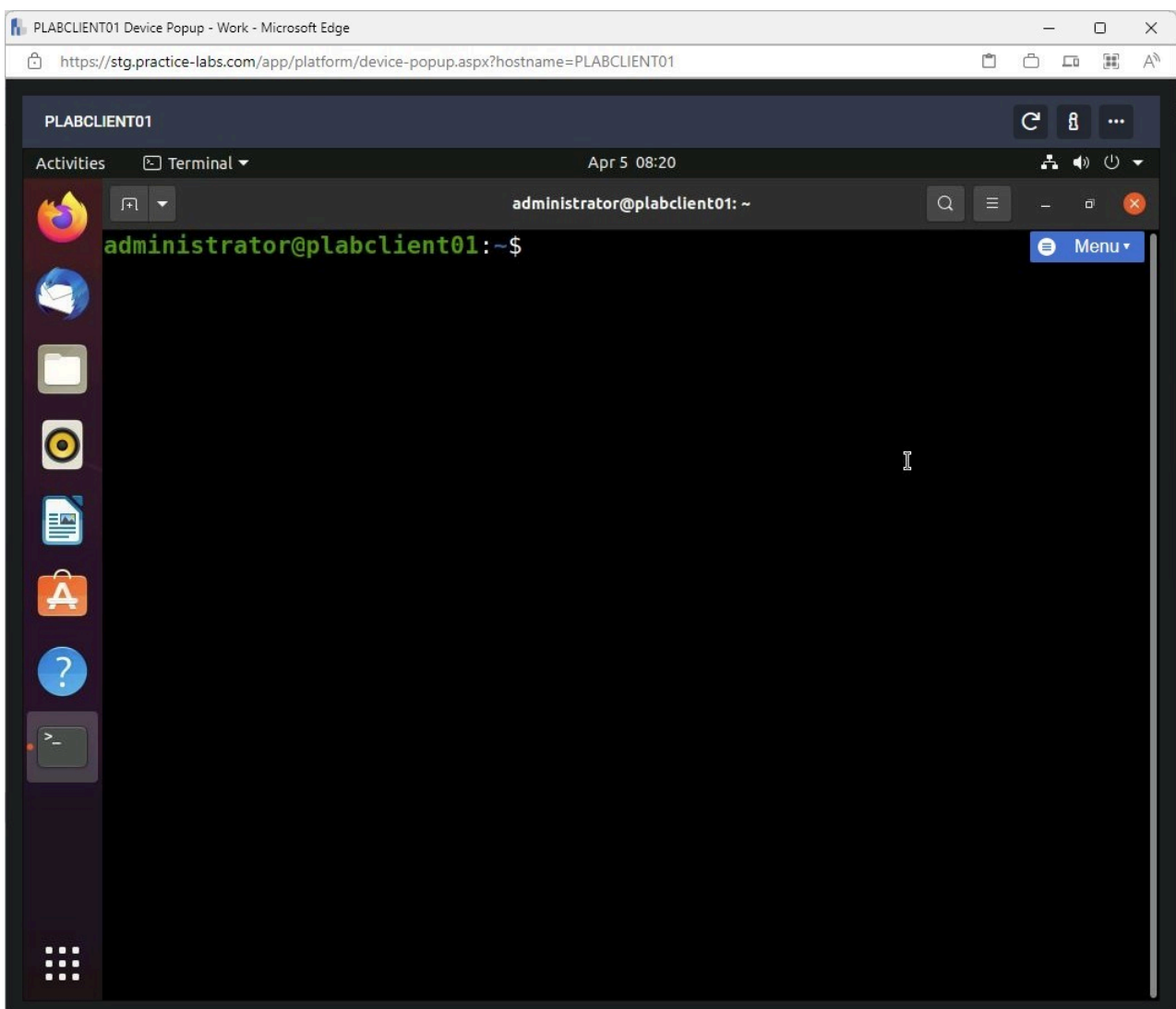


Figure 2.1 Screenshot of PLABCLIENT01: The Terminal window is displayed.

Note: The **administrator@plabclient01:~\$** prompt has the following parts:

administrator - Logged in username

plabclient01 - System name

~ - Home directory (**~** is an alias for **/home/administrator**)

\$ - Standard user prompt

A **root@plabclient01:/home/administrator#** * prompt will have the following parts:

root - Logged in username

plabclient01 - System name

/home/administrator - Home directory

- Root user prompt

* root login is disabled by default in Ubuntu but shown for differentiation between standard non-administrative user prompt (**~**) and root user prompt (**#**) in the GNOME Terminal.

Step 2

In the **Terminal** window, type the following command:

```
pwd
```

Press **Enter**.

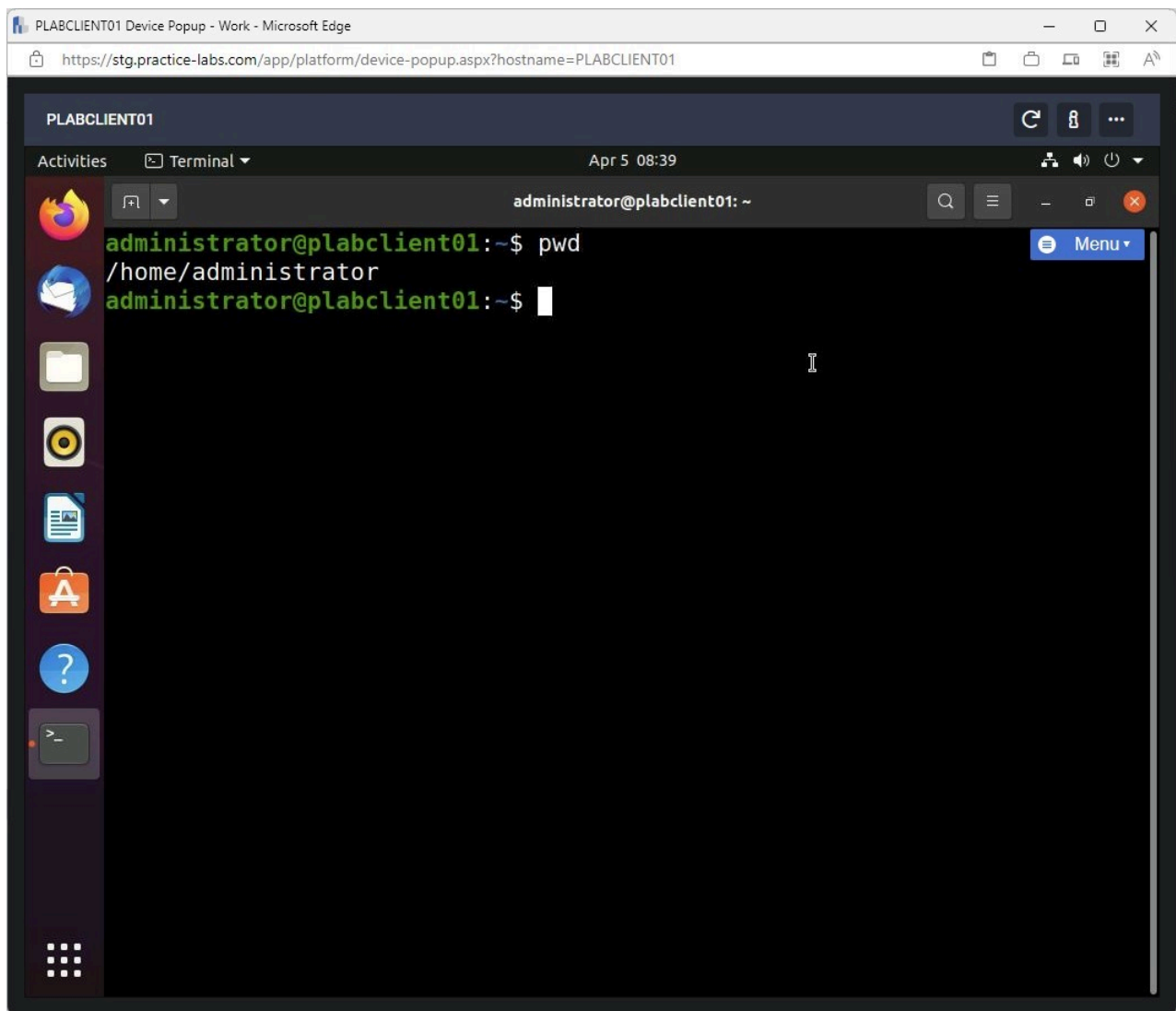


Figure 2.2 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **pwd** command will display the actual path of the directory.

/ is the root directory on the hard drive.

/home is a sub-directory named home under the root.

/home/administrator is a sub-directory of the home directory. Each user with a login account will keep their files separated into these directories with their names. This directory will appear as the **~** for a user command line prompt.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
ls
```

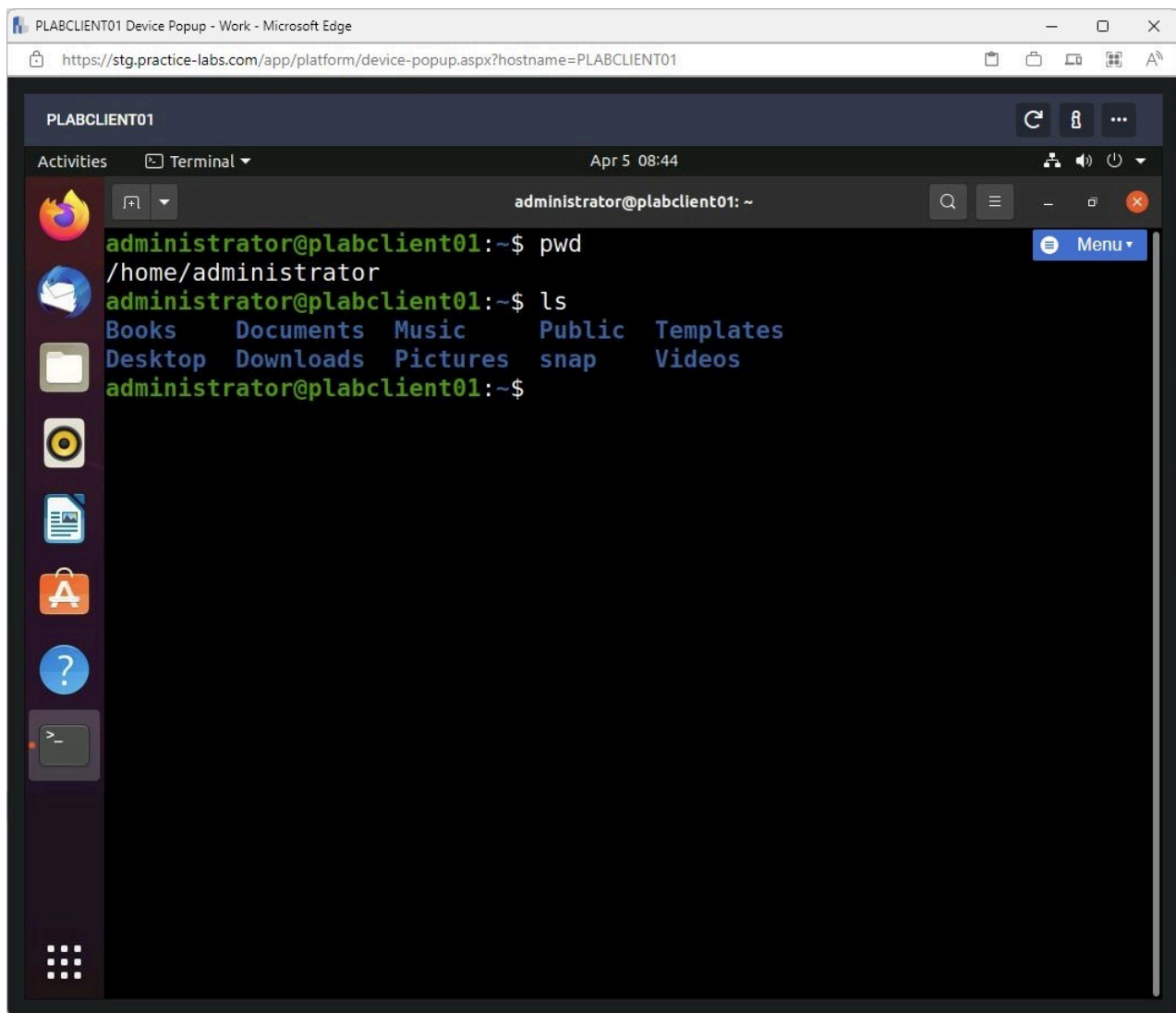


Figure 2.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *ls* command will list all the subdirectories that are available in the current directory. They are subdirectories within the user's home folder.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
cd Books
```

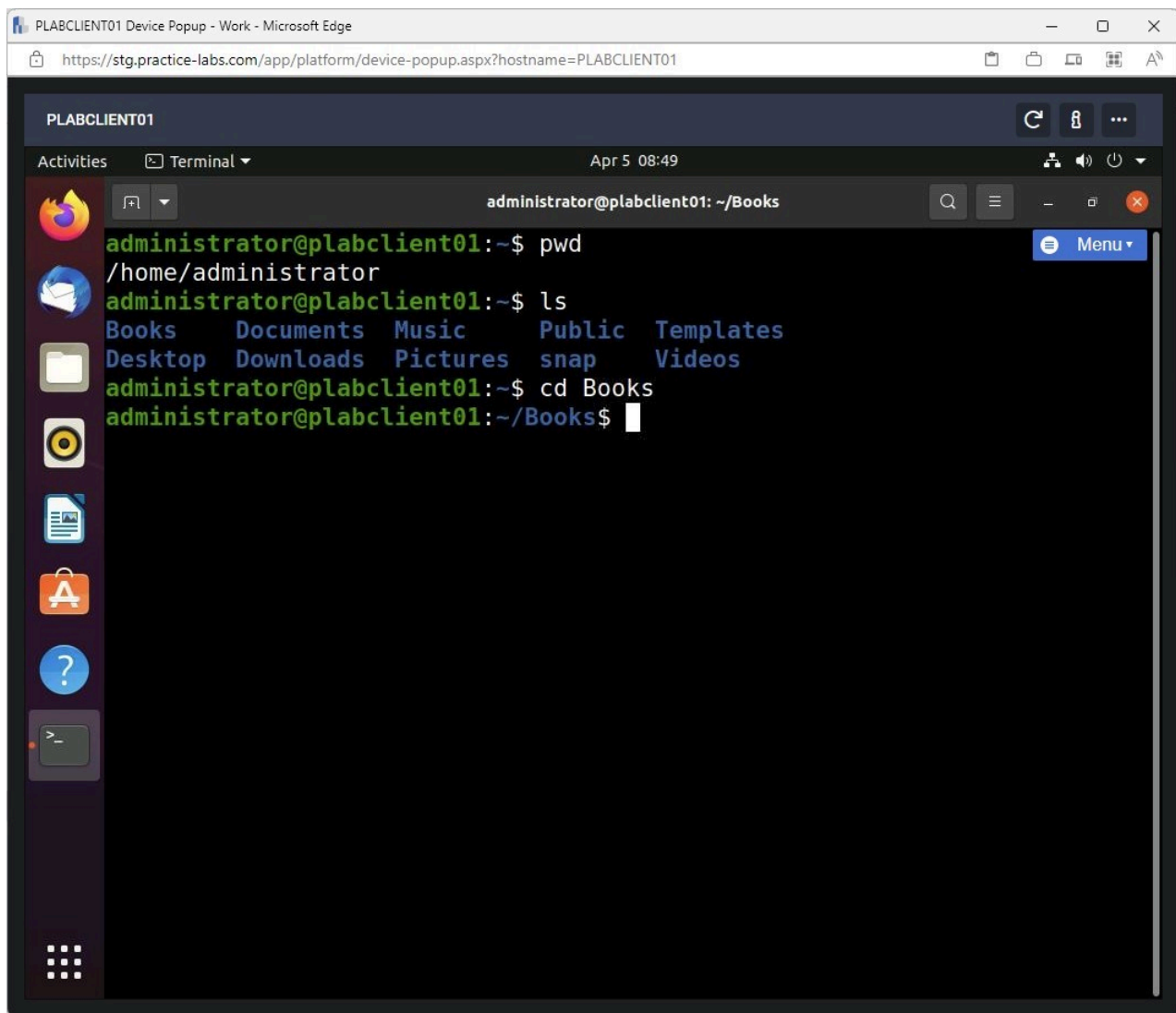


Figure 2.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **cd** command is used to change the directory from the home directory. Notice the change in the prompt.

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
pwd
```

Then, type the following command and press **Enter**:

```
cd /
```

Next, type the following command and press **Enter**:

```
pwd
```

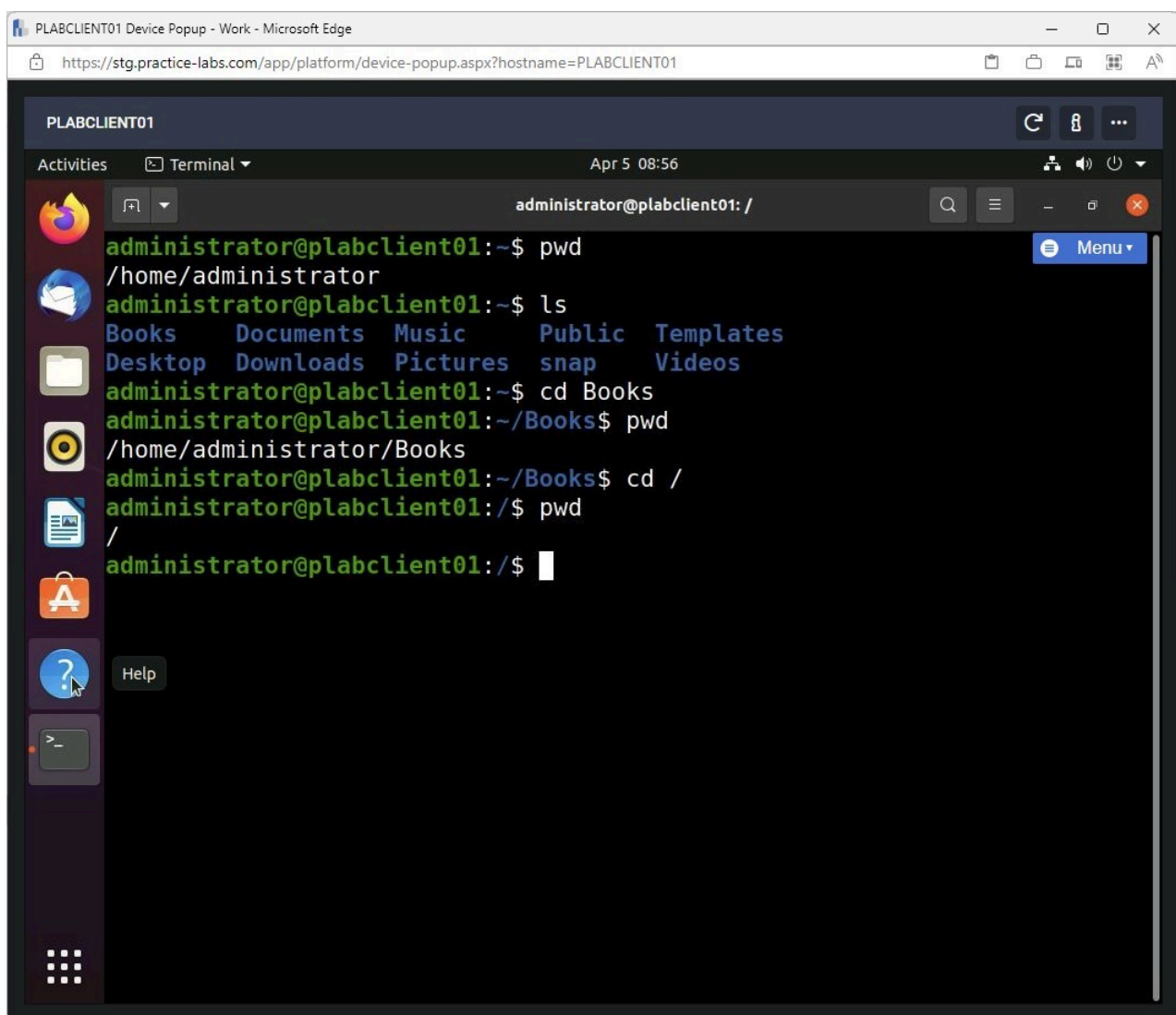


Figure 2.5 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Step 6

In the **Terminal** window, type the following command and press **Enter**:


```
cd /home/administrator/Books
```

Then, type the following command and press **Enter**:

```
pwd
```

Next, type the following command and press **Enter**:

```
cd ..
```

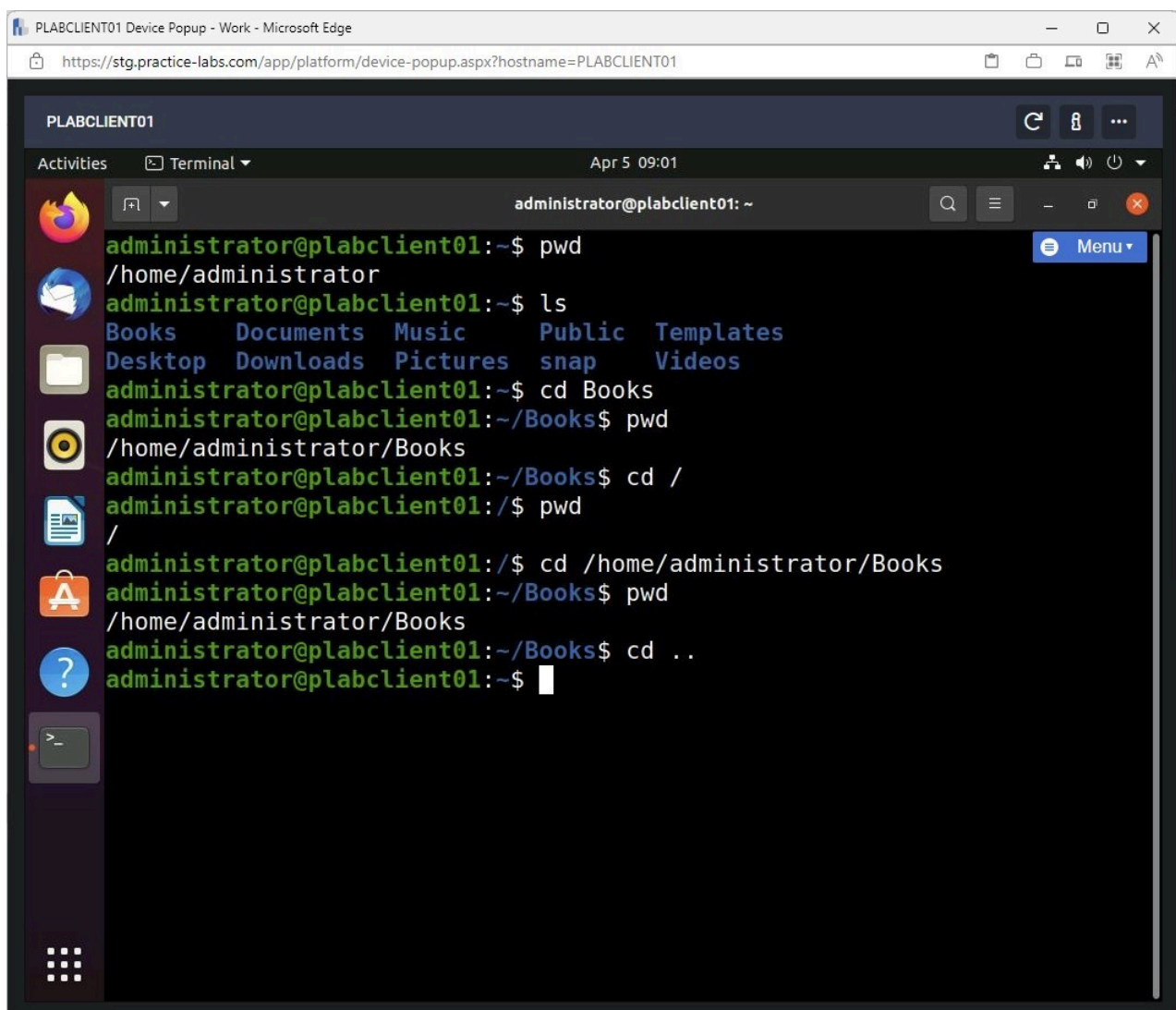


Figure 2.6 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: Notice you are now back in the home directory.

Step 7

In the **Terminal** window, type the following command and press **Enter**:

```
clear
```

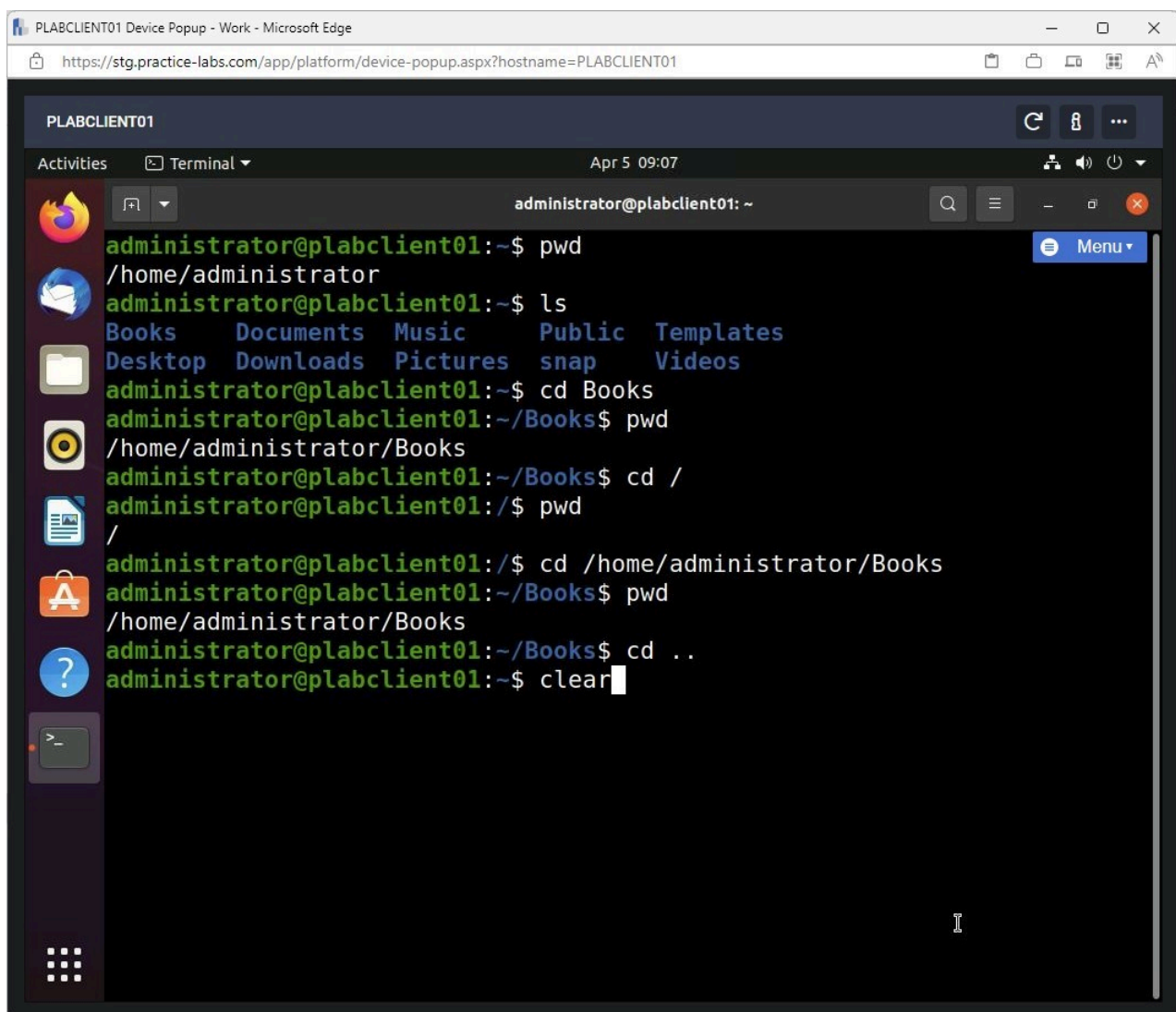


Figure 2.7 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: The *clear* command will clear the Terminal screen.

Keep the **Terminal** window open.

Task 2 - Create Directories in the Linux File System

In this task, you will create a new directory called **Contracts** in the home directory. You will then remove the Contracts directory.

Step 1

In **PLABCLIENT01**, the **Terminal** window is open. You will be in the home directory.

In the **Terminal** window, type the following command and press **Enter**:

```
mkdir Contracts
```

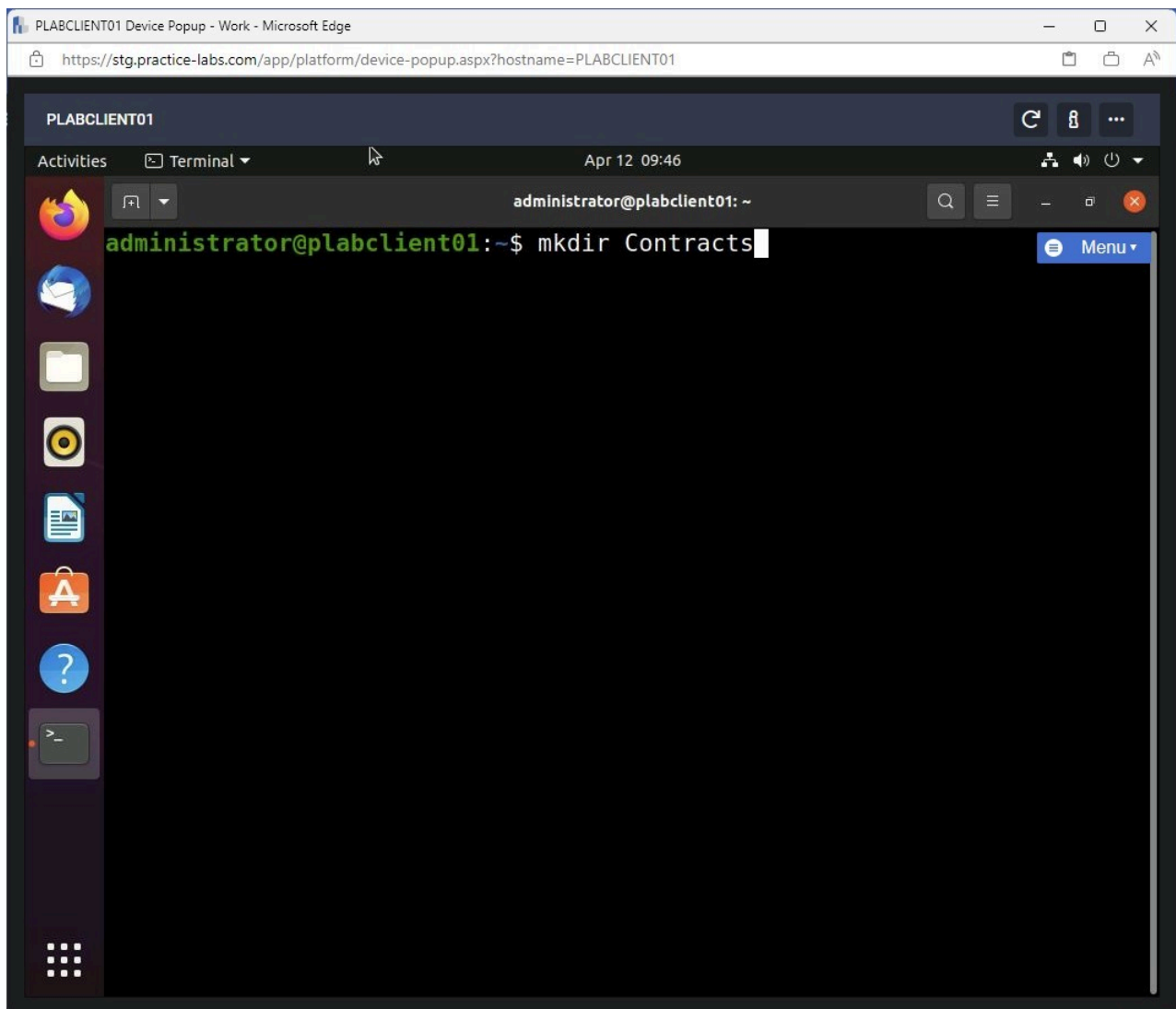


Figure 2.8 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: The **mkdir** command will create a new directory named **Contracts**.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
ls
```

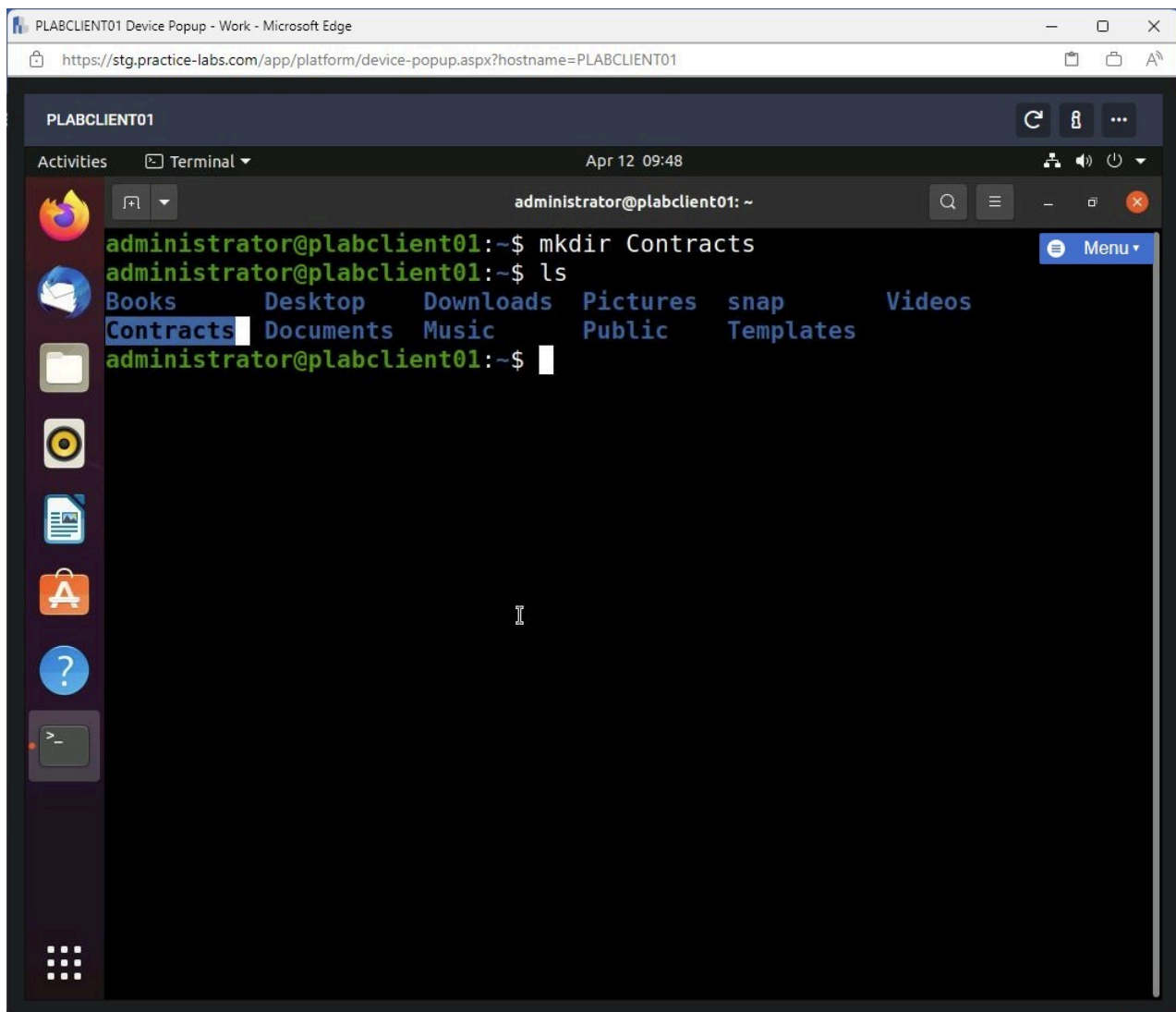


Figure 2.9 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: Notice the **Contracts** directory is now created.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
rmdir Contracts
```

Then, type the following command and press **Enter**:

```
ls
```

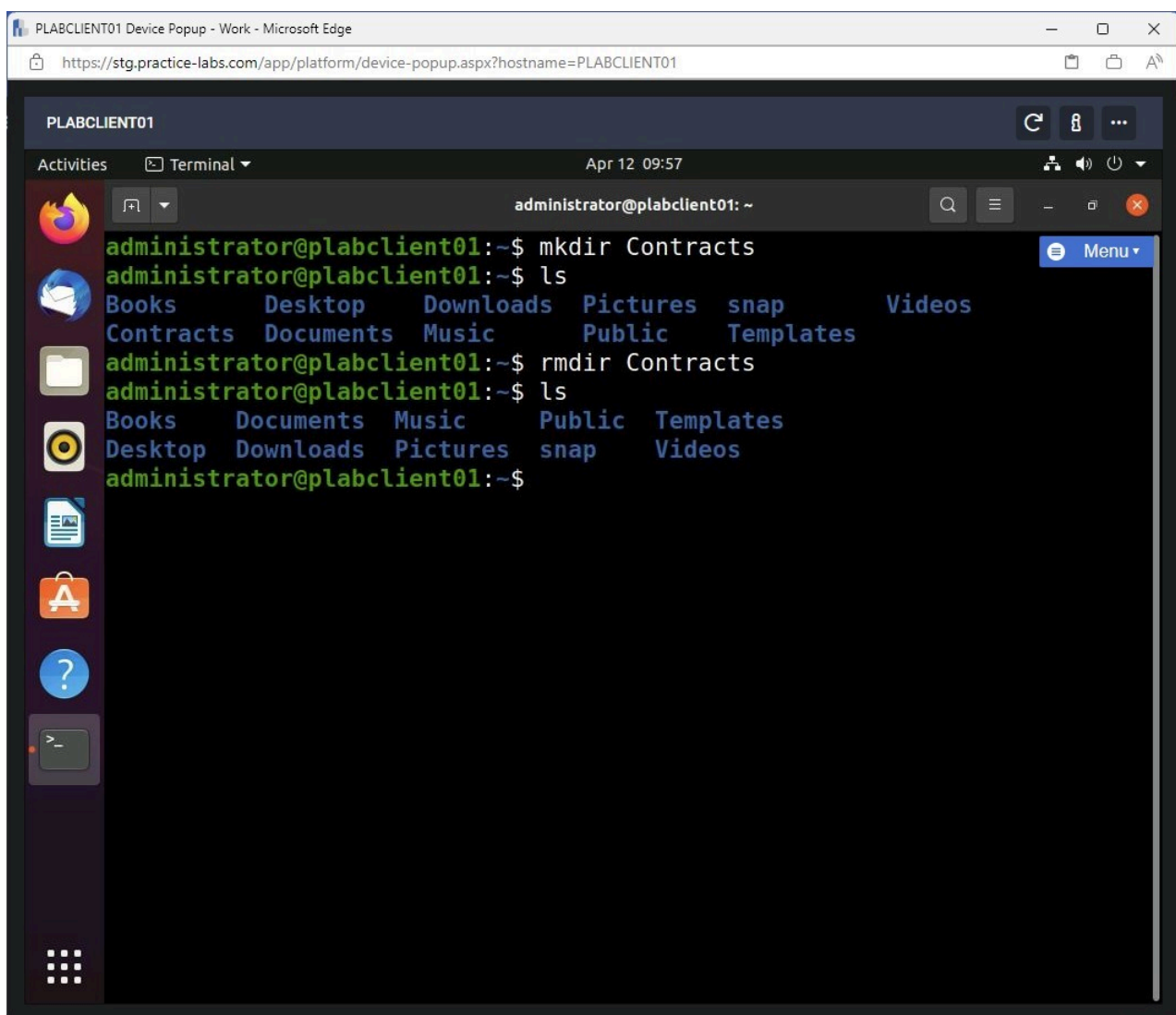


Figure 2.10 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **rmdir** command is used to remove a directory. Notice the **Contracts** directory is now removed.

Type the **clear** command to clear the Terminal window before proceeding to the next exercise.

Keep the **Terminal** window open.

Exercise 3 - Create Files in the Linux File System

In this exercise, you will use different Linux commands to create and view files in the Linux file system.

Learning Outcomes

After completing this exercise, you should be able to:

- Create and View Files using Linux Commands

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Create and View Files using Linux Commands

In this task, you will create and view files using the Linux commands.

Step 1

Ensure you are connected to **PLABCLIENT01**, where the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
sudo apt-get update
```

Type the following password:

Passw0rd

Next, type the following command and press **Enter**:

```
sudo apt-get upgrade
```

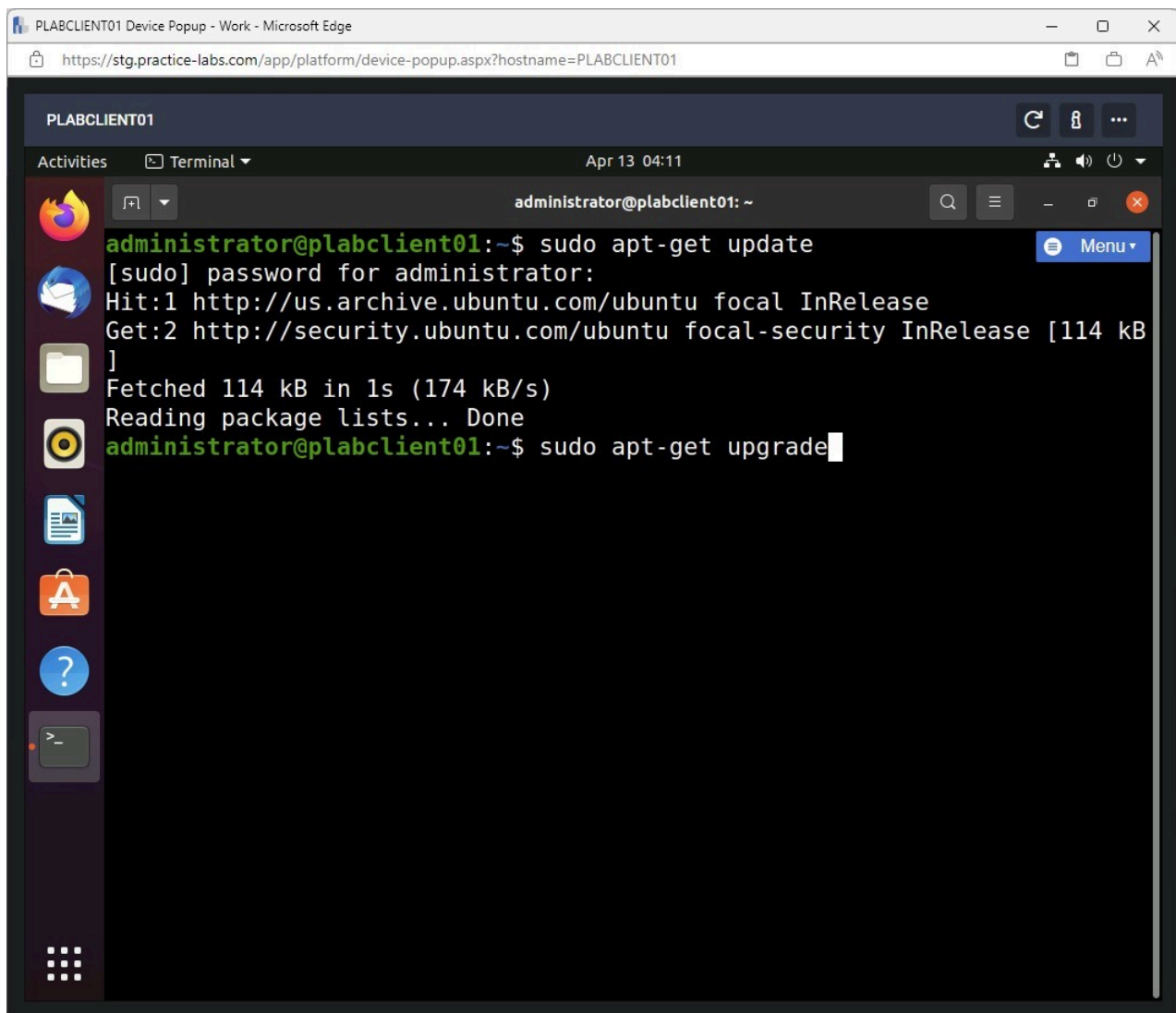


Figure 3.1 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The above commands will update Ubuntu. The **sudo** command will allow the user to perform the task with elevated privileges.

Step 2

In the **Do you want to continue?** prompt, type:

Y

Press **Enter**.

Note: If you encounter a **locked apt process** error while running an upgrade, please terminate the current process with the **Ctrl + C** command and follow these steps:

Check for active apt processes

Run the following command in the terminal:

```
ps -ef | grep apt
```

Identify the Process ID (PID)

Look at the output and note the PID of any active apt or dpkg processes.

To kill the locked process

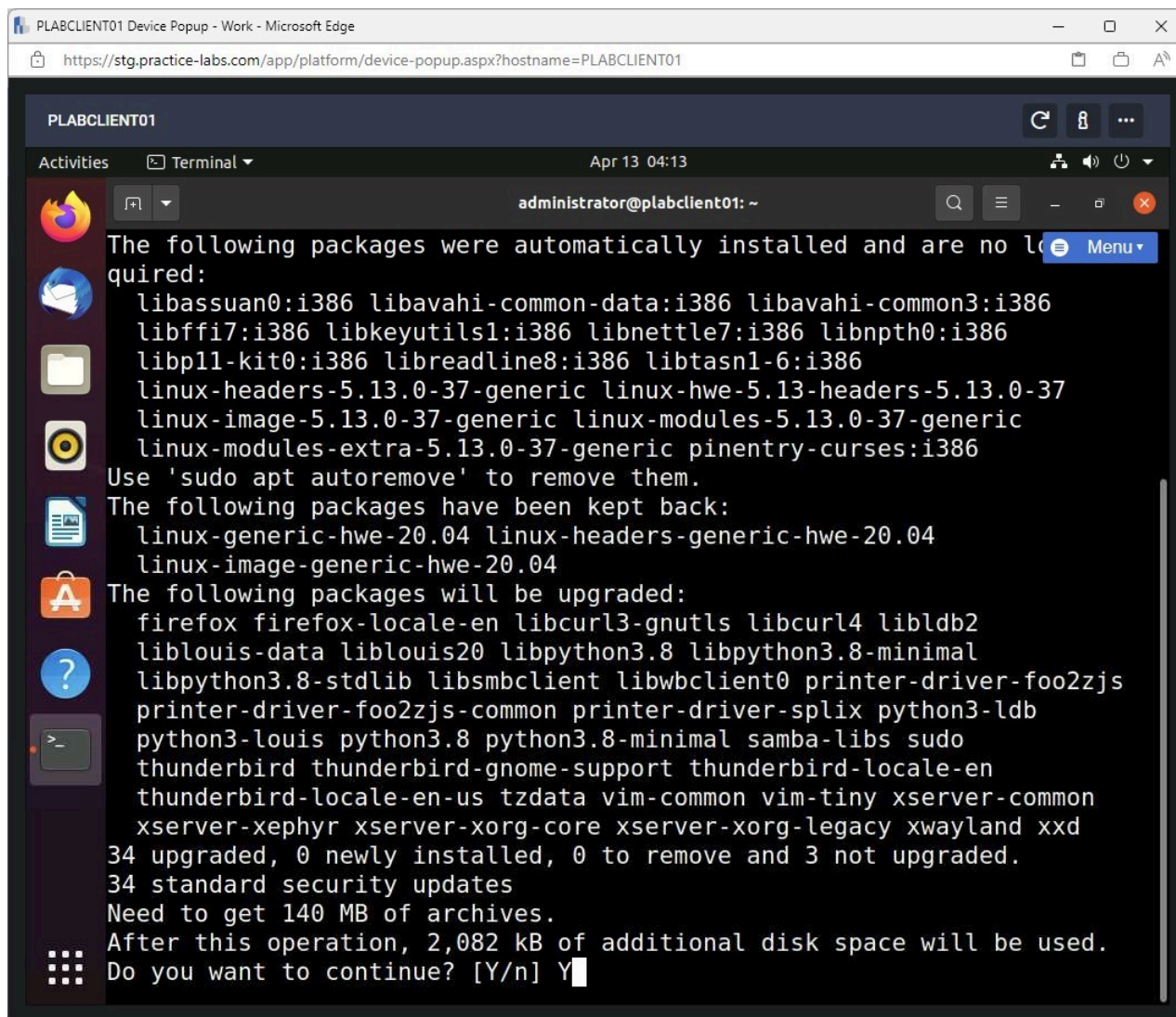
Run the command below, replacing <PID> with the actual number you found:

```
sudo kill <PID>
```

Retry the upgrade

Once the locked process is terminated, re-run the upgrade command:

```
sudo apt-get upgrade
```

```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Apr 13 04:13 administrator@plabclient01: ~
The following packages were automatically installed and are no longer required:
libassuan0:i386 libavahi-common-data:i386 libavahi-common3:i386
libffi7:i386 libkeyutils1:i386 libnettle7:i386 libnpth0:i386
libp11-kit0:i386 libreadline8:i386 libtasn1-6:i386
linux-headers-5.13.0-37-generic linux-hwe-5.13-headers-5.13.0-37
linux-image-5.13.0-37-generic linux-modules-5.13.0-37-generic
linux-modules-extra-5.13.0-37-generic pinentry-curses:i386
Use 'sudo apt autoremove' to remove them.
The following packages have been kept back:
linux-generic-hwe-20.04 linux-headers-generic-hwe-20.04
linux-image-generic-hwe-20.04
The following packages will be upgraded:
firefox firefox-locale-en libcurl3-gnutls libcurl4 libldb2
liblouis-data liblouis20 libpython3.8 libpython3.8-minimal
libpython3.8-stdlib libsmbclient libwbclient0 printer-driver-foo2zjs
printer-driver-foo2zjs-common printer-driver-splix python3-ldb
python3-louis python3.8 python3.8-minimal samba-lsmbd sudo
thunderbird thunderbird-gnome-support thunderbird-locale-en
thunderbird-locale-en-us tzdata vim-common vim-tiny xserver-common
xserver-xephyr xserver-xorg-core xserver-xorg-legacy xwayland xxd
34 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
34 standard security updates
Need to get 140 MB of archives.
After this operation, 2,082 kB of additional disk space will be used.
Do you want to continue? [Y/n] Y
```

Figure 3.2 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing Y entered next to the Do you want to continue? prompt.

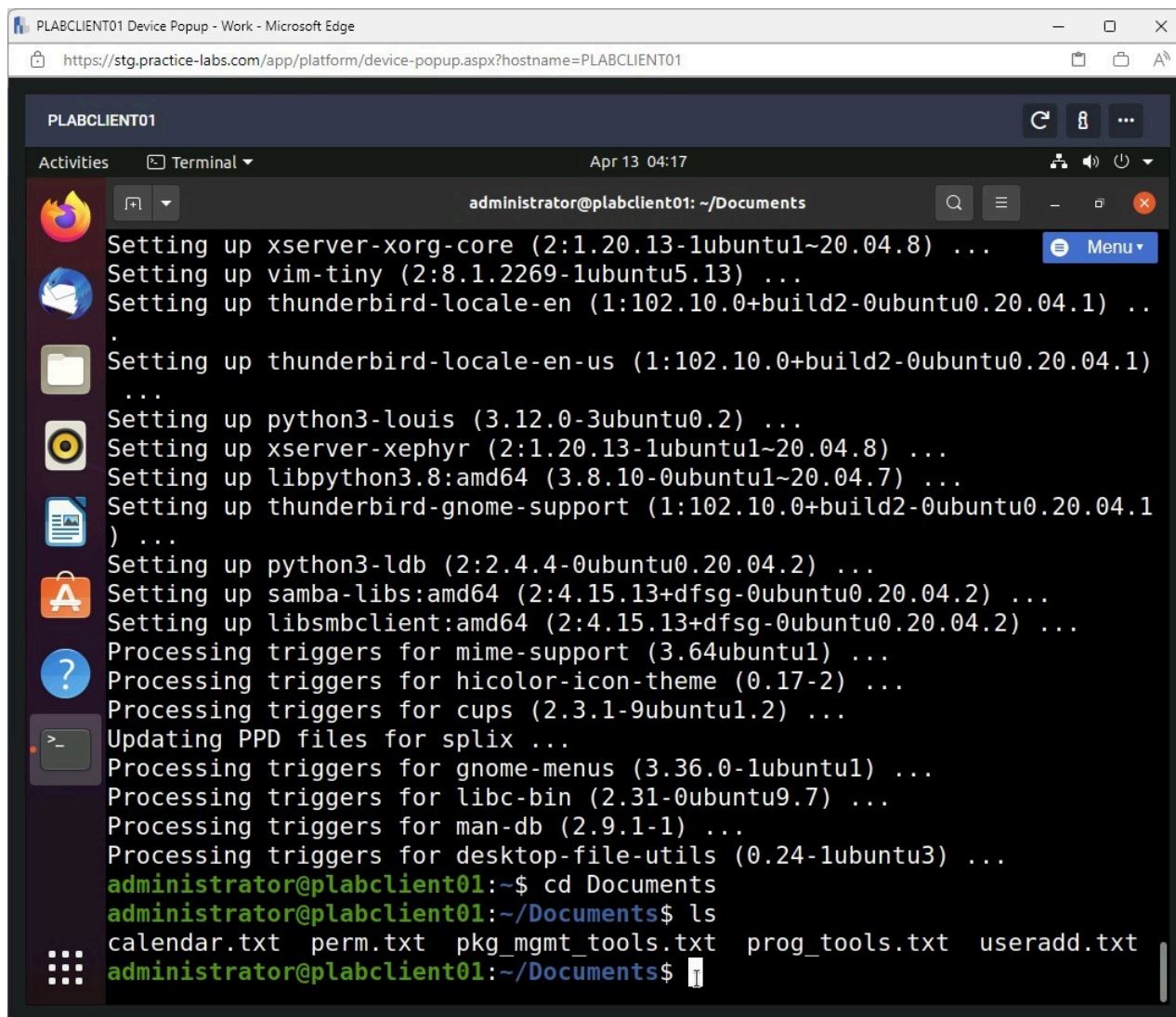
Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
cd Documents
```

Next, type the following command and press **Enter**:

```
ls
```

The screenshot shows a terminal window titled "PLABCLIENT01" with a URL bar displaying "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01". The terminal output shows the following commands and their outputs:

```
Setting up xserver-xorg-core (2:1.20.13-1ubuntu1~20.04.8) ...
Setting up vim-tiny (2:8.1.2269-1ubuntu5.13) ...
Setting up thunderbird-locale-en (1:102.10.0+build2-0ubuntu0.20.04.1) ..
.
Setting up thunderbird-locale-en-us (1:102.10.0+build2-0ubuntu0.20.04.1)
...
Setting up python3-louis (3.12.0-3ubuntu0.2) ...
Setting up xserver-xephyr (2:1.20.13-1ubuntu1~20.04.8) ...
Setting up libpython3.8:amd64 (3.8.10-0ubuntu1~20.04.7) ...
Setting up thunderbird-gnome-support (1:102.10.0+build2-0ubuntu0.20.04.1) ...
Setting up python3-ldb (2:2.4.4-0ubuntu0.20.04.2) ...
Setting up samba-lsmb:amd64 (2:4.15.13+dfsg-0ubuntu0.20.04.2) ...
Setting up libsmbclient:amd64 (2:4.15.13+dfsg-0ubuntu0.20.04.2) ...
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for cups (2.3.1-9ubuntu1.2) ...
Updating PPD files for splix ...
Processing triggers for gnome-menus (3.36.0-1ubuntu1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.7) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for desktop-file-utils (0.24-1ubuntu3) ...
administrator@plabclient01:~$ cd Documents
administrator@plabclient01:~/Documents$ ls
calendar.txt  perm.txt  pkg_mgmt_tools.txt  prog_tools.txt  useradd.txt
administrator@plabclient01:~/Documents$
```

Figure 3.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **cd** command is used to navigate to the **Documents** directory, and the **ls** command is used to list the contents of the **Documents** directory.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
touch example1.txt
```

Next, type the following command and press **Enter**:

ls

```

PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Apr 13 04:21 administrator@plabclient01: ~/Documents

Setting up thunderbird-locale-en-us (1:102.10.0+build2-0ubuntu0.2) ...
...
Setting up python3-louis (3.12.0-3ubuntu0.2) ...
Setting up xserver-xephyr (2:1.20.13-1ubuntu1~20.04.8) ...
Setting up libpython3.8:amd64 (3.8.10-0ubuntu1~20.04.7) ...
Setting up thunderbird-gnome-support (1:102.10.0+build2-0ubuntu0.20.04.1) ...
Setting up python3-ldb (2:2.4.4-0ubuntu0.20.04.2) ...
Setting up samba-libs:amd64 (2:4.15.13+dfsg-0ubuntu0.20.04.2) ...
Setting up libsmbclient:amd64 (2:4.15.13+dfsg-0ubuntu0.20.04.2) ...
Processing triggers for mime-support (3.64ubuntu1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for cups (2.3.1-9ubuntu1.2) ...
Updating PPD files for splix ...
Processing triggers for gnome-menus (3.36.0-1ubuntu1) ...
Processing triggers for libc-bin (2.31-0ubuntu9.7) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for desktop-file-utils (0.24-1ubuntu3) ...
administrator@plabclient01:~$ cd Documents
administrator@plabclient01:~/Documents$ ls
calendar.txt  perm.txt  pkg_mgmt_tools.txt  prog_tools.txt  useradd.txt
administrator@plabclient01:~/Documents$ touch example1.txt
administrator@plabclient01:~/Documents$ ls
calendar.txt  perm.txt  prog_tools.txt
example1.txt  pkg_mgmt_tools.txt  useradd.txt
administrator@plabclient01:~/Documents$

```

Figure 3.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **touch** command is used to create a new file named **example1.txt**. Notice the file is now listed in the **Documents** directory.

Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
nano example2.txt
```

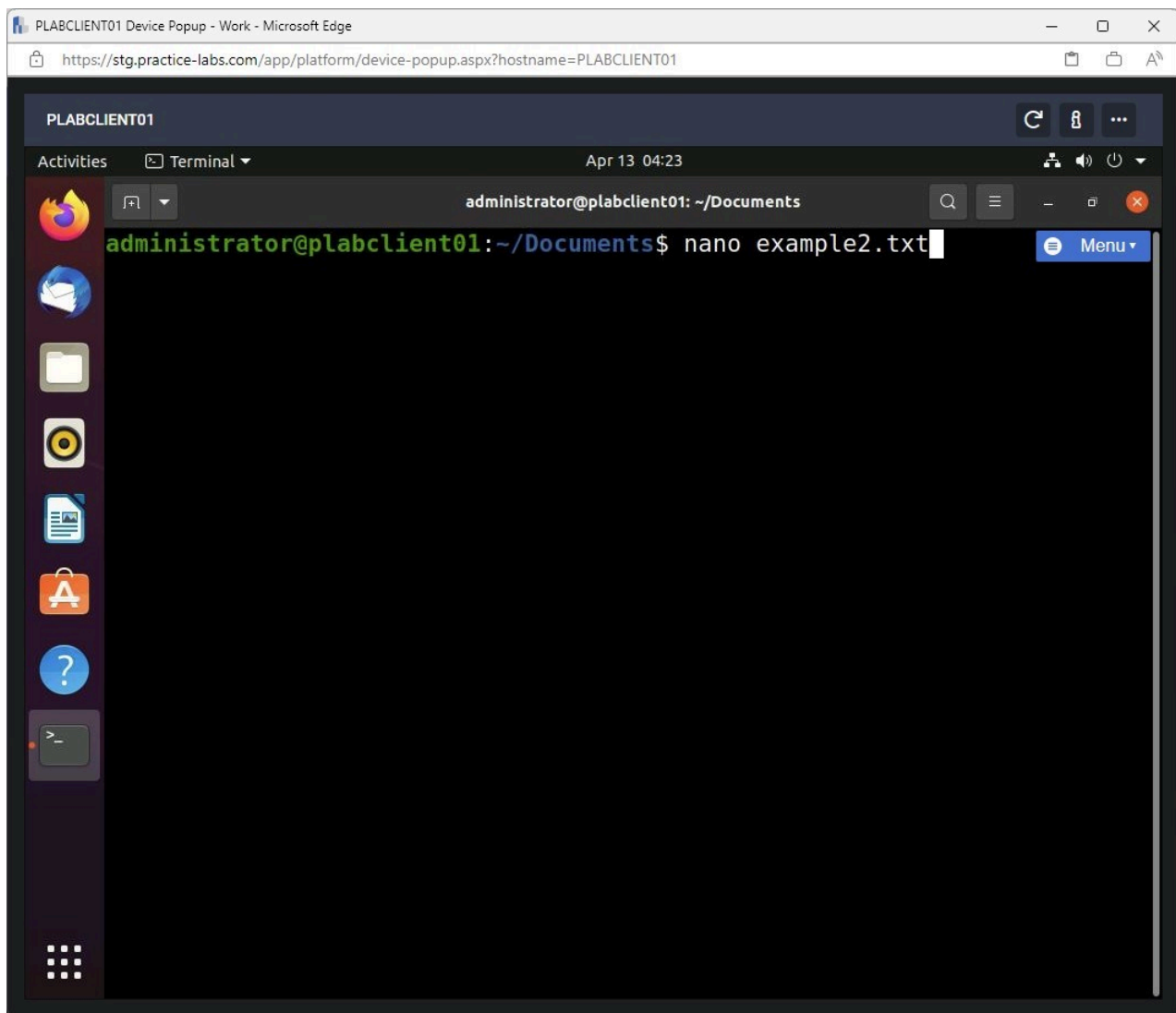


Figure 3.5 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Note: You have now used the **nano** text editor to create a file named **example2.txt**.

Step 6

In the **example.txt** file, type the following:

I have created this file using nano. I will use the shortcuts Ctrl+O and Ctrl+X to save the file and close nano.

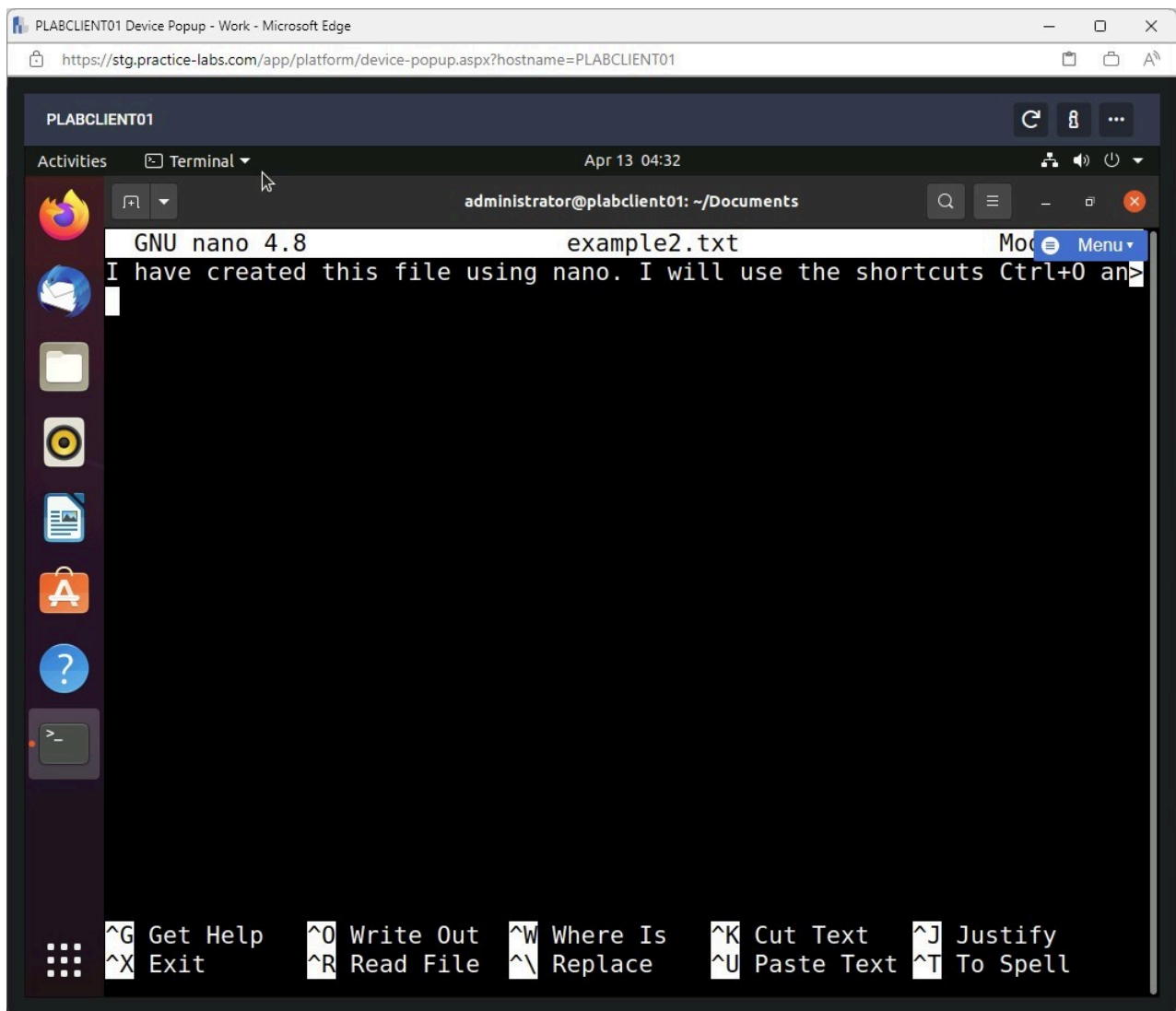


Figure 3.6 Screenshot of PLABCLIENT01: example.txt file is displayed, showing the required text entered.

Step 7

Press the **Ctrl+O** keys on the keyboard, then press **Enter** to save the file.

Next, press the **Ctrl+X** keys to exit the file.

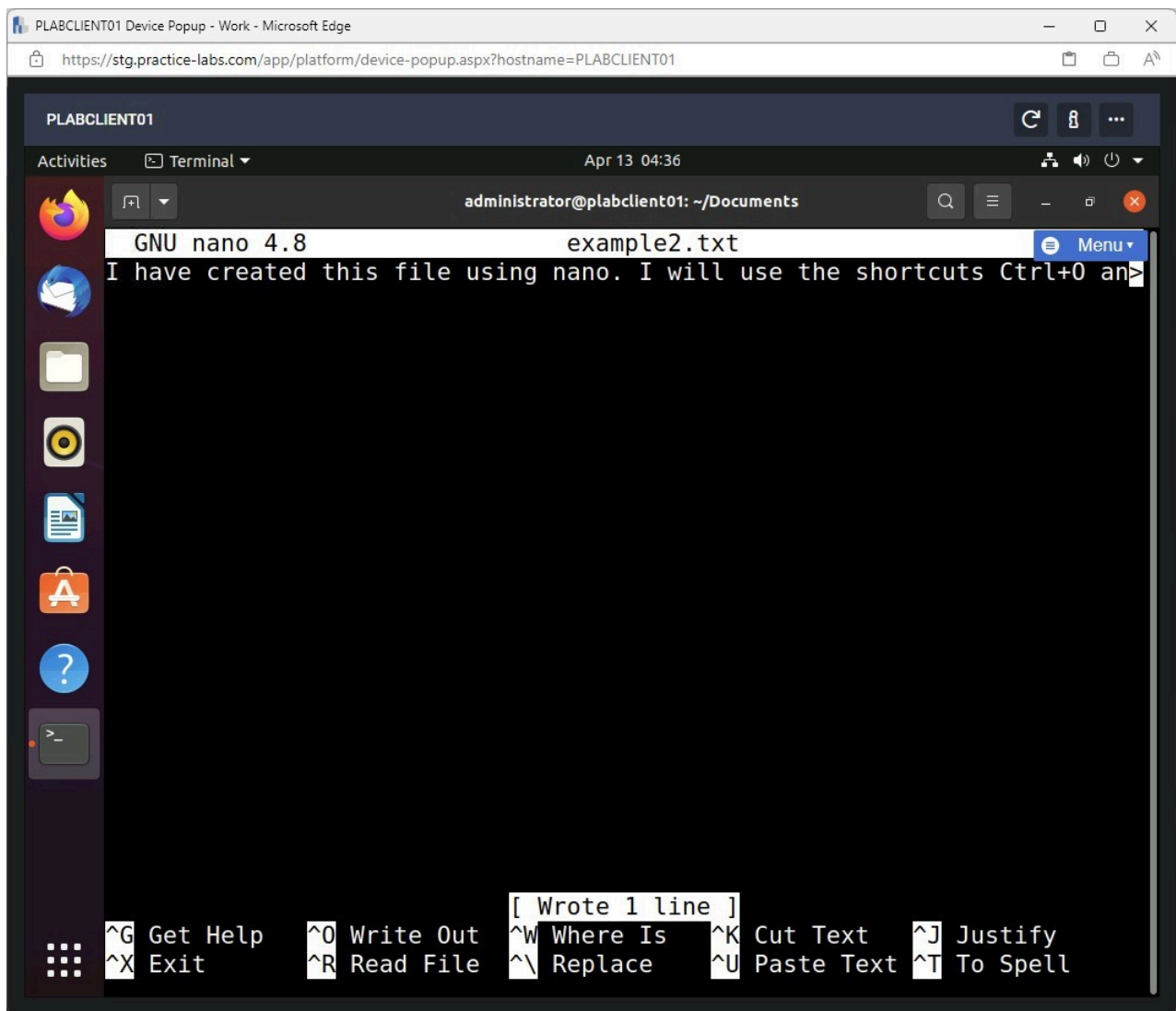


Figure 3.7 Screenshot of PLABCLIENT01: example.txt file is displayed, showing the required keys pressed.

Step 8

You will now create a file using the vim editor. Vim is not installed by default on Ubuntu. To install vim, type the following command and press **Enter**:

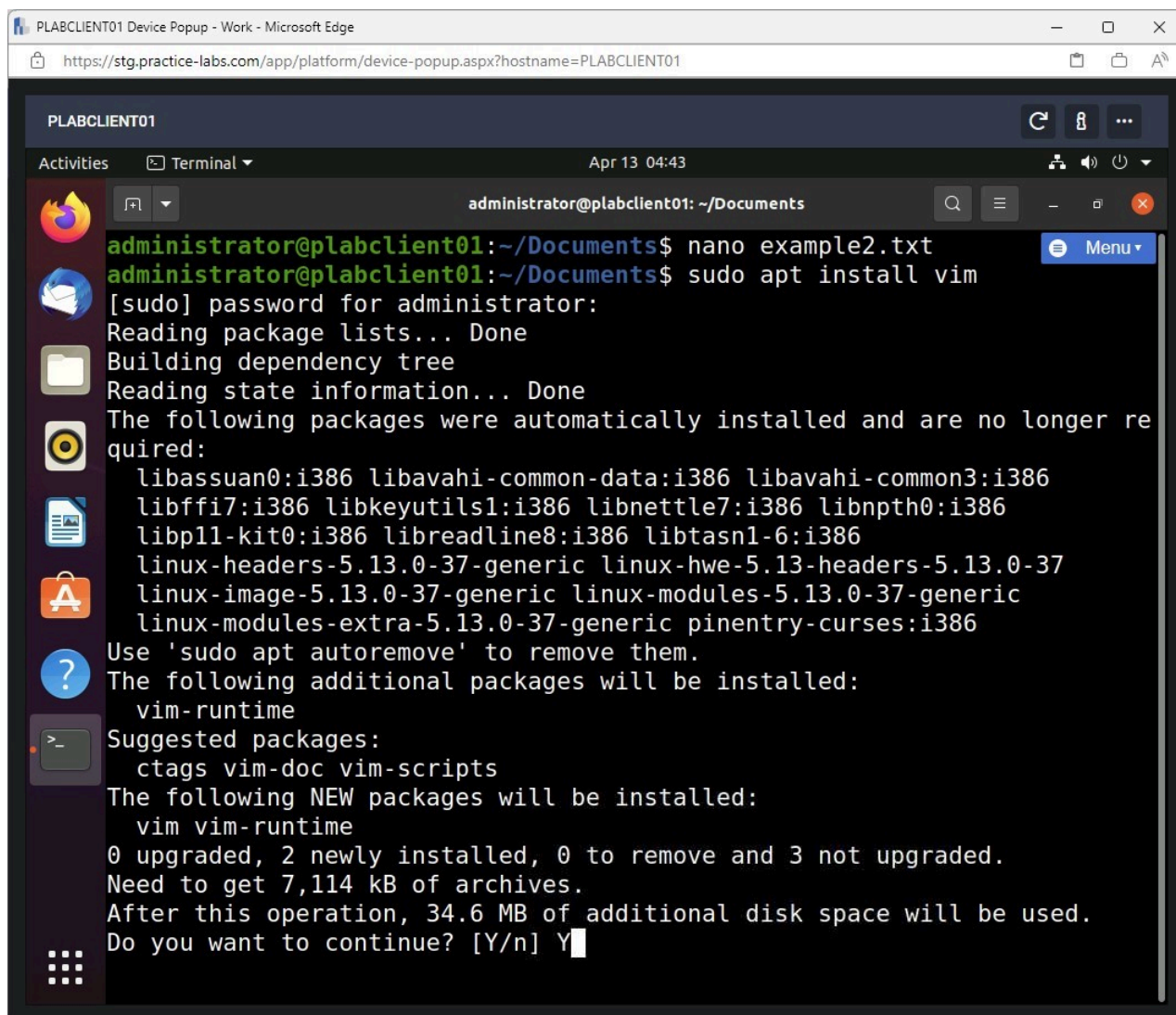
```
sudo apt install vim
```

Type the following password:

Passw0rd

In the **Do you want to continue?** prompt, type:

Y



The screenshot shows a terminal window titled 'PLABCLIENT01' with a dark background. The user is 'administrator@plabclient01' in the directory '~/Documents'. The terminal shows the following commands and output:

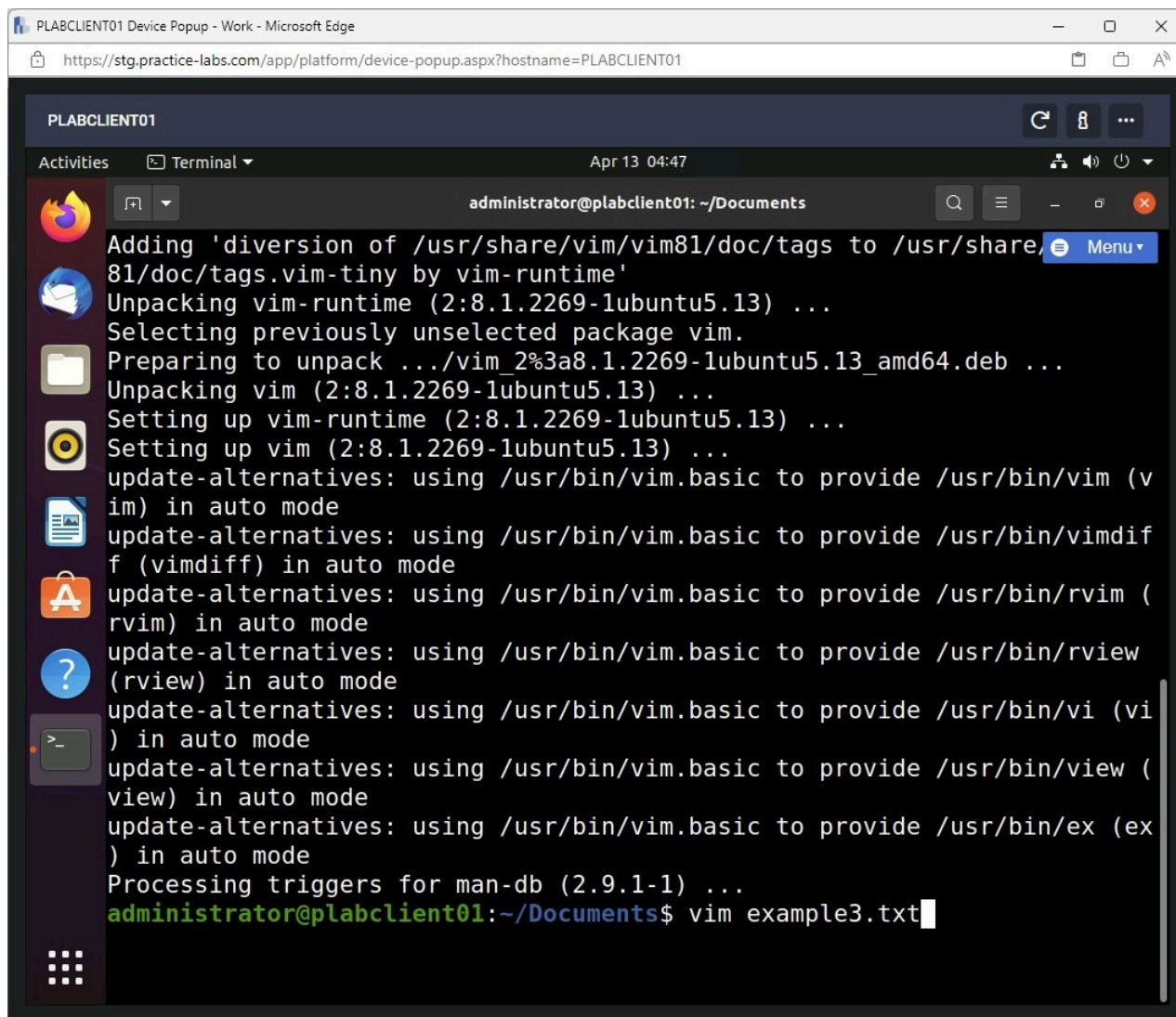
```
administrator@plabclient01:~/Documents$ nano example2.txt
administrator@plabclient01:~/Documents$ sudo apt install vim
[sudo] password for administrator:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libassuan0:i386 libavahi-common-data:i386 libavahi-common3:i386
  libffi7:i386 libkeyutils1:i386 libnettle7:i386 libnpth0:i386
  libp11-kit0:i386 libreadline8:i386 libtasn1-6:i386
  linux-headers-5.13.0-37-generic linux-hwe-5.13-headers-5.13.0-37
  linux-image-5.13.0-37-generic linux-modules-5.13.0-37-generic
  linux-modules-extra-5.13.0-37-generic pinentry-curses:i386
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  vim-runtime
Suggested packages:
  ctags vim-doc vim-scripts
The following NEW packages will be installed:
  vim vim-runtime
0 upgraded, 2 newly installed, 0 to remove and 3 not upgraded.
Need to get 7,114 kB of archives.
After this operation, 34.6 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
```

Figure 3.8 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Step 9

In the **Terminal** window, type the following command and press **Enter**:

vim example3.txt



The screenshot shows a terminal window titled 'PLABCLIENT01' with a dark background. The terminal output shows the installation of vim: 'Adding diversion of /usr/share/vim/vim81/doc/tags to /usr/share/vim81/doc/tags.vim-tiny by vim-runtime', 'Unpacking vim-runtime (2:8.1.2269-1ubuntu5.13) ...', 'Selecting previously unselected package vim.', 'Preparing to unpack .../vim_2%3a8.1.2269-1ubuntu5.13_amd64.deb ...', 'Unpacking vim (2:8.1.2269-1ubuntu5.13) ...', 'Setting up vim-runtime (2:8.1.2269-1ubuntu5.13) ...', 'Setting up vim (2:8.1.2269-1ubuntu5.13) ...'. It then lists several update-alternatives for vim variants (vimdiff, rvim, rview, vi, view, ex) in auto mode. Finally, it shows 'Processing triggers for man-db (2.9.1-1) ...' and the prompt 'administrator@plabclient01:~/Documents\$ vim example3.txt' with a cursor at the end of the command.

Figure 3.9 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Step 10

In the **example3.txt** file, type the following so that vim is in **Insert** mode and you can edit the file:

i

Next, type the following text:

This is an example of a file created using the vim editor.

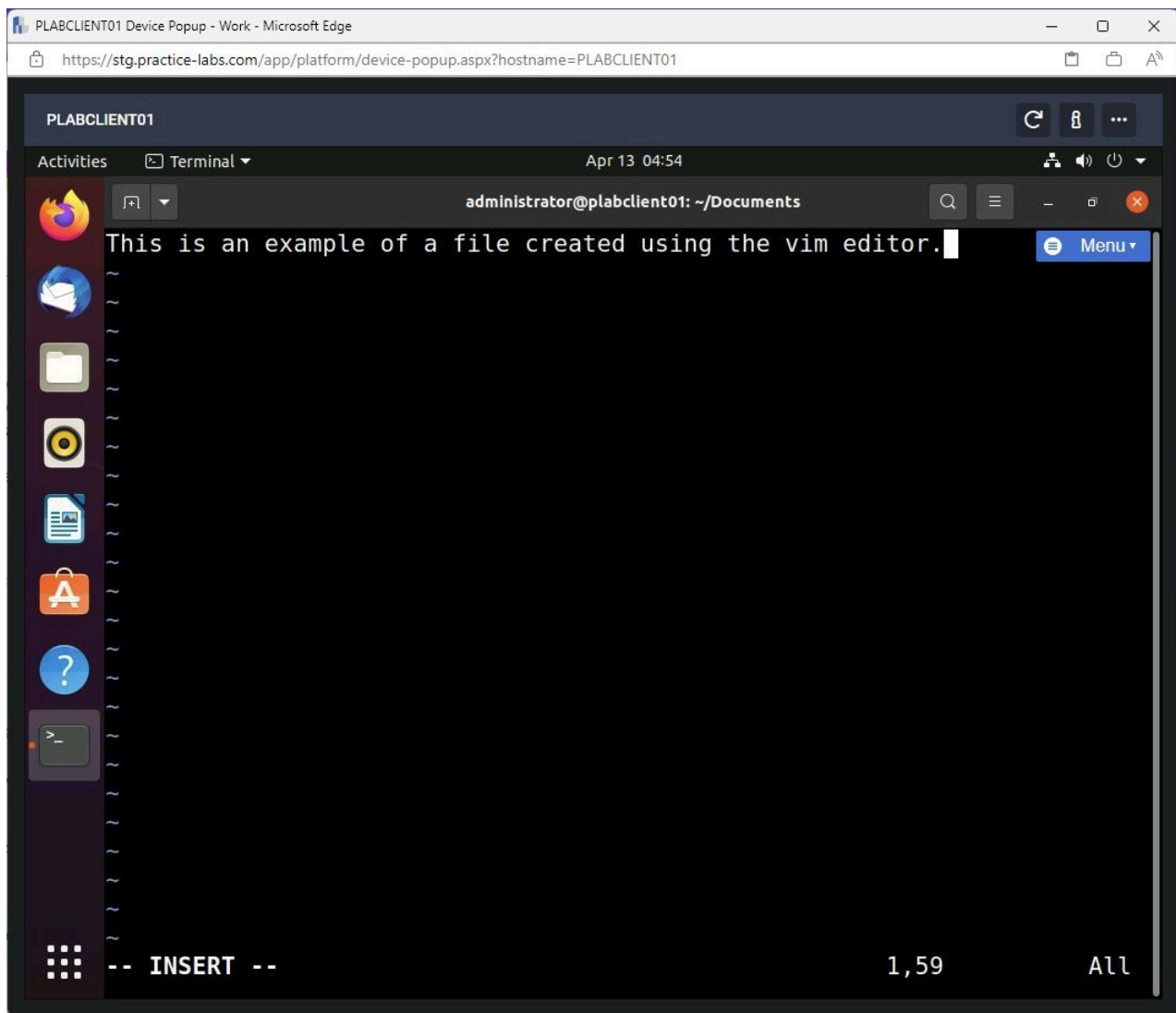


Figure 3.10 Screenshot of PLABCLIENT01: example3.txt file is displayed, showing the required text typed in.

Step 11

Press the **Esc** key on the keyboard.

Next, type the following and press **Enter** to exit the vim editor window:

```
:wq
```

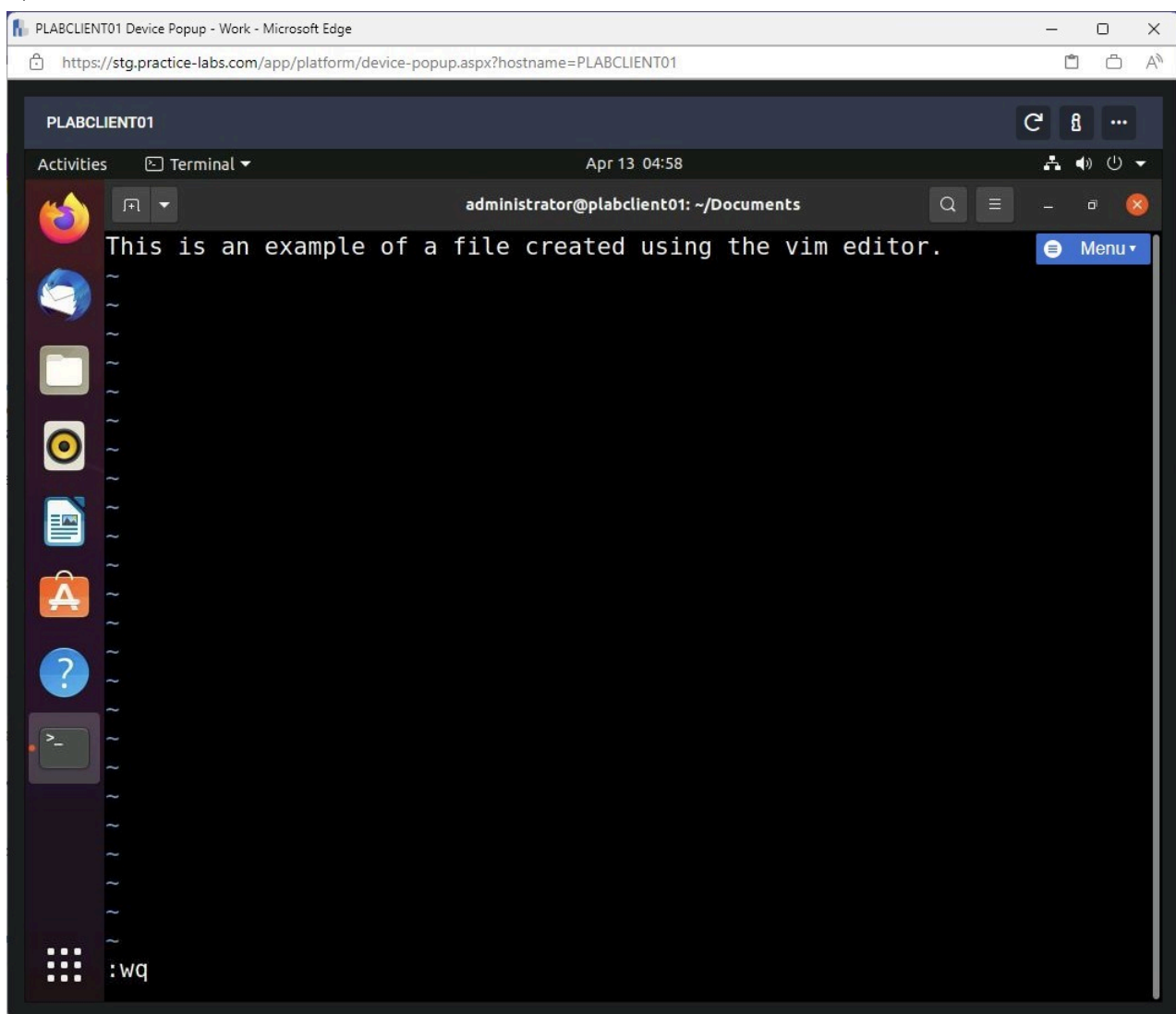
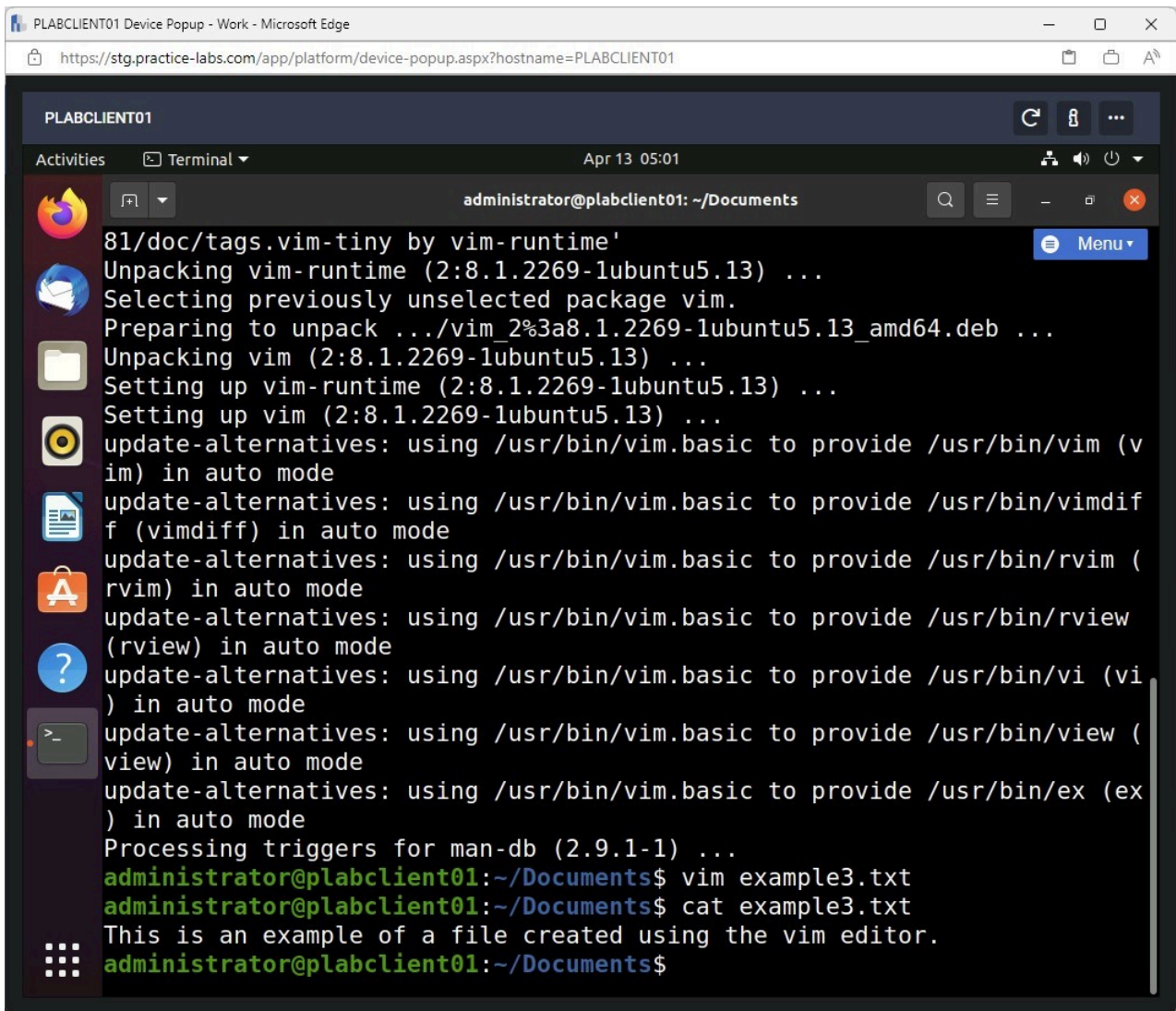


Figure 3.11 Screenshot of PLABCLIENT01: example3.txt file is displayed.

Step 12

In the **Terminal** window, type the following command and press **Enter**:

```
cat example3.txt
```



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Apr 13 05:01 administrator@plabclient01: ~/Documents
81/doc/tags.vim-tiny by vim-runtime'
Unpacking vim-runtime (2:8.1.2269-1ubuntu5.13) ...
Selecting previously unselected package vim.
Preparing to unpack .../vim_2%3a8.1.2269-1ubuntu5.13_amd64.deb ...
Unpacking vim (2:8.1.2269-1ubuntu5.13) ...
Setting up vim-runtime (2:8.1.2269-1ubuntu5.13) ...
Setting up vim (2:8.1.2269-1ubuntu5.13) ...
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/vim (v
im) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/vimdif
f (vimdiff) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/rvim (
rvim) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/rview
(rview) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/vi (vi
) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/view (
view) in auto mode
update-alternatives: using /usr/bin/vim.basic to provide /usr/bin/ex (ex
) in auto mode
Processing triggers for man-db (2.9.1-1) ...
administrator@plabclient01:~/Documents$ vim example3.txt
administrator@plabclient01:~/Documents$ cat example3.txt
This is an example of a file created using the vim editor.
administrator@plabclient01:~/Documents$
```

Figure 3.12 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The *cat* command will display the contents of the **example3.txt** file.

Type the **clear** command to clear the Terminal window before proceeding to the next exercise.

Keep the **Terminal** window open.

Exercise 4 - Move Files and Create Links

In this exercise, you will move files between directories and create hard and soft links.

When a selected file is hard linked to another file, the hard linked file is a copy of the selected file. It will have access to all the data available in the original file. If the selected file is deleted,

the hard linked file will still have access to its data.

When a selected file is soft linked to another file, the soft linked file will act as a reference to the selected file. It will not have access to the data in the original file. If the selected file is deleted, the soft link will be referenced to a file that does not exist.

Learning Outcomes

After completing this exercise, you should be able to:

- Move and View Files using Linux Commands
- Create Hard links and Soft links

Your Devices

You will be using the following device in this lab. Please power this on now.

- **PLABCLIENT01** - Ubuntu 20.04 - Stand-Alone Ubuntu Workstation (GUI)

Task 1 - Move and View Files using Linux Commands

In this task, you will move and view files using Linux commands.

Step 1

Ensure you are connected to **PLABCLIENT01** and the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

```
cd ~/Books
```

Next, type the following command and press **Enter**:

```
ls
```

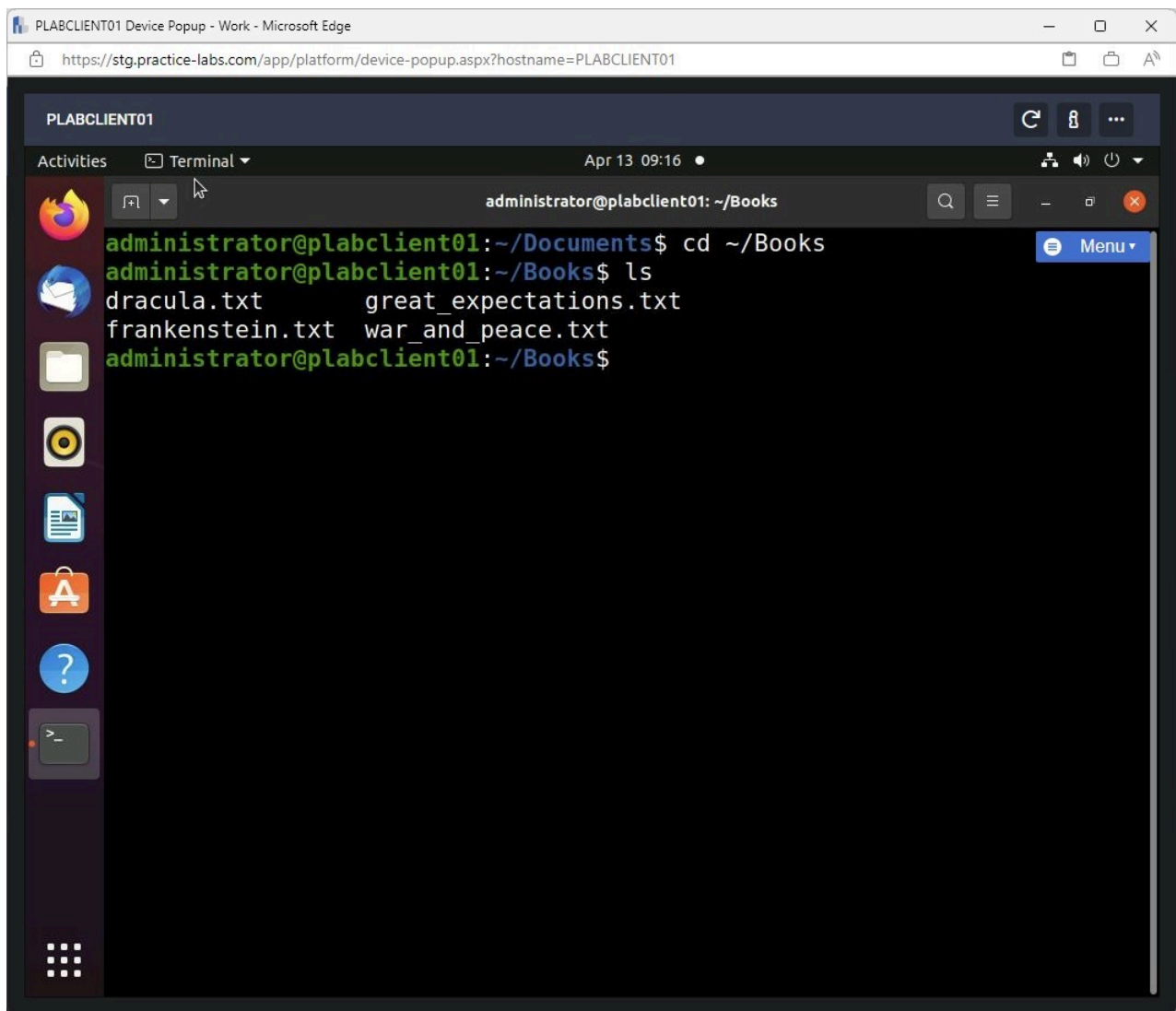


Figure 4.1 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **cd** command is used to navigate to the **Books** directory, and the **ls** command is used to list the contents of the **Books** directory.

Step 2

In the **Terminal** window, type the following command and press **Enter**:

```
cp dracula.txt Dracula.txt
```

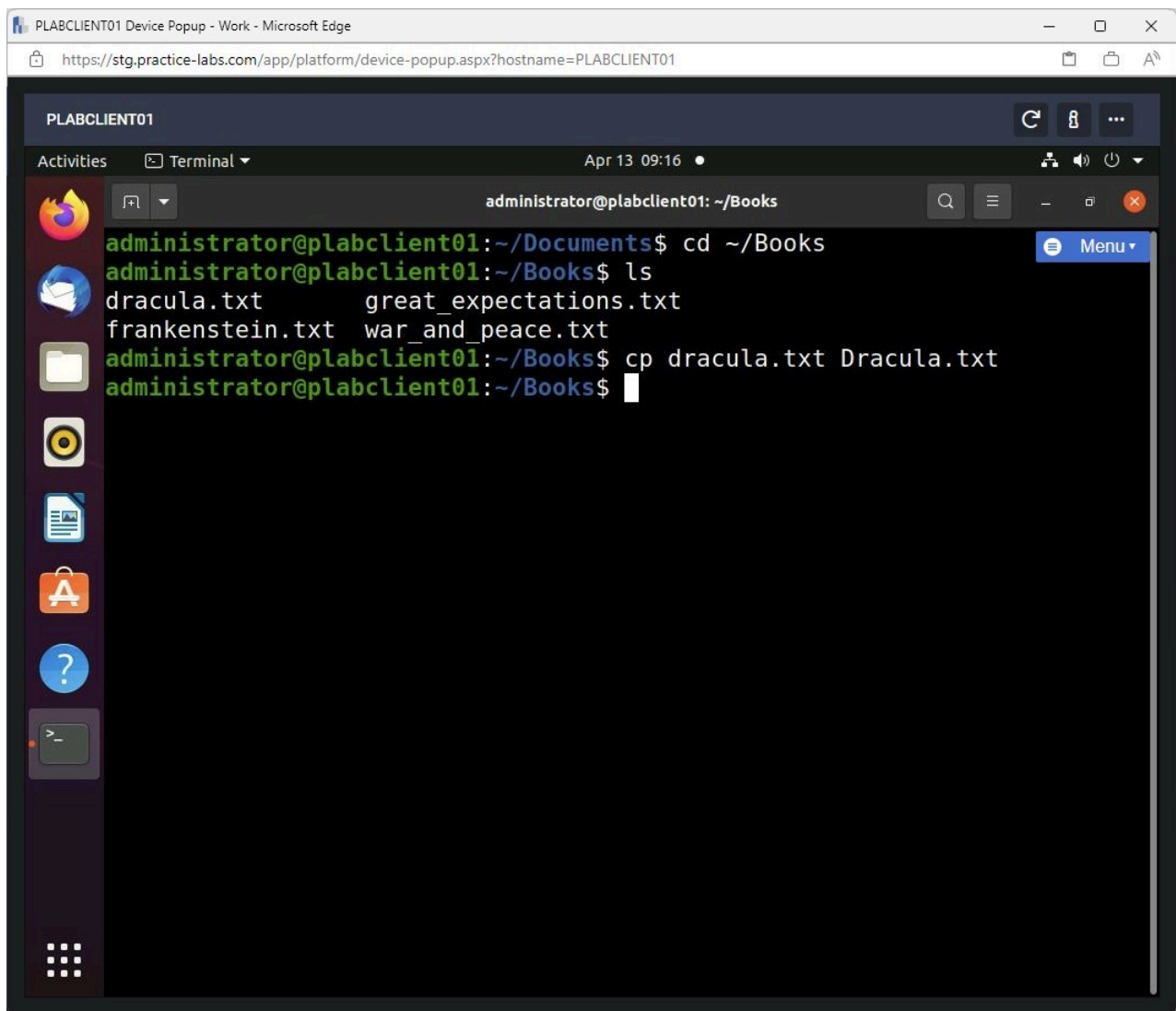


Figure 4.2 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **cp** command will create a copy file named **dracula.txt** and rename it to **Dracula.txt** within the same directory.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
cp frankenstein.txt ~/Documents/Frankenstein.txt
```

Next, type the following command and press **Enter**:


```
cd ~/Documents/
```

Then, type the following command and press **Enter**:

```
ls
```

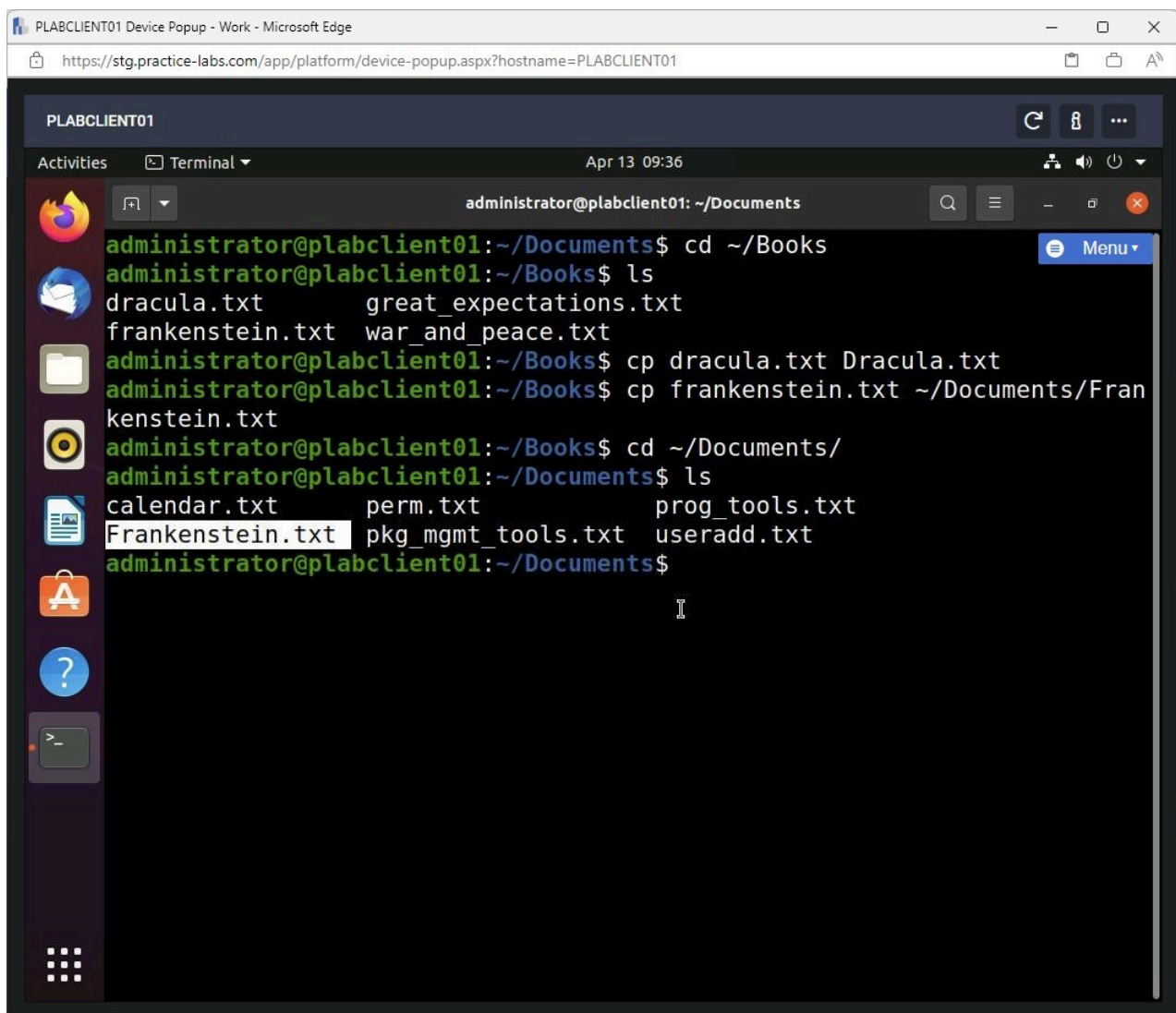


Figure 4.3 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **cp** command will create a copy file named **frankenstein.txt** and rename it to **Frankenstein.txt** in the **~/Documents** directory.

Step 4

In the **Terminal** window, type the following command and press **Enter**:

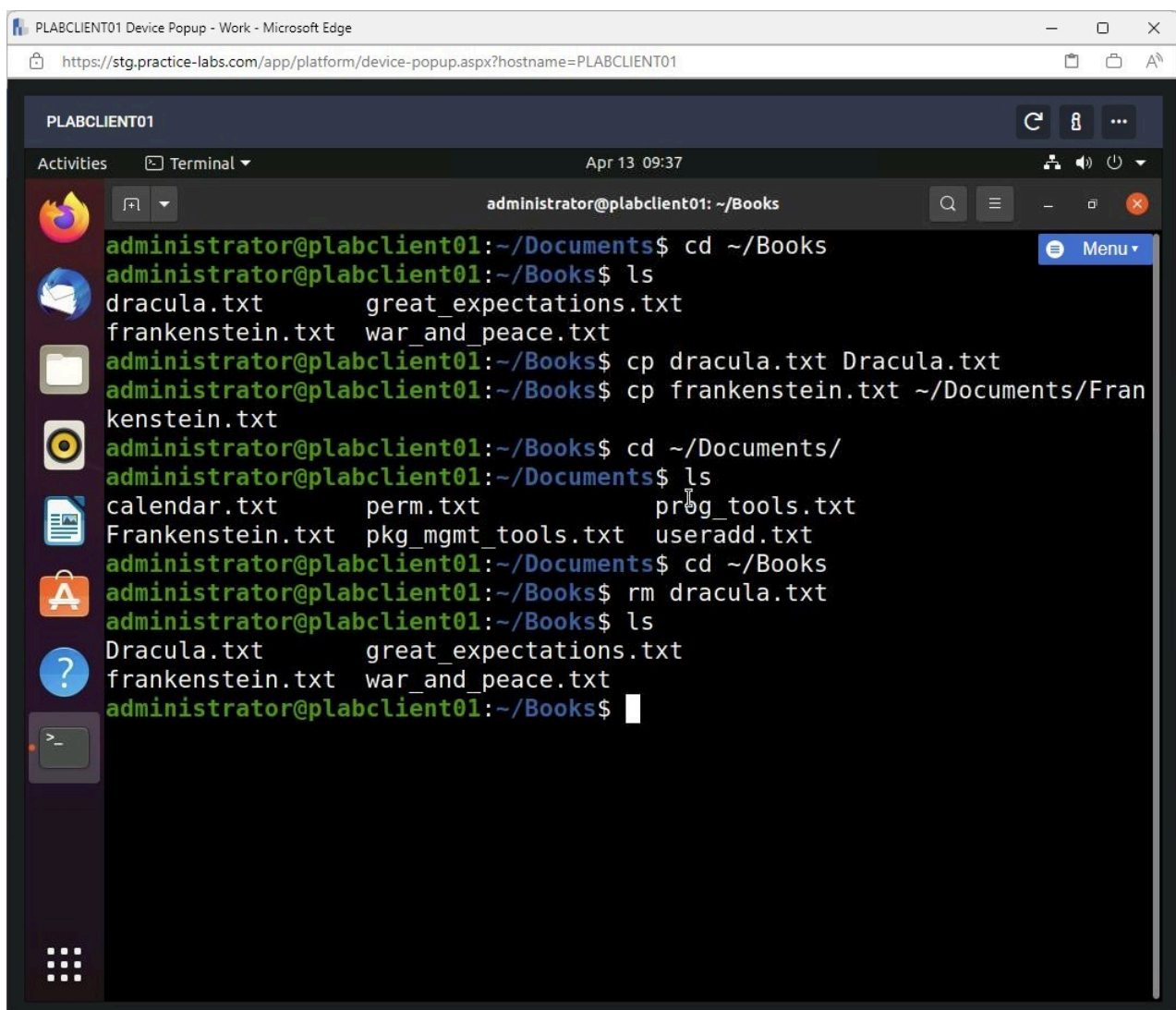
```
cd ~/Books
```

Next, type the following command and press **Enter**:

```
rm dracula.txt
```

Then, type the following command and press **Enter**:

```
ls
```



The screenshot shows a terminal window titled "PLABCLIENT01 Device Popup - Work - Microsoft Edge" with the URL "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01". The terminal interface includes a top bar with "PLABCLIENT01", a search icon, and a "Menu" button. Below the top bar is a status bar showing "Activities", "Terminal", and the date/time "Apr 13 09:37". The main terminal area shows the following commands and outputs:

```
administrator@plabclient01: ~/Documents$ cd ~/Books
administrator@plabclient01: ~/Books$ ls
dracula.txt          great_expectations.txt
frankenstein.txt     war_and_peace.txt
administrator@plabclient01: ~/Books$ cp dracula.txt Dracula.txt
administrator@plabclient01: ~/Books$ cp frankenstein.txt ~/Documents/Fran
kenstein.txt
administrator@plabclient01: ~/Books$ cd ~/Documents/
administrator@plabclient01: ~/Documents$ ls
calendar.txt         perm.txt             prog_tools.txt
Frankenstein.txt     pkg_mgmt_tools.txt  useradd.txt
administrator@plabclient01: ~/Documents$ cd ~/Books
administrator@plabclient01: ~/Books$ rm dracula.txt
administrator@plabclient01: ~/Books$ ls
Dracula.txt          great_expectations.txt
frankenstein.txt     war_and_peace.txt
administrator@plabclient01: ~/Books$
```

Figure 4.4 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **rm** command is used to remove the **dracula.txt** file. Notice the **dracula.txt** file is now removed from the **Books** directory.

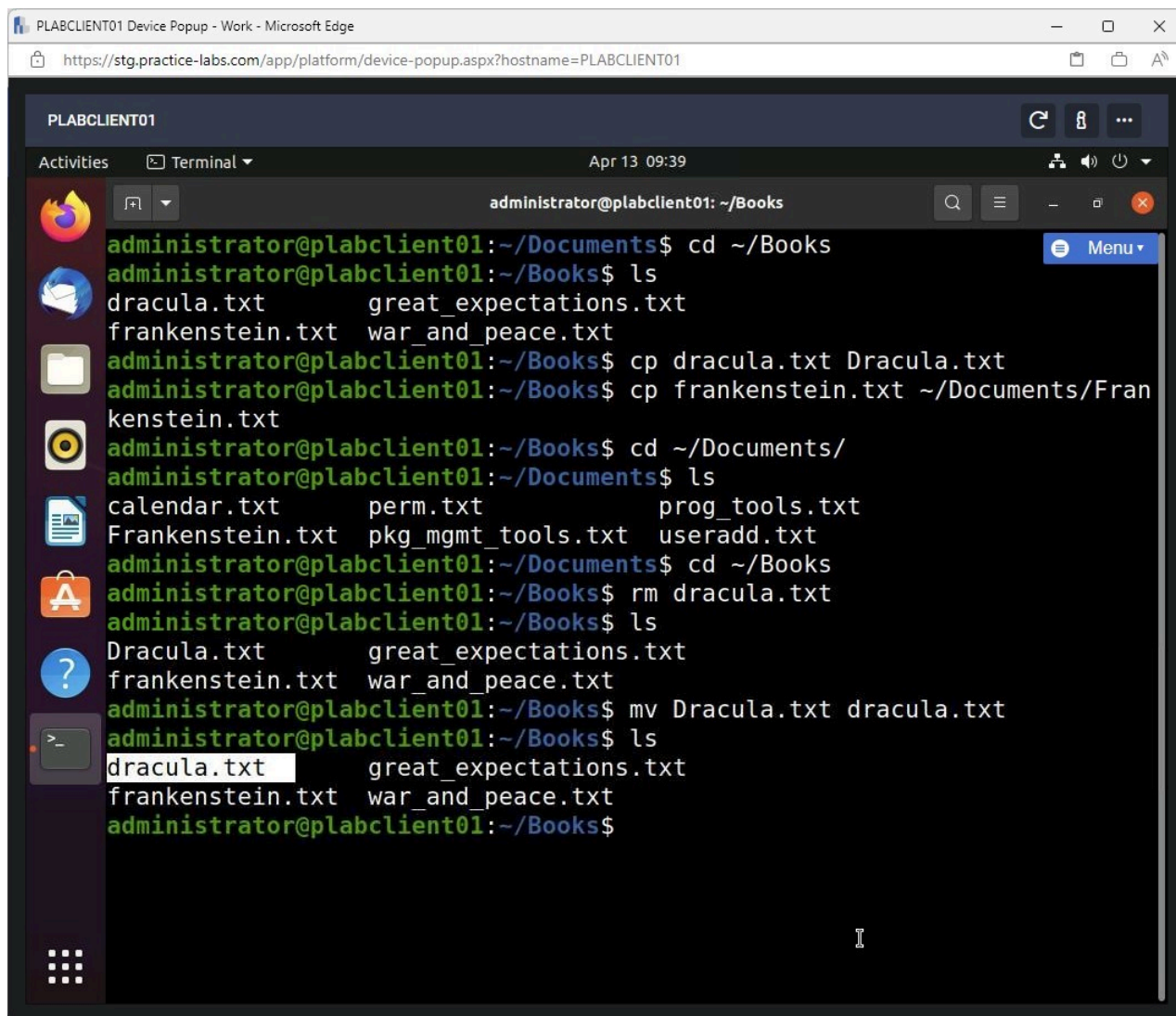
Step 5

In the **Terminal** window, type the following command and press **Enter**:

```
mv Dracula.txt dracula.txt
```

Then, type the following command and press **Enter**:

```
ls
```



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Apr 13 09:39 administrator@plabclient01: ~/Books

administrator@plabclient01:~/Documents$ cd ~/Books
administrator@plabclient01:~/Books$ ls
dracula.txt          great_expectations.txt
frankenstein.txt     war_and_peace.txt
administrator@plabclient01:~/Books$ cp dracula.txt Dracula.txt
administrator@plabclient01:~/Books$ cp frankenstein.txt ~/Documents/Fran
kenstein.txt
administrator@plabclient01:~/Books$ cd ~/Documents/
administrator@plabclient01:~/Documents$ ls
calendar.txt          perm.txt              prog_tools.txt
Frankenstein.txt     pkg_mgmt              tools.txt            useradd.txt
administrator@plabclient01:~/Documents$ cd ~/Books
administrator@plabclient01:~/Books$ rm dracula.txt
administrator@plabclient01:~/Books$ ls
Dracula.txt          great_expectations.txt
frankenstein.txt     war_and_peace.txt
administrator@plabclient01:~/Books$ mv Dracula.txt dracula.txt
administrator@plabclient01:~/Books$ ls
dracula.txt          great_expectations.txt
frankenstein.txt     war_and_peace.txt
administrator@plabclient01:~/Books$
```

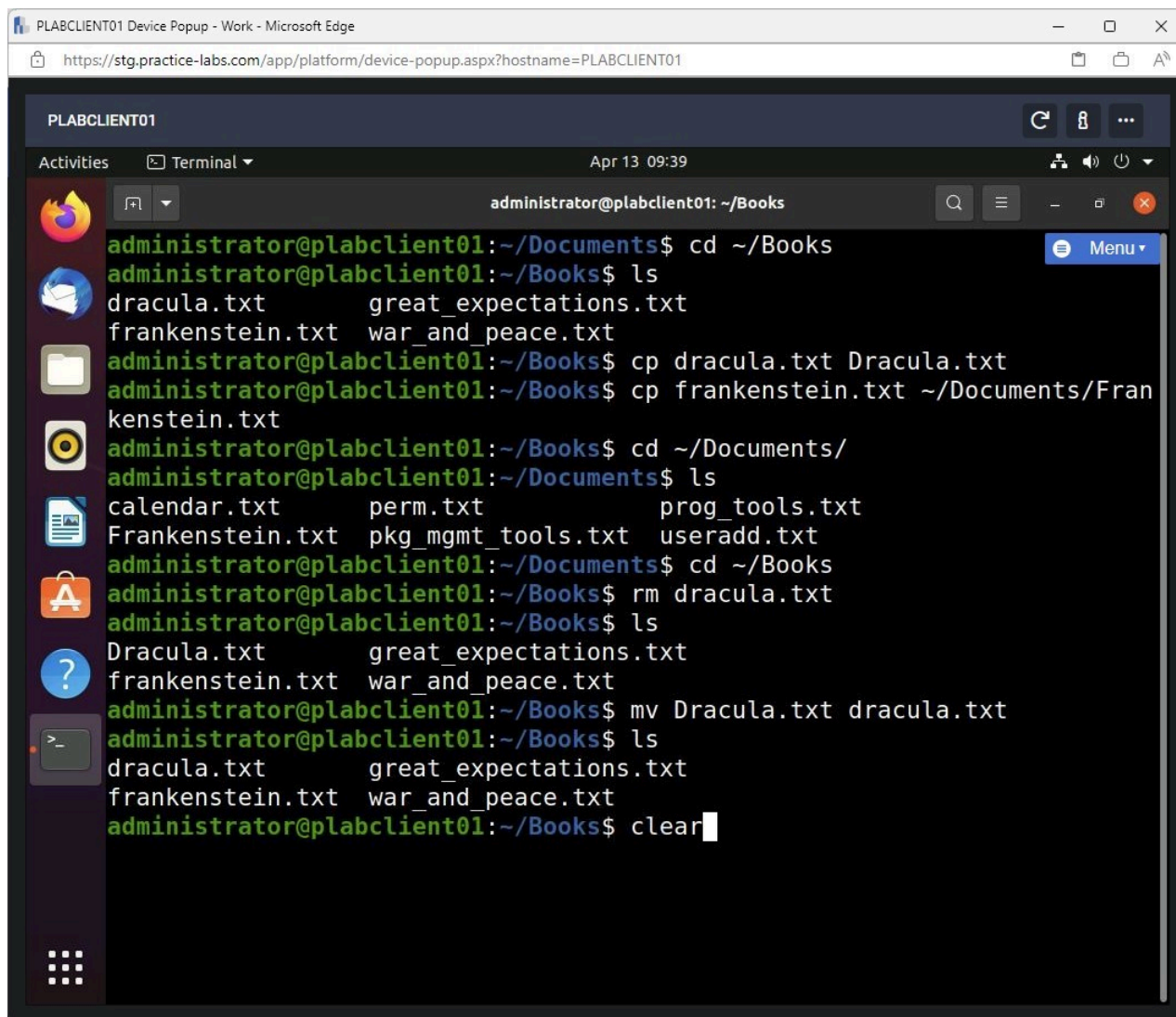
Figure 4.5 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered and executed.

Note: The **mv** command creates a copy of **Dracula.txt** and renames it to **dracula.txt**, then deletes **Dracula.txt**.

Step 6

In the **Terminal** window, type the following command and press **Enter**:

```
clear
```

```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Apr 13 09:39 administrator@plabclient01: ~/Books

administrator@plabclient01:~/Documents$ cd ~/Books
administrator@plabclient01:~/Books$ ls
dracula.txt          great_expectations.txt
frankenstein.txt     war_and_peace.txt
administrator@plabclient01:~/Books$ cp dracula.txt Dracula.txt
administrator@plabclient01:~/Books$ cp frankenstein.txt ~/Documents/Fran
kenstein.txt
administrator@plabclient01:~/Books$ cd ~/Documents/
administrator@plabclient01:~/Documents$ ls
calendar.txt         perm.txt             prog_tools.txt
Frankenstein.txt     pkg_mgmt             tools.txt            useradd.txt
administrator@plabclient01:~/Documents$ cd ~/Books
administrator@plabclient01:~/Books$ rm dracula.txt
administrator@plabclient01:~/Books$ ls
Dracula.txt          great_expectations.txt
frankenstein.txt     war_and_peace.txt
administrator@plabclient01:~/Books$ mv Dracula.txt dracula.txt
administrator@plabclient01:~/Books$ ls
dracula.txt          great_expectations.txt
frankenstein.txt     war_and_peace.txt
administrator@plabclient01:~/Books$ clear
```

Figure 4.6 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required command entered.

Task 2 - Create Hard Links and Soft Links

In this task, you will create hard links and soft links.

Step 1

In **PLABCLIENT01**, the **Terminal** window is open.

In the **Terminal** window, type the following command and press **Enter**:

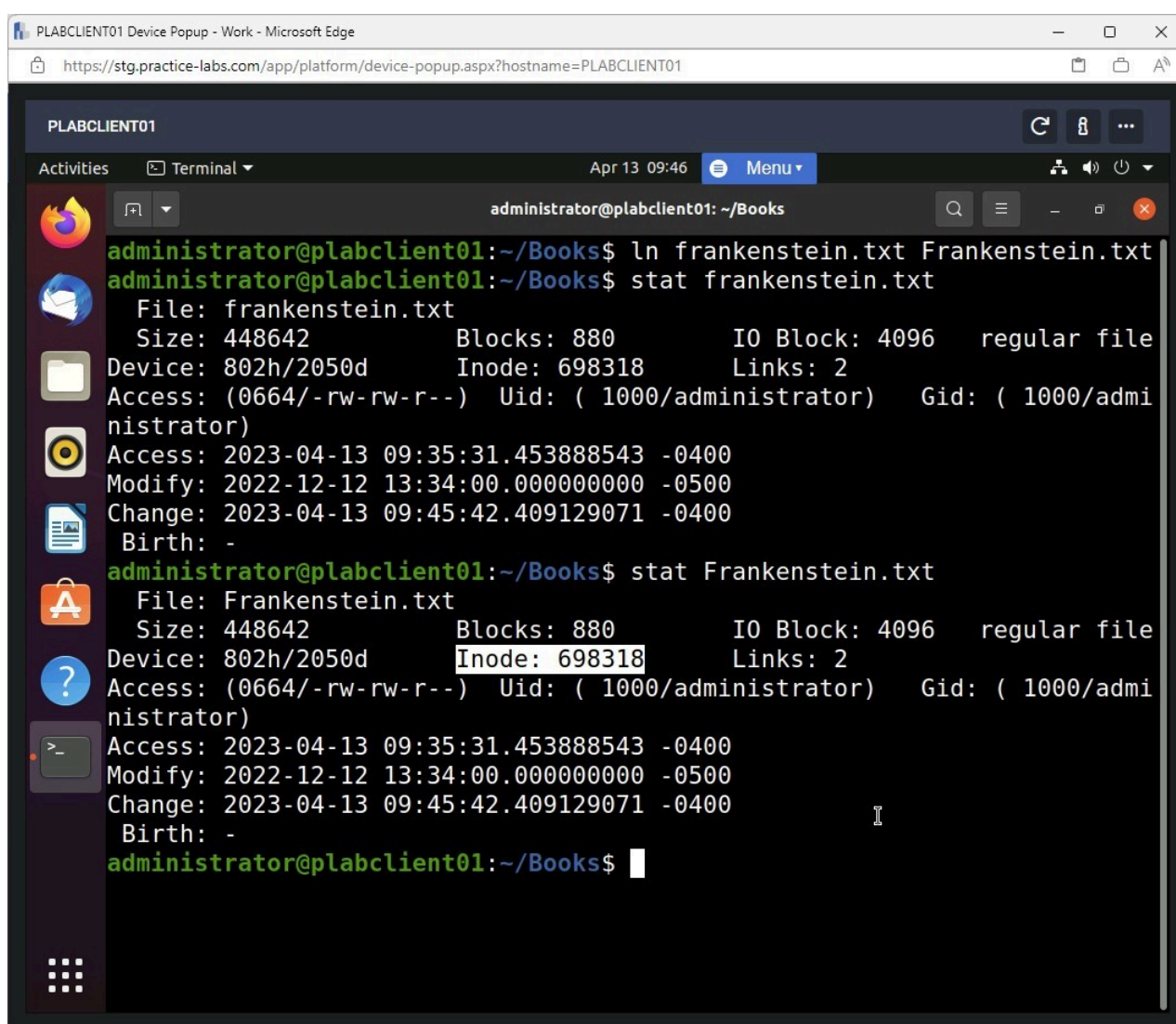
```
ln frankenstein.txt Frankenstein.txt
```

Next, type the following command and press **Enter**:

```
stat frankenstein.txt
```

Then, type the following command and press **Enter**:

```
stat Frankenstein.txt
```



The screenshot shows a terminal window titled 'PLABCLIENT01' with a dark background. The prompt is 'administrator@plabclient01: ~/Books'. The user enters the command 'ln frankenstein.txt Frankenstein.txt', which creates a hard link. Then, the user enters 'stat frankenstein.txt', displaying file statistics for 'frankenstein.txt'. The output shows a regular file with size 448642, 880 blocks, and inode 698318. Next, the user enters 'stat Frankenstein.txt', displaying statistics for 'Frankenstein.txt'. The output is identical, with the 'Inode: 698318' field highlighted. The terminal window includes a sidebar with application icons and a top bar with system information like 'Apr 13 09:46' and a 'Menu' button.

```
administrator@plabclient01:~/Books$ ln frankenstein.txt Frankenstein.txt
administrator@plabclient01:~/Books$ stat frankenstein.txt
  File: frankenstein.txt
  Size: 448642          Blocks: 880          IO Block: 4096   regular file
Device: 802h/2050d     Inode: 698318        Links: 2
Access: (0664/-rw-rw-r--)  Uid: ( 1000/administrator)   Gid: ( 1000/administrator)
Access: 2023-04-13 09:35:31.453888543 -0400
Modify: 2022-12-12 13:34:00.000000000 -0500
Change: 2023-04-13 09:45:42.409129071 -0400
 Birth: -
administrator@plabclient01:~/Books$ stat Frankenstein.txt
  File: Frankenstein.txt
  Size: 448642          Blocks: 880          IO Block: 4096   regular file
Device: 802h/2050d     Inode: 698318        Links: 2
Access: (0664/-rw-rw-r--)  Uid: ( 1000/administrator)   Gid: ( 1000/administrator)
Access: 2023-04-13 09:35:31.453888543 -0400
Modify: 2022-12-12 13:34:00.000000000 -0500
Change: 2023-04-13 09:45:42.409129071 -0400
 Birth: -
administrator@plabclient01:~/Books$
```

Figure 4.7 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **ln** command will create a hard link for the **frankenstein.txt** file to **Frankenstein.txt**. This allows you to refer to the same file contents on the system regardless of how the file name is typed. Notice that the **Inode** number is the same. Files

*that are hard linked will have the same **Inode** number. Another file with the same content wasn't created.*

Step 2

In the **Terminal** window, type the following command and press **Enter**:

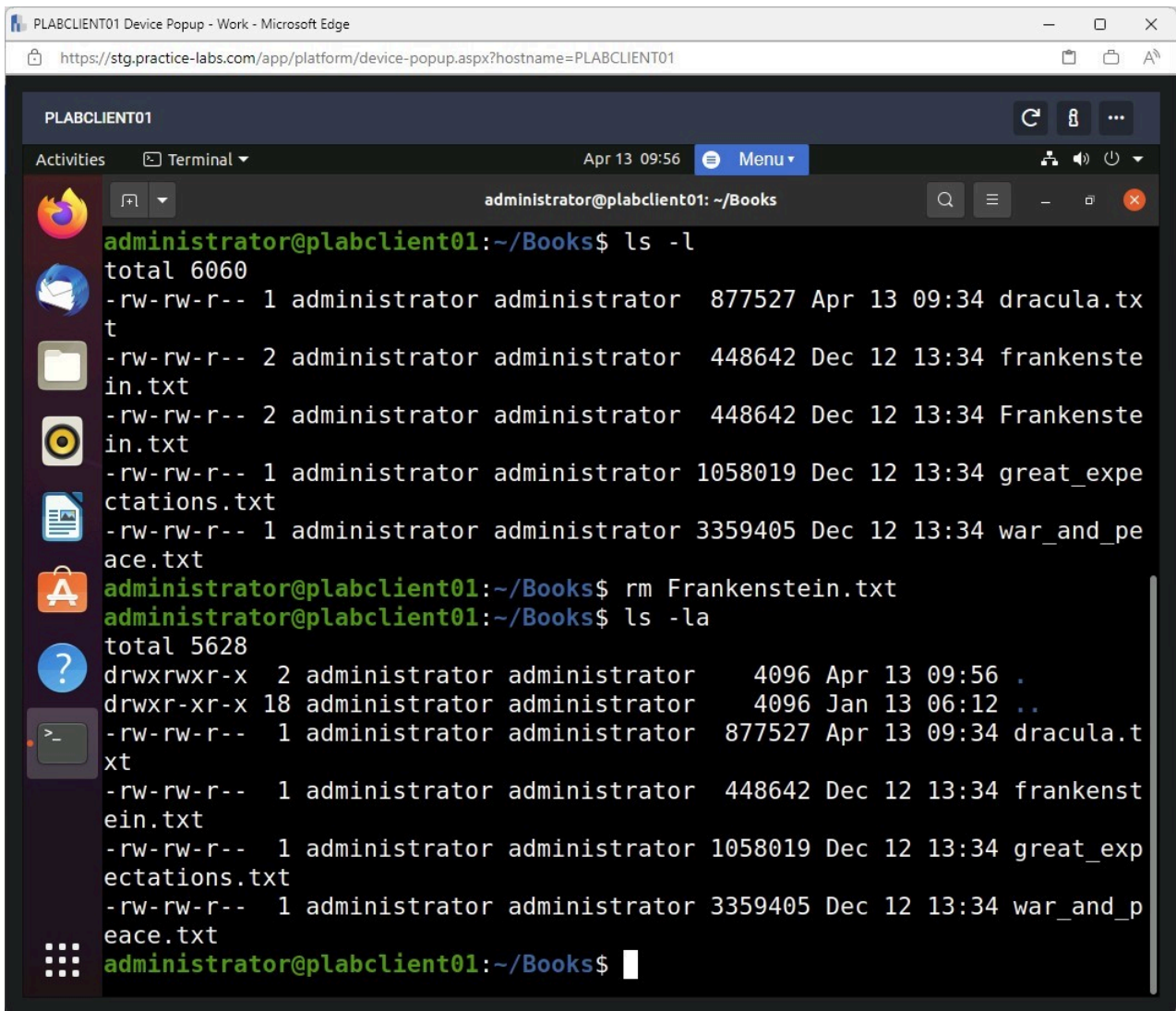
```
ls -l
```

Next, type the following command and press **Enter**:

```
rm Frankenstein.txt
```

Then, type the following command and press **Enter**:

```
ls -la
```

A screenshot of a terminal window titled 'PLABCLIENT01 Device Popup - Work - Microsoft Edge'. The browser address bar shows 'https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01'. The terminal window has a dark theme and shows the following commands and output:

```
administrator@plabclient01: ~/Books
administrator@plabclient01:~/Books$ ls -l
total 6060
-rw-rw-r-- 1 administrator administrator 877527 Apr 13 09:34 dracula.txt
-rw-rw-r-- 2 administrator administrator 448642 Dec 12 13:34 frankenstein.txt
-rw-rw-r-- 2 administrator administrator 448642 Dec 12 13:34 Frankenstein.txt
-rw-rw-r-- 1 administrator administrator 1058019 Dec 12 13:34 great_expectations.txt
-rw-rw-r-- 1 administrator administrator 3359405 Dec 12 13:34 war_and_peace.txt
administrator@plabclient01:~/Books$ rm Frankenstein.txt
administrator@plabclient01:~/Books$ ls -la
total 5628
drwxrwxr-x 2 administrator administrator 4096 Apr 13 09:56 .
drwxr-xr-x 18 administrator administrator 4096 Jan 13 06:12 ..
-rw-rw-r-- 1 administrator administrator 877527 Apr 13 09:34 dracula.txt
-rw-rw-r-- 1 administrator administrator 448642 Dec 12 13:34 frankenstein.txt
-rw-rw-r-- 1 administrator administrator 1058019 Dec 12 13:34 great_expectations.txt
-rw-rw-r-- 1 administrator administrator 3359405 Dec 12 13:34 war_and_peace.txt
administrator@plabclient01:~/Books$
```

Figure 4.8 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **rm** command will remove the hard link to **Frankenstein.txt**. The **ls -la** command is used to verify the removal of the hard link **Frankenstein.txt** file. Notice that the hard link file is now removed.

Step 3

In the **Terminal** window, type the following command and press **Enter**:

```
ln -s /home/administrator/Books/dracula.txt
/home/administrator/Books/Dracula.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```

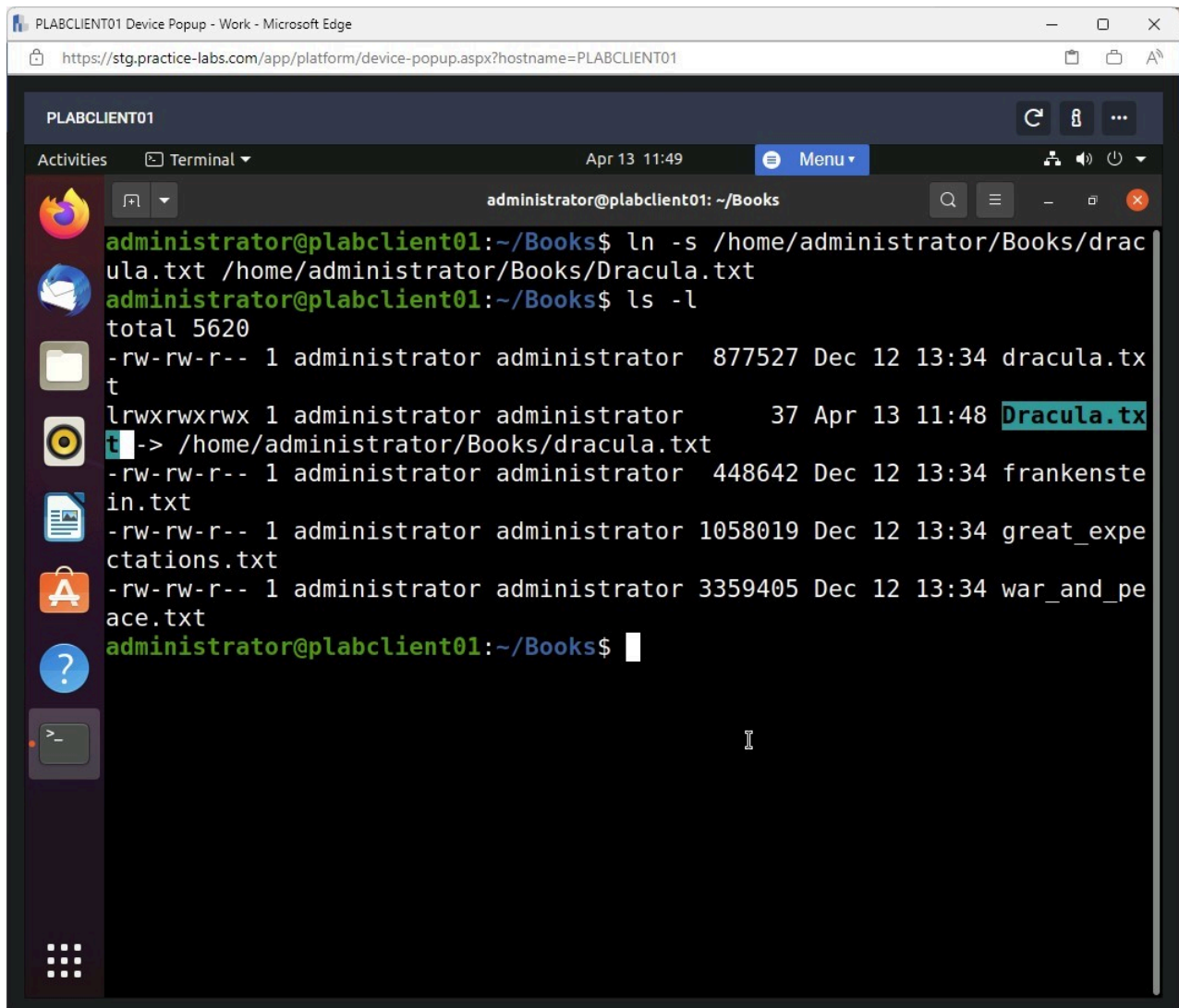


Figure 4.9 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **ln -s** command will create a soft link for the **dracula.txt** file to **Dracula.txt**. Notice the line entry for **Dracula.txt** once the **ls** command is executed.

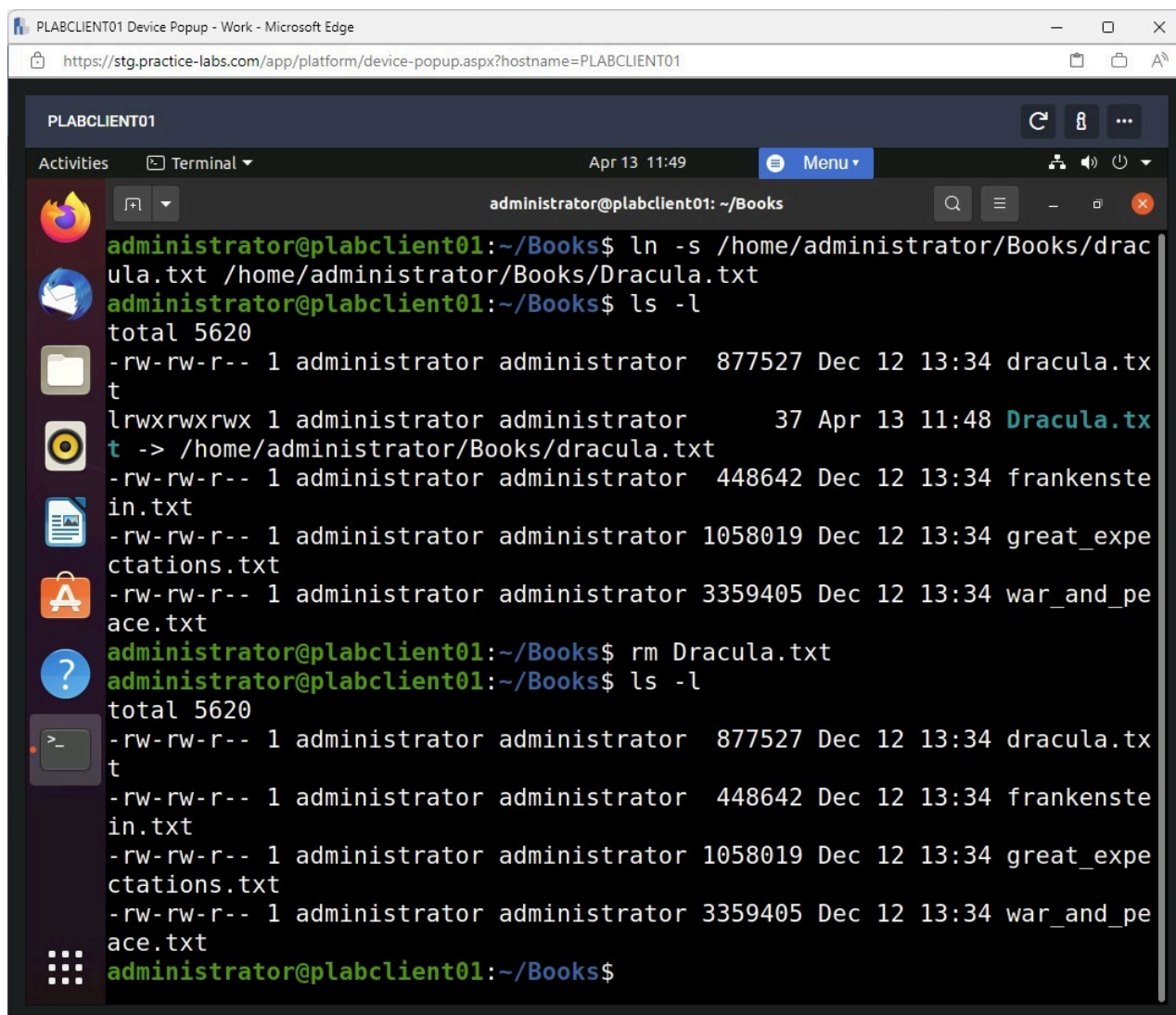
Step 4

In the **Terminal** window, type the following command and press **Enter**:

```
rm Dracula.txt
```

Next, type the following command and press **Enter**:

```
ls -l
```



```
PLABCLIENT01 Device Popup - Work - Microsoft Edge
https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01

PLABCLIENT01
Activities Terminal Apr 13 11:49 Menu
administrator@plabclient01: ~/Books
administrator@plabclient01:~/Books$ ln -s /home/administrator/Books/dracula.txt /home/administrator/Books/Dracula.txt
administrator@plabclient01:~/Books$ ls -l
total 5620
-rw-rw-r-- 1 administrator administrator 877527 Dec 12 13:34 dracula.txt
lrwxrwxrwx 1 administrator administrator 37 Apr 13 11:48 Dracula.txt -> /home/administrator/Books/dracula.txt
-rw-rw-r-- 1 administrator administrator 448642 Dec 12 13:34 frankenstein.txt
-rw-rw-r-- 1 administrator administrator 1058019 Dec 12 13:34 great_expectations.txt
-rw-rw-r-- 1 administrator administrator 3359405 Dec 12 13:34 war_and_peace.txt
administrator@plabclient01:~/Books$ rm Dracula.txt
administrator@plabclient01:~/Books$ ls -l
total 5620
-rw-rw-r-- 1 administrator administrator 877527 Dec 12 13:34 dracula.txt
-rw-rw-r-- 1 administrator administrator 448642 Dec 12 13:34 frankenstein.txt
-rw-rw-r-- 1 administrator administrator 1058019 Dec 12 13:34 great_expectations.txt
-rw-rw-r-- 1 administrator administrator 3359405 Dec 12 13:34 war_and_peace.txt
administrator@plabclient01:~/Books$
```

Figure 4.10 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **rm** command will remove the soft link to **Dracula.txt**. The **ls -l** command is used to verify the removal of the soft link **Dracula.txt** file. Notice that the soft link file is now removed.

Step 5

In the **Terminal** window, type the following command and press **Enter**:

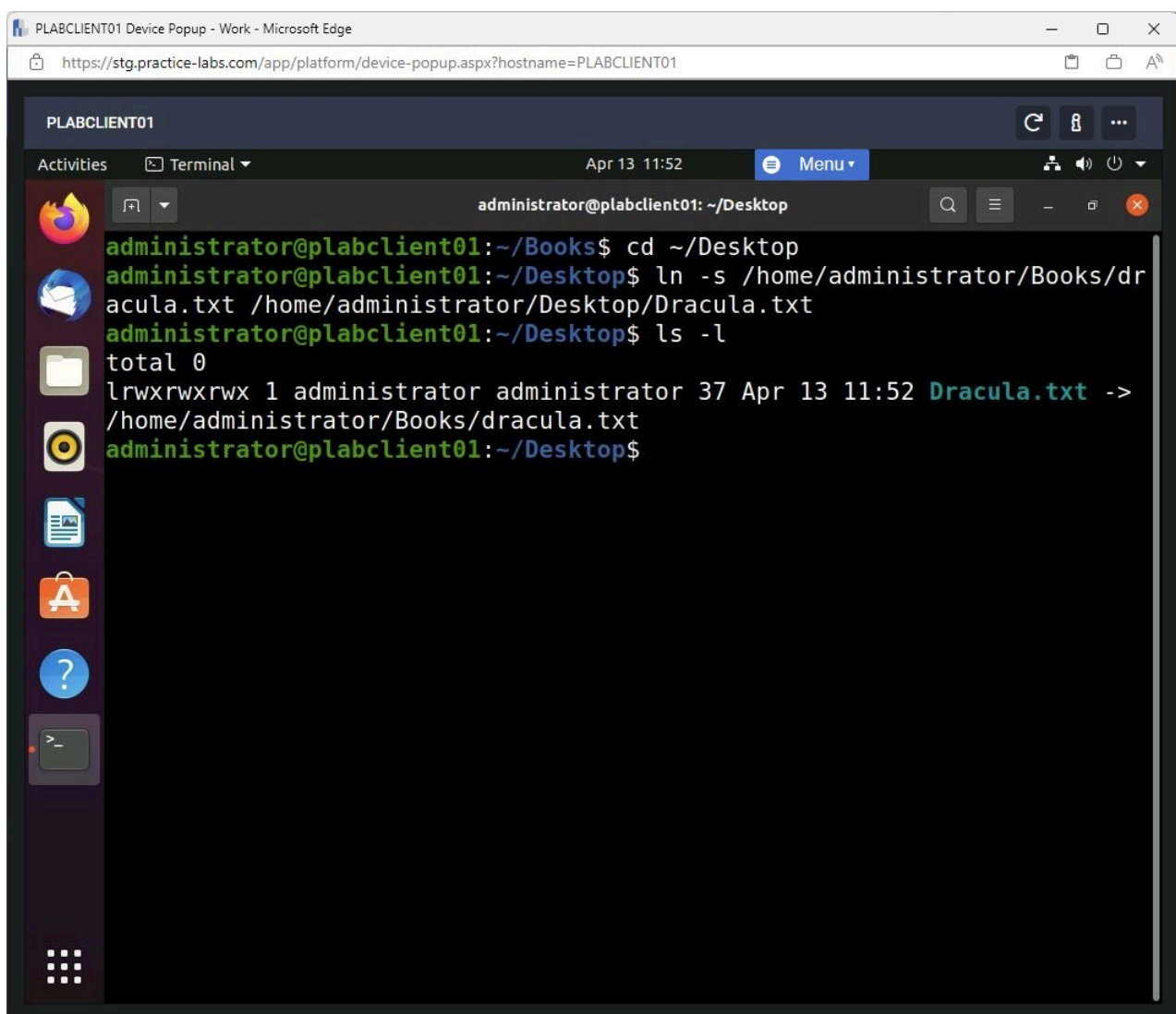
```
cd ~/Desktop
```

Next, type the following command and press **Enter**:

```
ln -s /home/administrator/Books/dracula.txt  
/home/administrator/Desktop/Dracula.txt
```

Then, type the following command and press **Enter**:

```
ls -l
```



The screenshot shows a terminal window titled "PLABCLIENT01 Device Popup - Work - Microsoft Edge" with the URL "https://stg.practice-labs.com/app/platform/device-popup.aspx?hostname=PLABCLIENT01". The terminal prompt is "administrator@plabclient01: ~/Desktop". The commands and their output are as follows:

```
administrator@plabclient01:~/Books$ cd ~/Desktop  
administrator@plabclient01:~/Desktop$ ln -s /home/administrator/Books/dracula.txt /home/administrator/Desktop/Dracula.txt  
administrator@plabclient01:~/Desktop$ ls -l  
total 0  
lrwxrwxrwx 1 administrator administrator 37 Apr 13 11:52 Dracula.txt ->  
/home/administrator/Books/dracula.txt  
administrator@plabclient01:~/Desktop$
```


Figure 4.11 Screenshot of PLABCLIENT01: The Terminal window is displayed, showing the required commands entered and executed.

Note: The **ln -s** command will create a new soft link file named **Dracula.txt** to the **Desktop**.

Step 6

Minimize the **Terminal** window.

Notice the soft linked file **Dracula.txt** now appears on the **Desktop**.

Double-click on the **Dracula.txt** file to verify the link works.

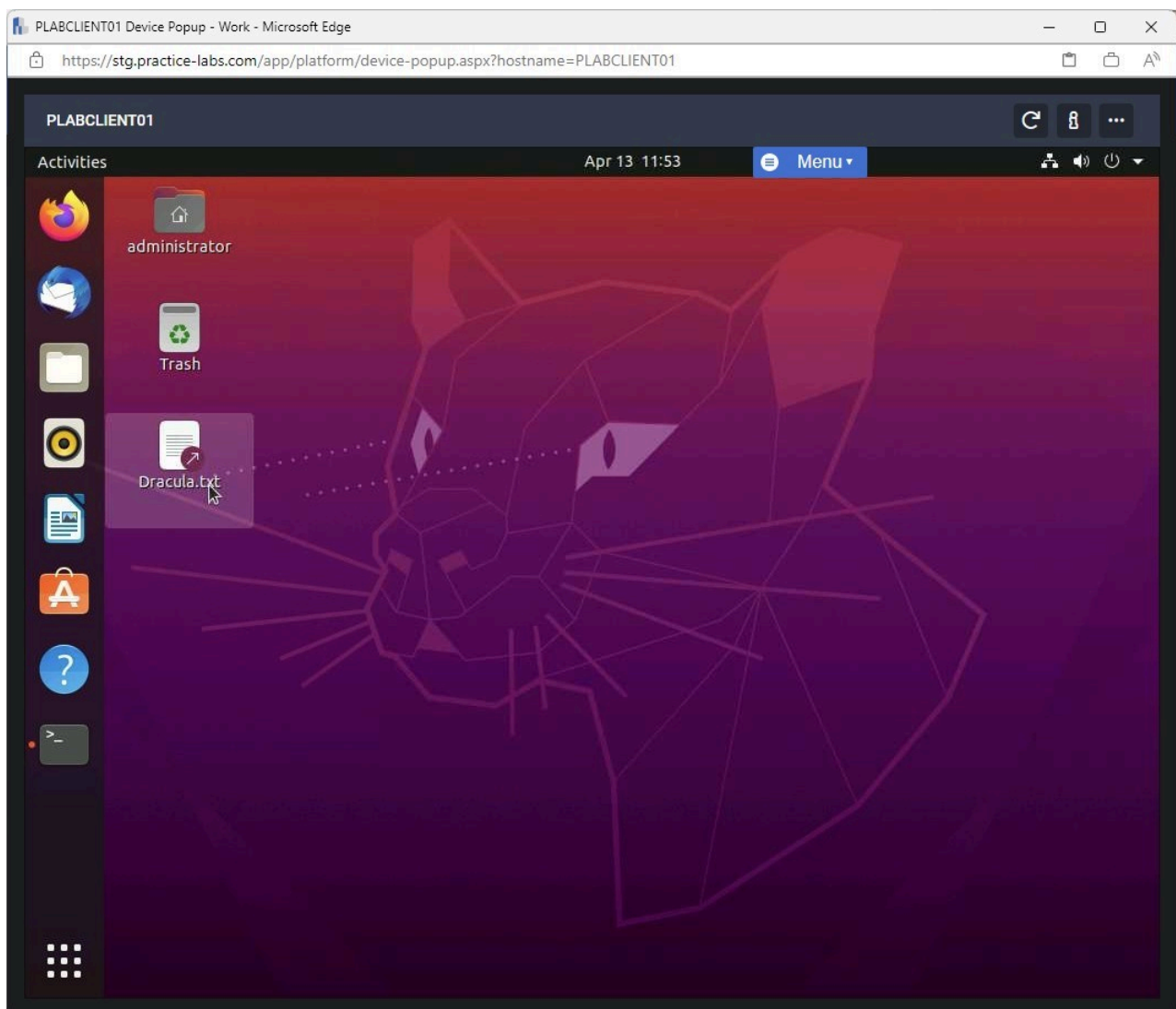


Figure 4.12 Screenshot of PLABCLIENT01: Ubuntu desktop is displayed, showing the soft linked Dracula.txt file.

Step 7

The **Dracula.txt** file is displayed in the **Text Editor** window.

Close the **Text Editor** and the **Terminal** windows.

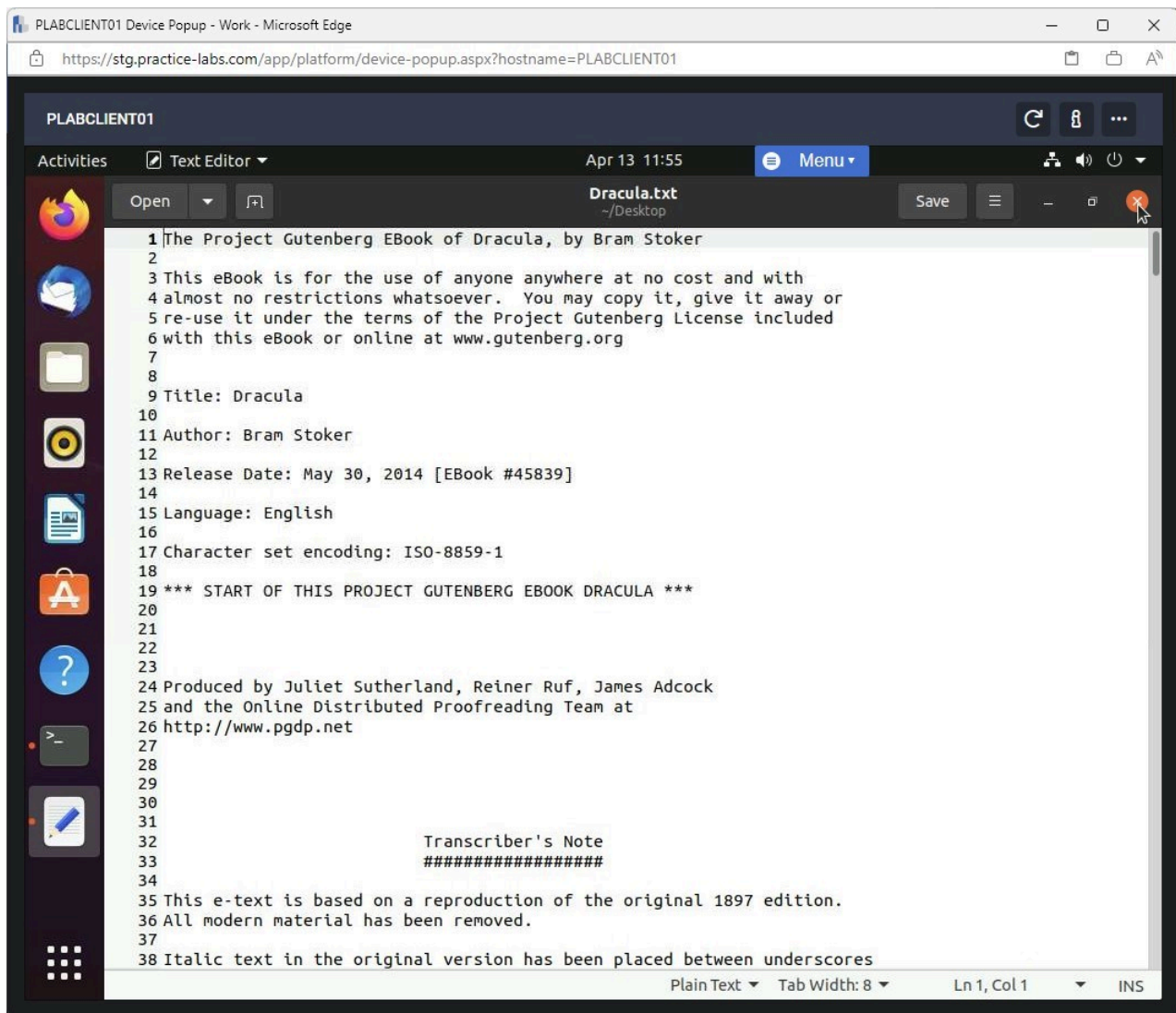


Figure 4.13 Screenshot of PLABCLIENT01: The Dracula.txt file is displayed in the Text Editor window. Close icon on the top-right corner of the window is selected.

Keep all devices that you have powered on in their current state and proceed to the review section.

Review

Well done, you have completed the **File and Directory Management in Linux** Practice Lab.

Summary

You completed the following exercises:

- Exercise 1 - Using the Linux Terminal
- Exercise 2 - Navigate the Linux File System
- Exercise 3 - Create Files in the Linux File System
- Exercise 4 - Move Files and Create Links

You should now be able to:

- Launch the GNOME Terminal
- Customize the Terminal
- Use Basic Commands to Navigate the Linux File System
- Create Directories in the Linux File System
- Create and View Files using Linux Commands
- Move and View Files using Linux Commands
- Create Hard Links and Soft Links

Feedback

Shutdown all virtual machines used in this lab. Alternatively, you can log out of the lab platform.